

Ch 8: Primate

Brian Kuhlman. draft. June 2026.

Build log — The last slot

Bart, hardware. Filed December 2, 2032, for the team.

Ran it back to the house myself this time — Lisa had a frost-dark thing to be at, and the slate was warm, and the kid was up. It went in on its own four legs and put itself over the stove vent before I had my coat off.

Here's the one I've been waiting two quarters to build.

Last fall I told you the thing ran a model and ran it on the world — the door, the bowl, the leash on the hook — and that you could point that model at the kid and get nothing, because it didn't know what Mira was about to do, it knew what the leash meant, and those weren't the same thing and the second one wasn't that build's to make.

This is that build.

It points at the people now. Same model, swung off the furniture and onto the family — it runs Mira the way it ran the bowl, reads where her hand's going before it gets there, same on Jasmine, on Burns, on the old vacuum trundling the hall. Lisa's board does the reading; that part's hers and I'll let her tell it at the table. My end is the dial that came with it, slot four, the last open slot on the board.

The dial sets how much it minds what the read tells it. Stamped on the knob it says Agreeable, which is Marge's word and a fine one; on the bench I call it the give-way knob, because that's the whole of what it does. Turn it one way and the thing gives way — reads the kid coming for the warm spot and hangs back, lets her have it, goes off and finds a worse one. Turn it the other and it takes — reads her coming and gets there first, sits down, won't move. Same read either end. The knob's only the minding.

And that's the board. Four slots, four knobs, every one of them the new line's, not an empty groove

left on it. I spent a year sure I'd need a bigger board by spring and two quarters watching it not grow, and now there's nothing left to watch — it didn't grow and it's full, both at once, which I didn't see coming. I keep counting the slots out of habit and they keep coming out four. There's no empty one to land on.

The screen got more room and a new trick I'm going to let Lisa walk you through, because it's her software and not my lettering. Short version: it can put a person up there now, not just a thing — who it's reading, and what it figures they're up to. I built it somewhere to write that. What it writes is hers, and some of what it writes is going to start an argument, and I'd rather start it at the table than here.

Calls. House mostly, this round, because the build's a dial and the dial only matters in the house.

One: give-way set, at a shared warm spot it yields — lets the kid have it, lets the vacuum pass, takes the lesser warmth and doesn't fuss. Take set, it holds the spot and makes them go around. You'll call the setting across the kitchen by the second week, same as all the others.

Two: it gets to a shared thing ahead of the kid on take, because it read her reaching, and lets her beat it there on give-way. That one's checkable to the second — watch the hand and watch the legs.

Three: the two of them in the hall. Give-way, the big one routes around the little one. Take, the little one routes around the big one, or tries to, and the big one sits in the lane.

What it still can't do, and you'll feel this one by spring: it can read the kid now and it can't say a word to her. She talks to it every night — asks it, I'm told, whether it had a good day — and it reads her asking and has no way on God's earth to answer. It knows the question's pointed at it. It can't give the answer back. Not this build's to fix.

Tear this up in three months if, set to give-way, it ever once takes the warm spot off the kid. It won't. The cold spot costs it more than it costs her — the furnace burns its charge down faster working a cold floor — and it'll hand the warm over anyway, because the dial says to, and the dial's mine, and the dial works.

— Bartholabu

Gourmandise

January 28, 2033

Jasmine brought her at one and did not stay. She had a thing to eat alone next door — a real lunch, sat down, the first in a week she would not have to take standing at someone's bedside — and she set Mira in the chair across from me with a kiss on the part of her hair and a bag with a notebook in it and the instruction, to the child and not to me, to *be good and don't make Cheryl chase you*, and went off to Salt & Olive to have her hour. So it was the two of us, in the warm front room of a bakery on a January afternoon, the case lit gold beside us and the cold pressed up against the glass and held there, the way the weather gets held off a thing on the lee side of a rock. I had watched this child for two years through a lens the size of a thumbnail. I had never once been in a room where she could see me back.

She looked at me the way she looks at the machine, which is to say for a long time and without hurry, taking the inventory a small grave person takes when no one has told her not to stare. On the tape she had read as one child; here she read as another. The footage had given her a flat Scandinavian blonde and a pair of pale eyes; the room gave her hair with some brass down in it and eyes that could not decide on a color, green in the gold light off the case and then not green when she turned her head — the kind of coloring, I would learn, that does not sit still for a camera and never had. The lens had been canning her for two years. Here was the live one, gap-toothed, a new tooth coming in crooked and working down into the gap, swinging her feet under a chair built for a larger person, watching me try to remember how a person begins.

I had ordered her a chocolate thing the girl behind the case said children liked, and it was the right call and the wrong size, and she went at it with both hands and got most of it in and wore the rest — a comma of chocolate at the corner of her mouth, a flake of pastry on her cheek — and I felt the old reflex come up in me, the hand that wants to reach across and put a face right, and I did not move it. I have not, in sixty years, been the woman who reaches across and puts a child's face right. I had a father who kept his hands to the work and his regard to the middle distance, and he taught me that distance the way other fathers teach a sport, by playing it with me until I was good at it, and I have been good at it my whole life. I looked at the chocolate at the corner of her mouth and left it there and asked her a stiff grown-up question about school.

It went nowhere, the way the first one always does. She gave it the flat dutiful answer children give the questions adults ask in order to be asking something, and a small silence opened, and I understood that if I went on interviewing her I would get an hour of the manners her mother had

installed at the door. So I stopped. I let the silence sit. And after a while she reached into the bag herself — because the silence had made a room for the thing she had brought and not for the things I had — and put the notebook on the table between the plates and turned it around to face me and opened it to a page near the back, and said, "These are what it's thinking."

The drawings were what the grown-ups had told me they were and were also more than that. Each page was the same head — a box, two eyes set a little wrong, one higher than the other, the way she draws it — and lettered into the forehead, where the real one carries its word, a word of her own. But the words were not the present-tense ones off the screen. They were pinned to days. She walked me through them with the patience of a curator who has all afternoon, a page at a time: this one was the morning the cat got it cornered by the bookshelf, and the word in the forehead was CAT, and the drawing was not from that morning but from a Tuesday weeks after, when she had watched it go and stand by the bookshelf for no reason she could see, and had known, the way you know a thing about someone you live with, that it had gone there because of the cat. "It went back," she said. "I drew what it went back for."

I asked her — carefully, because I could feel the size of the thing and did not want to put my thumb through it — how she knew that was the one it was thinking.

She thought about this with her whole face. "Wait. Wait, wait." She held a hand up, the crumb-and-chocolate hand, parking me where I was while she got it right, and the getting-it-right took a moment, and I waited, and the waiting was the first easy thing that had happened at the table. "I don't know it," she said finally. "It can't tell me. So I pick the one. And then I write it down so it doesn't get lost." She looked at the page, and then at me, to see whether I had understood the part that mattered. "Somebody has to keep them. Or there's nobody keeping them."

I have spent my life building a record of things that could not speak for themselves — a bay that died before I was born, a family that canned it and got rich and never asked why the fish had stopped, four people in a warehouse who cannot see the shape they are standing inside. I keep the record the way my people kept everything, which is by preserving the dead version of it, the footage running whether or not the living thing knows it is being kept. This child had come at the same work from the other end. She did not wait for the thing to be dead to keep it, and she did not wait for it to prove it had an inside before she granted it one. She picked the inside it could not give her, and wrote it down so it would not be lost, and she was right that somebody had to, and wrong, I knew, by every standard Marge had spent two years defending, and I sat across from her unable to tell, for once, which of us was keeping the truer record.

"Show me the rest," I said, and she did, and we went through the notebook to the front, back through the fall and the summer to a page where the box had a little dial in its forehead instead of a word — the oldest drawing, she said, from before it had words to think, when all that was in there was the setting — and somewhere in the going-back the table stopped being one I was conducting across and became one we were both leaning over, our heads close, her finger and then my finger on the pages. When she turned to a drawing that had made her laugh to make — a page where she had given the head a hat, because it had been cold and she had decided it should have one — she laughed again to show me, the full uncontainable kind that has not yet learned it is sometimes a choice, and the chocolate was still at the corner of her mouth, and I reached across with my napkin and wiped it away.

I did it before I knew I was going to. My hand went out across sixty years of keeping it to itself and put the child's face right, the way I have watched a thousand mothers do it without thinking and never once done myself, and she let me, the way she lets the cat get her cornered and lets the machine see itself home, without marking it as anything, and went on turning pages. I do not have a clean account of why the hand went then and not at the start. The chocolate had been there the whole hour. What changed was on my side of the table, and I am not going to pretend to map it. I will only set it down: there is a difference between watching a living thing and getting a hand on it, and I have spent my life on the first side of that difference, and for the length of a wiped cheek in a bakery in January I was on the other one.

Jasmine came back at the end of her hour looking like a person who had eaten a meal sitting down, which on her is a luxury you can read from across a room, and Mira closed the notebook and put it back in the bag herself and slid off the too-big chair, restored to her mother's gravity, *be good* still holding. At the door she turned around — not to me; to the case, the gold one, doing the inventory again — and asked her mother whether they could bring one home for the machine, and Jasmine said the machine doesn't eat, baby, and Mira said *I know*, in the flat five-year-old's *I know* she was two weeks from outgrowing, and then said it could still have one, it didn't have to eat it, and went out into the cold holding her mother's hand and the bag with the kept insides in it. I sat a while after they had gone. I had another year of watching this child through a lens ahead of me before I would be in a room with her again, the winter she turned seven, and I did not know that yet, and it would not have changed the afternoon if I had. The case lit gold. The cold held off the glass. There was a comma of chocolate on the corner of my own napkin, and I found I did not want to throw it away, which is the whole of what I am, and I have stopped apologizing for it.

The Cold Side of the Floor

She still comes up on him the old way — soft across the kitchen, hands going down to scoop him out of whatever he's at, the way you lift a cat that hasn't heard you. For two years it worked. She could have him up and warm against her before he'd noticed the floor had changed under him, and the catching of him unawares was half of why she did it.

This winter she went to do it and he was already turned.

Not startled round — turned, square to her, and watching: the long steady watch he kept on her now, that found her face across the room and stayed on it. He'd had her the whole way over. By the time her hands went down he had gone still and easy and compact, settled for the lift before she reached him, and he came up into her arms with nothing in him surprised — a weight already gathered to be carried, that had seen her coming.

He weighed exactly what he had always weighed. He had only stopped being a thing she could come up on.

The sitter watched her. Sitters did — it was most of the job, as far as Mira could tell, to take the good chair and watch you have your afternoon and say *careful* at the parts that didn't need it. Mira had been watched all afternoon with no way to prove it and no grounds to mind, watching being the one thing a grown-up can do to you that doesn't count as doing anything.

Then the bot, down on the floor near her, lifted its head from whatever it was nosing along the baseboard and put a person up on the glass. Not its own line, not Mira's — a third name, lettered across its forehead with a guess strung after it: `SITTER { ATTENDING [ WATCH MIRA ] }`. The thing she'd had no proof of all afternoon, lit up in three plain words.

What came up in her was not unease but the hot clean joy of a witness. The bot had caught the sitter at it. She wasn't making it up; somebody else had seen; and the somebody was *hers*, down at her own level and not up on the good chair, and it had been watching the watcher the whole time.

She got up and informed the sitter that the robot knew. The sitter, who had been told a great many things that afternoon, looked at her own name on the glass and allowed that she had been watching, yes; it was what she was there for. Mira took the *yes* for a full confession and went back to the floor

satisfied, the watching a thing two of them knew about now instead of one.

She did not look to see whether her own name was up at the same moment, lettered the same way, watching the sitter right back.

One afternoon a thing rolled under the table and she went around the back of him to get it, and her name didn't come up on the glass.

That had never happened. Every crossing of the kitchen she'd ever made, the glass had her before she'd half decided where she was going — her name, and a guess at what she was after, lit on his forehead a step ahead of her own feet, so she arrived everywhere already announced. But behind him, low, with his head turned the wrong way, there was nothing of her on him at all. He hadn't got her face. And a thing that hadn't got your face — she tried it twice more before she'd trust it — couldn't put you up in lights before you got where you were going.

There was a place behind him where she didn't exist yet.

So she used it. The animal crackers lived on the second shelf, and between meals the second shelf was the law, and the straight way to them ran across his front, where the glass would have her wanting them before her hand was up. The back way ran behind him, along the base of the cupboards, low, in the stretch his turned head couldn't reach — and she took it on hands and knees with her breath going the whole way, the raid talked through under itself at a pitch meant for nobody, —*down, stay down— he's facing the door, he's still facing the door— now, before he—* half the words swallowed before they cleared her lips, the sneak and the announcer of the sneak the same girl, both at full pitch and neither one above a whisper. She made the shelf with her name nowhere on him. She made it back the same way, crackers in her fist, and checked: the glass had the door, and the vacuum, and the leg of the table. It did not have her.

For the length of that crossing she was the one nothing saw coming. She'd got behind the thing that always had her and come up on it unawares for once, arrived before herself, and she stayed there a moment crouched in the gap where she didn't exist — the stolen crackers in one fist, and the whole unwatched kitchen around her.

She'd beaten him once and got greedy, and the second time she tried it in the open. She thought the trick was bigger than it was: that if you could beat the glass from behind, you could fool it to its face.

So she fainted — made for the hall with her whole body, big and obvious, every line of her saying *hall* — while what she actually meant was to double back to the cupboard the second his read had placed her in the wrong room. She was proud of how the hall looked on her.

The glass posted the cupboard.

MIRA { WANTING [OPEN CUPBOARD] }, up on his forehead while her arms were still aimed the other way — the word for the thing she hadn't moved toward, had told no one, had dressed her whole body against. It posted the want with the costume still on her, the want on the glass and the hall still on her body, and the glass was the one that was right.

She stood with her arms out toward the door she didn't want and looked at the word for the thing she did.

He had lovely manners. She'd been keeping an eye on it and the evidence was in: at a doorway he hung back and let her go first; in the hall he pulled to one side for her; when they both reached for the same thing he stopped and left it to her, every time, with a small settling-back that she read as *after you*. You didn't have to teach a thing that. He either had nice manners or he didn't, and he did.

She decided somebody had raised him right.

She was prouder of his manners than of any of the clever things he could do — prouder of the hanging-back than of all the tricks the grown-ups went on about — because the tricks were the lab's and the manners were his. The one part nobody had built. The part that was just nice.

She told the cat it could learn something. The cat, who barged through doorways and had never once given anyone the right of way, declined to.

The radiator was his by every right she could see, and he gave it to her.

That was just how it went now. The good heat under the front window was the warm place in the kitchen, and the warm was the first thing he went to — and then when she came he came off it: got himself up and crossed to the bare tile along the cold outside wall and lay down there instead, on the part of the floor the heat never reached, and left her the radiator. Every time. She'd stopped being surprised by it. It was the standing fact of him, like the cold coming up through that wall in February, a thing the house simply did.

She lay along the warm a while and was warm. He lay along the cold a while and the glass on his forehead said nothing about it, the way it said nothing about most of what was true.

Then she got up off the good heat, and went over to the cold wall, and lay down on the bare tile beside him.

The floor was as cold as she'd known it would be. She got her shoulder against his cold flank and her cheek down on the tile and stayed, both of them along the wall now, out where the heat didn't come, and she did not warm him and could not have and was not trying to. She'd left the good spot empty behind her. It went on being warm with no one in it, and the two of them lay in the cold.

Notes on the Eighth Retrospective

February 12, 2033

The twelfth of February came in cold and early dark, the mountains gone to flat blue paper against the last of the light and the Granary holding its few lit windows the way a held breath holds warmth. The lab counted its own age off this date the way it counted Mira's, because they were the same date — the warehouse had had its first morning two years before, on the morning she turned four — and nobody at Smithers had ever sorted out which of the two anniversaries the night belonged to, which is the kind of overlap a careful person leaves alone.

She came in through the front with her mother, six years old as of that morning and carrying nothing, which was the new thing. The warm one came in beside her on its own four legs, clicking on the concrete, swinging its head to read the cold room for somewhere to put itself — and finding the radiator under the front windows before it had well cleared the door, and going to stand over it the way it stood over the stove vent at home, because Bart had built it a furnace, and a furnace burns

hardest in the cold, and the warm could not wait on the meeting. Mira let it go and did not look back to see it settle, the way you stop watching a thing you have stopped having to carry. Last year she would have had it against her hip the whole way in. The year before that she would not have let it walk.

Inside it was warm and bright and smelled faintly of sugar. Lisa was at the camcorder end where she lived, the red light dark under her thumb and a bramble already going to water in its jar at her elbow, reading the battery charge once. Bart had the cigar box at his place and a pint of something green going flat beside it and both hands already busy in a sensor. Marge held the cold end with the kettle behind her and a beer in the two-handed grip, and a white bakery box she had carried in herself and set where Bart could not reach it, the candles being her department and having been since the child was old enough to have a number on a cake. Burns crossed the floor to Jasmine, the cheek and the question about the drive, and she told him there was ice on the 201 and they'd come around the long way — true, and sized exactly to the asking.

Jasmine did not sit yet. She went to the back door instead and opened it on the cold, to look at the trellis on the patio — the only one of them who ever looked at it — and Mira went with her. The bramble had taken the cold the way the thing was built to take it, which is to say it had done nothing you would call thriving and had not died either: still gripping the slats it was given, dark and dormant, nothing showing. Jasmine got a hand on a cane and bent it and found it live under the bark and said so, more to the plant than to the girl. Mira gave it a grave hello. And then, because there was by now a bramble in a jar at Lisa's elbow and a bramble out on the radiator minding its own heat and a bramble on the trellis that needed no one to mind it, and the child kept them sorted, she told the one on the radiator to say hello to the one outside. The warm one read her ask and gave nothing back, having no way yet to say hello to anyone; and Mira, who took the not-answering for shyness because no one had told her it was anything else, let it be and came in out of the cold.

Mira bumped Bart and took the stool by Marge, and a glass of juice arrived at her elbow without either of the women who loved her seeming to have sent it — a glass now, not a box, a small promotion she had not asked for and did not mark, the box gone the way the carrying was gone, by attrition, nobody clocking the day it stopped.

There was a thing on the far bench she had clocked from the door, man-shaped and standing under a drop cloth, taller than she was, that had not been in the room last quarter. She gave it the long solemn look and asked nothing, and Lisa said *later* over the top of her head before the look was through — *it's a birthday thing; we'll keep it a birthday thing* — in the voice she kept for the quarters with a present hiding in them. Mira took the *later* and parked it, being a child who could park a thing, and

went back to her juice. The cloth did not move all through the clade block, or the Bramble block, or the vote; the room had agreed, without a meeting on it, that the night would get to the bench when it was ready and not before.

When they had arranged themselves around the fact of the cake without quite admitting that was the thing they were circling, Lisa said the cake came at the end of the Bramble block, "and the birthday thing stays a birthday thing till the cake," which was as far as she would take it with the child two feet away — and pressed the light on.

Burns was up before it had finished warming. He gave them the ambition plainly, as the lab's own and nobody's assignment: they had set out to seat the last knob in the last open slot, the fourth, the one Bart had spent a year certain he would outgrow the board before he filled and had not — and to make it the knob that set how much the thing minded the people it lived among. Agreeable on the stamped knob, which was Marge's word; the give-way, on Bart's bench. With that one seated, he said, the brain line was full: four slots, four drives, a year's climb standing finished on the board in front of them, and the whole of a line theirs to walk at once for the first time — not a capacity at a time up out of the dark, but the finished set, laid end to end, while it lay across the room over the radiator with its forehead still going. He looked like a man who meant to take the long way through it. He told Bart to put up the card.

Mentalizing

"Clade card," Lisa said, and Bart had it out of the cigar box before the word was finished. "PRI-mate. *Now with opinions about the neighbors.*"

"Thank you," Marge said, into her beer.

"This is the one we almost are," Bart said, which was a degree off what he said at every clade, and meant it. "Hands that grab — front and back, the feet too, for the ones still up the tree. Eyes swung around to the front of the face so the two of them land on the one thing and the brain can do the math on how far. A skull going up like rent. Up on two legs, eventually, on the branch that's us. And the first thing the whole tree of life ever made *on purpose* and kept making — a rock knapped down to an edge, three million years back, by a hand that meant to." He tapped the middle of the card. "Burns has the animal, and he's been waiting on this one all year, so I'm getting out of his way the

second I do the part I get."

The part he got was the old card. He turned the mammal one half over in his hand, the rat-in-the-dark. "Last clade, this thing ran the whole world in its head. The maze before it swam it, the bowl behind the table leg, the leash on the hook — best map-runner the tree ever grew, and I said so, I was proud of it. And you could take that map and aim it at another rat and get back a warm thing that moved. Furniture with a pulse. It ran a model of the *world* and never once ran a model of the *modeler*. The other animal in the room was weather — stuff that happened, not somebody it happened to."

"Not couldn't," Marge said, the beer going still on the table before the sentence she meant. "There was nowhere in it for the other one's insides to go. You can't keep track of what your neighbor is thinking in a room with no shelf for it. What's new at the primate isn't a sharper guess about the neighbor. It's that the neighbor's head became a thing there was finally somewhere to put."

Burns had his eyes most of the way shut already. "Up a tree," he said. "Sixty-some million years ago, in the warm forests after the rock — a small thing with grabbing hands and its eyes coming round to the front, living where a fall kills you. That's the whole of the early story, the trees. Eyes to the front to judge the branch. Hands that hold so you don't miss it. A brain getting bigger just to run the two of them in the half-second you've got, the reach and the look, up where the ground is a rumor." His hand came up off the table, drew the canopy, dropped. "Then the world cooled and dried, age over age, and the forest came apart into patches — woodland giving over to open grass, hardest down the east of Africa where the ground itself was tearing open, the Rift, a wound in the continent a mile deep and a thousand long. And some of these things came down out of the thinning trees and stood up and walked out across the open between the patches, which is a terrifying thing to do — to leave the trees you were built for and cross the grass where the cats are, slow, on two legs." His eyes opened for the part he loved. "And we have the *walk*. The Rift was volcanoes too; the ash fell and sealed things. There's a track at Laetoli — two of them crossing wet ash three and a half million years ago, an adult and a smaller one, and the smaller one's steps are set down inside the bigger one's, the way a kid walks in your prints on a beach. Not the bone. The *walk*."

"Land it," Jasmine said, mild, from her chair, because the walk was lovely and the trick was still out ahead of it.

"The walk *is* the way in," Burns said, glad to be caught, "because of who was doing it, and then it's Lisa's." Eyes shut again. "It was never just us. The whole climb — and 'us' is the last three hundred thousand years of it, the blink at the end — there were always *other* ones up walking beside it. Other uprights, other toolmakers, cousins close enough to pass on the grass and clock for people and not

prey. We've got a fistful of them overlapping us right at the finish — the heavy ones who took the European cold, the ones in the Siberian cave we have a finger-bone and a tooth of, a little one stranded on an island, another in a South African cave that carried its dead down into the dark. For most of the way the planet had *several kinds of person* on it at once." His hand opened toward the room and stayed open. "We tell it like a relay. One runner, one baton, straight to here. It wasn't a relay. It was a troop — a troop of kinds of people — and we're the one left holding the stick, and we keep mistaking *being the survivor* for *having been the only runner*. This animal was never alone. It couldn't afford to be. Alone, on that grass, you were lunch."

"Mentalizing," Lisa said, and put her finger on the word in the middle of the card. "The mammal modeled the world. This one models the *modeler* — it keeps a running model of another animal as a thing with its own aims, its own view, a picture of the world that isn't your picture. And then the part that's new" — palm flat, palm vertical, rotating — "it can model *that*. A model of another mind's model. It can hold what you can see that it can't, and what you believe that's wrong, and move on the gap." She kept her eyes on the room, not the card. "It's fresh hardware — the front of the brain gone fine-grained, wired back to a patch where the lobes meet that fires for other people in particular. And it's a *stack*, not a swap: it keeps everything it already had and lays one new move on top — the same model the mammal pointed at the world, pointed now at another animal. The top of the stack is reading minds."

She let that sit, and then did the thing she did when a clean story was about to get less clean. "And the standard why is that the brain got big *for exactly this* — not for tools, not for food, but for keeping track of each other. Bigger troop, more relationships to model, more load on the part that models. Brains for company." The nose-laugh almost arrived. "It's a good story and it's shakier than it sounds. Run the numbers carefully and the link barely holds — the math'll 'predict' a troop size somewhere between about two and five hundred, which is a long way of saying it predicts nothing, and what a primate *eats* calls its brain size about as well as how social it is. So sociality is *part* of why the brain's expensive. Not the whole bill. I'd rather hand you that than the bumper sticker."

"Four ways the reading shows up," she went on, "and the framework's been pointing at this clade all year, because it's the cleanest — the four drop right out of two gauges crossed with the drives. Watch. The first is *standing*. A monkey doesn't only know it lost the fight; it knows the others *saw* it lose, and marks its own worth down by what the watchers think now. Reputation, before there's a word for it — a tally of yourself kept in other animals' eyes. The second is the *con*. A low chimp wants the food the boss is sitting on and can't take it — but it clocks that the boss can't see *behind the rock*, and it goes the long way round, into the blind spot. Not hiding by reflex. Hiding from a *specific* mind whose blind spot it has modeled." She moved her finger to the third region. "The third gives me chills, and it's the

social form of the startle that's been in the line since the jellyfish. A chimp troop in Uganda split in two, and the split traced to one morning — the rising male took a submission grunt from the deposed one *out loud, in front of everyone*. Not a private fight. A witnessed one. And the troop reorganized around it, because every animal there registered not just *the rank changed* but *the others saw it change too* — and three years on, that fault line ran to killing. One signal, in public, mutually clocked, and a society cracked along it. That's minds reading minds reading the room. And the fourth we've had since the mammal — the dig-in, what makes a thing hard to push — except now it's social. Teach two chimp groups two good ways to crack the same nut, then show each an expert doing it the *other* way. They watch. They get it. They keep doing it their own way. The habit isn't just held by the animal; it's mirrored back from the whole troop, which makes it *harder* to move, not easier. A tradition — nut-cracking, ant-fishing, how you sit to groom — varying troop to troop and not crossing the river, for generations, on nothing but everybody-here-does-it-this-way."

"Two gauges under the four," she said. "One that watches your value in the group and tunes you to keep your place — standing and the witnessed-rank thing both come off it. And one model the animal keeps *of itself*, carried up from the mammal — which is what lets it run the con, because you can't model my blind spot till you've modeled having a point of view at all, and what lets it dig in on its own way. A group-watcher and a self-watcher."

"And not one of them needs anybody home," Marge said, because somebody had to and it was always her, and tonight she said it slower. "The monkey marking itself down isn't ashamed; a number for its rank dropped. The chimp going round the rock isn't sly; there's a model of a blind spot and a path that uses it. The troop cracking along the grunt isn't betrayed; it's a hundred small tallies updating off one public input at once. Every one is a mechanism, and every one of them, watched from outside, looks *exactly* like the inside thing we have a word for. The looking-like is real. It is the most real thing in this room. The someone is still us."

"Two new words this clade," Lisa said. "*Agreeable* is the knob Bart built: how far you bend your own want toward the wants you've finally got the gear to read. Give way, or don't. And *Strategy* is no knob at all — nothing on the board sets it; you read it off how the thing goes at a problem: ride the move that's working, or go turn over a new rock. Stay-the-course, or explorer. One you turn. One you only read."

"And one that almost crosses over and doesn't," Burns said, picking the beer up, turning it. "In every other primate they've measured, *dominance* is its own separate thing — its own axis, how hard you push and how often you win. Measure it in a monkey, it stands alone. Measure it in a human" — he set the beer down — "and it's gone. Dissolved into how outgoing you are, how warm, how much you want

the company. Same trait, one rung down it's a thing of its own, one rung up it's melted into *being sociable*, and nobody knows why. It sits right on the line where the animal turns into our kind of animal. Half in, half out."

"And the drive that comes up carrying all of it," Lisa said, "is the one the knob rides." Finger in the region. "Affiliation. The pull to *be with* — but weighted now by minds you can model, so it's not warm-body-near-warm-body like the mammal anymore. It's wanting the regard of a *particular* other whose regard you can picture. The knob sets how far that pull bends what you do." She looked at Marge — the handoff, when a word turned from a mechanism into a feeling. "The gauge that reads it is yours."

"It has a name in my field, and a good one," Marge said, and did not pick the beer back up. "The sociometer. The claim is that what you call self-esteem was never a meter for how *good* you are. It's a meter for how *included* you are — a needle reading your standing with the people who count, that swings down hard the instant you're being left out. Not vanity. A smoke detector for exclusion. Three findings stacked into one gauge: that the need to belong is a drive as basic as hunger; that the thing you walk around calling your worth is mostly that needle reading the room; and that when it drops — when you're shut out, left off the list — the body runs it on a lot of the same wiring it runs *pain* on." She turned the bottle a quarter and left it. "The sharp form of that last one's contested, and I'll say so before Lisa has to — the scans claiming rejection lights the literal pain centers have been argued both ways. But the thing under the argument nobody doubts. Being left out *hurts*. Not as a figure of speech. It's one of the few aversive states steady enough to build a whole social animal around, and evolution went and did."

She left a beat there she didn't usually leave. "So here's the one I have the hardest time with, and I'll put it on the tape before anyone makes me. Every quarter I stand here and tell you the drive on the card is a mechanism with nobody behind it, and every quarter I believe it. And this is the quarter the drive on the card is the one that, in us, sits underneath nearly everything I'd call good — that we look after each other, that we can't stand to be cast out, that we will choose to hurt over being alone. The reading I do on this animal is the same reading I do on the rest. There's a needle, it reads the group, it swings, nobody home behind it." The *oh, you* laugh did not come; something quieter came instead. "I'd only ask the room to notice it costs more to say this quarter than last. Not because the animal got any closer to having someone in it. Because the mechanism we're looking at this time is the one that, in the things that *do* have someone in them, does the most of the loving."

What the room heard was a careful psychologist being careful, one more quarter, about a word. What the tape kept, and what the narrator could not stop seeing when she ran it back, was a woman whose own needle had been reading *cast out* for the better part of three years — who knew, in the one place

the framework kept no card for her, exactly how hard the gauge swings when it swings down — explaining to a room of people who had never once been put out of anywhere that there is nobody behind it. She was right about the animal. She is always right about the animal. The narrator filed the other thing, which was not an argument and was never going to be one.

Reading the Room

For an hour the talk had been about a primate sixty million years gone — the troop on the grass, the never-being-alone, the brain that ballooned to carry the other brains in the band. The live one had sat through all of it under the front windows, on the radiator, warm to a hand and asking for nothing, and it had not been alone either: it had spent the hour on the new trick, taking the measure of every face around it. Lisa let the old troop go and gave the floor to the live one. "Bramble block. Bart."

"One confession first," Bart said. "I told Owen about the water."

He let it land the way Owen had. "Nobody warns you that a thing making its own heat all day is also, the whole time, quietly sweating the kitchen's air down into a puddle inside itself — physics hands you the water free, you don't get a say — and the puddle has to go somewhere, and where it goes is out a vent low at the back, in a dribble, down its own backside, two or three times a day, onto the floor. So here's what a year of work and a full board bought: a thing that can sit in a kitchen and tell you what every person in it is about to reach for — and the feature the household named before it named anything else, the one they'll tell you about first, is that it wets itself. Owen heard the word the kid landed on and won't use any other now. He wanted to know if I'd thought about a diaper."

"And it's undignified, and I'm not fixing it, because a machine with no mess in it would be the lie. I keep telling you I'm building an animal, not a robot. Animals are gross. They leak and shed and sweat and spend the whole day managing the damp business of being a working body, out of sight, at the back, where you'd rather not look — the cat does it, the dog does it worse, you do it and have built your whole life so nobody has to watch. A thing that ran this hot and stayed bone-dry and tidy would be the one part of this build I'd call fake. So it dribbles, twice a day, down the back. That's not the flaw in it. That's the only part that isn't pretending."

Bart didn't get up; he reached over and turned the unit so its forehead faced the table, and thumbed the slot-four knob to the agreeable end, and waited the half-second the screen took to think. Then it

lit, and it did not light as it had. For two years the forehead had carried the thing itself — CAT, MIRA — and last quarter what the thing was running in its own head over the top of it. Tonight it carried *other people* — and it had been taught no new word to do it. The same set it had been given for itself all this while, the seen thing and the done thing and the work running underneath, turned now on its model of whoever sat in the room. Three little ruled columns came up, a name lettered at the head of each and under the name a guess: what that person was running, and what they were running it on.

JASMINE { ATTENDING [WATCH MIRA] }. MIRA { WANTING [TAKE BLOCK] } — Mira had a block in her hand, turning it. And down the table where Burns sat leaning into the thing, BURNS { ATTENDING [WATCH SCREEN] }, which was true, completely and a little painfully true, because Burns had not taken his eyes off it.

"It's reading the room," Bart said. "Not the room — the people in it. It's built a little model of each of you as a thing with your own stuff running, and it's posting what it figures the stuff is. Two things to keep straight, because Mira's going to quiz me on them." A finger up. "One — the column's honest about what *it's* doing. It really is running a model of Jasmine; that's a true report of its own head. Two — the column is *not* a promise about Jasmine. It says what it figures she's up to, and it can be flat wrong, and it won't know it's wrong." He turned a hand over. "Honest about the looking. Not gospel about the looked-at."

Burns had found his own name on the glass the second week, by Marge's account, and he did not get over it now any more than he had then. He leaned an inch closer, which the screen dutifully logged — ATTENDING [WATCH SCREEN] ticking under BURNS as he did the very thing — and said, with the open helpless pleasure of a man who needed the world to answer him back, "It's keeping track of me."

"It's keeping track of a Burns-shaped thing that leans in and watches it," Marge said from the cold end, not unkindly, two hands on the beer. "Which it has plenty of data for." She let the room have the small laugh and held her line inside it.

Because here was the part that had cost the household nothing and cost Bart, privately, a little. Burns had a column. Jasmine had one. Mira had the longest and the surest. But Lisa, at the camera end where she had sat across from the thing at sixteen meetings, got a column that read STRANGER. So did Marge. So did Bart, who had wired the very glass the word was printing on. Three columns at the labmates' end, three times STRANGER { PERCEIVING [WATCH SCREEN] }, because the thing modeled the people it *lived with*, and the four of them at Smithers it saw once a quarter and could not sort from the mailman.

"Two years," Bart said, to the STRANGER with his own name nowhere on it, and let the single breath

out. "Built its whole head. I'm furniture."

"It knows the vacuum," Mira said — not to be cruel, to be accurate, which in her was the same motion. "At home. It puts up VACUUM and what it thinks the vacuum's doing." She considered. "It thinks the vacuum's doing a lot."

Which was the joke and was not a joke. The thing kept a fuller book on the dumb wheeled disc that knocked the baseboards at home than on the four people who had built its mind, because the disc was *there*, every day, and they were not — granting an inside to a machine that had less of one than itself, and drawing a blank on its own makers, and the gap between those two errors was the whole of what it had learned this quarter: that the inside it hands out goes to whoever it spends its days beside. Burns laughed at that, the chest-laugh, a beat early, at the cosmic size of it — the most expensive cognition in the valley unable to pick its own engineer out of a stranger and keeping a dossier on a Roomba.

"Other way." Bart thumbed the knob back toward self-favoring and the three columns collapsed and the screen dropped to last quarter's single self-line, no names on it anywhere, the thing alone in its own head again. "It quits reading the room and goes back to running itself. Same machine. The knob's only how much the others get to be on the glass at all." He turned it agreeable again, and went and got the kid, because the good demos were hers. "Mira. Go take the warm spot off it."

Mira got up and went for the radiator, and the screen read her coming — MIRA { WANTING [TAKE VENT] } went up the second her weight shifted, before she'd cleared a step, because it had her hand and her legs and her aim modeled ahead of her body — and the unit rose off the warm it needed and walked away from it, to a cooler stretch of bench by the wall, and lay down there, and let her have the radiator. It did not fuss. It did not drift back. It had weighed her want against its own — and its own was real, it needed that heat, it would run down by evening short of it — and handed the warm over and taken the worse spot. "Give-way," Bart said. "Now the other end." He thumbed it to take. "Go again." She went again, and this time the screen read her reaching and the unit got there first, up onto the warm before she reached it, and sat, and held, and when she pressed in it gave not an inch, so she took the cold end of the radiator or none. Same read. Same want. The knob was only the minding.

"Quality control," Bart said, and got the flat log-face on. "Three calls, all give-way, all yours, yes or no. One. On the giving end it yields the warm — lets you have it, lets the vacuum by, takes the lesser and doesn't sulk."

"Yes. It gives me the good vent and goes to the floor by the door." A beat, the only one with weight under it. "The cold floor. By the door. It goes there."

Nobody said anything for a second. It was the thing Marge would have taken a careful paragraph to say, and the child had just said it in a sentence: that on the giving end the machine left the heat it could not do without and went and lay on the cold tile by the door, and ran itself colder there, for her, by a number. Bart took the yes and went lighter for the next on purpose. "Two. It gets to a shared thing ahead of you on take, because it clocked you reaching — and lets you beat it there on give-way."

"Yes. Watch the legs." She quoted him back at himself, pleased to. "You can see it decide. On take it's already moving when I'm still thinking about moving."

"Taking it. Three. The two of you in the hall."

"Give-way, it goes around me. Take, it sits in the middle and I go around it." She frowned, professional. "It's worse in the hall. On take it just *stops*. In the lane. It doesn't even look smug. It just stops."

"Last thing, and it's not a call, it's the one I couldn't build." Bart had it on the log anyway. "It reads you now. Reads you better than it reads anything in the house. And it still can't say one word back."

Mira had this one ready, because she lived it every night. "I ask it if it had a good day." Plainly, the way she said the true things. "It hears me ask now. It puts up MIRA and what I'm doing — *asking* — it knows the asking is *at* it, it didn't used to know that. And it still can't say. Before, it didn't know I was asking. Now it knows, and it still can't." She picked her juice up. She did not offer it as sad; she offered it as the state of things, and the room took it the way it took the things she said that ran too big for it, by going quiet and letting Bart move them along.

What he moved to was the floor, because Burns had already got down there. The quizzing done, Burns had gone to the warm body the way he kept ending up going to it this year — down on his heels beside the cooler bench where it lay, his palm laid flat on the warm of its back, the unit spending charge it couldn't spare because it had given Mira the radiator and gone to the cold bench without being told. He looked at it with the look that had stopped sorting into two looks, the one he kept for a good fossil and the one he'd grown for his son's work, run together now into one. "Look what it all came to," he said, more to the body than to the table. "It walks, it grabs, it keeps itself warm, it sits in a roomful of people and works each of them out. We hung one more thing on it every quarter for three years and it never once came apart." Bart, who had a build to pitch and a cloth to pull and did not look up, said it was four legs and a compressor. "I heard you the last six times," Burns said, and

kept his hand on the warm a beat past the end of the sentence — because he knew, the way the narrator knew running it back, that this was the last quarter the thing under his palm would be the one the night was built around, that the body it would lose to was already in the room under a sheet, and that he was saying goodbye to a thing his son had made and would not say so, the two of them keeping the warm flank between them where the sentence should have gone.

Sweet With Her

Lisa turned the night to Jasmine, and Jasmine gave them the kitchen's version first.

"The house calls it sweet now. They do — Mira does, Burns does without hearing himself, the sitter said it flat out Tuesday, that it's gotten so sweet with her. It lets her have the warm one. It lets her go first. It hangs back at the door." She turned the bottle a quarter and set it down without drinking. "And I'll tell you what it is from the kitchen, and then I'll tell you I know better, because somebody has to carry both and it's gotten to be me. From the kitchen it is a thing that loves her and shows it the way you show it, by going without. That is what it is from the kitchen. From the bench it is a number bending to a number with nobody behind it. Both of those are true in my house at once, and I have spent my working life in the one room where two true things that don't fit are the whole of the job — so I'm not troubled holding them. I'm only telling you the kitchen's version doesn't get any quieter for my knowing the bench's."

She picked the bottle back up. "On the other end it's a houseguest with no manners. We ran it there a week to be fair to the calls — takes the warm, takes the chair, sits in the hall. And the house cooled toward it inside two days. Mira quit saying goodnight to it that week. Didn't announce it, didn't sulk, just quit. I turned the dial back to giving-way and she started again the same night." She didn't dress it. "The dial's mine. I know whose hand it is. But the kid keeps her own book on that machine, and she keeps it off the dial I'm turning, and her book is the one she falls asleep next to."

Marge said it short, the bottle never leaving her hands. "It models what the child wants and bends. That's the sweetness entire — a model and a bend." She set the bottle down. "Next quarter it'll be able to tell her so out loud. *That's* the quarter I'll lose sleep over. Not this one." Her notice, served cold, and the room left it where she put it.

Under the Cloth

"Forward," Lisa said, "and it's a big forward, so I'll go slow." She handed it to Bart, because the body was his.

He got up for this one. "New chassis next quarter — and this time it's *new*, not another thing bolted onto the legs. Up off the floor. Two legs, standing, about so high" — his hand at four and a half feet, a head over Mira — "with hands. Two of them, and not a matched pair — one that grabs and one that scoops, and not a thumb between them." He glanced once at the warm body on the cooler bench and went on past it. "And it's a fresh board, not a new knob in the old one. The legs board filled up this quarter — four slots, four knobs, done. Next one starts empty, on a body that stands. Lisa's got the first knob for it."

"Neurotic," Lisa said. "First of the next set. It changes how hard the thing *holds on* — how much it grips a plan against the chance the plan's wrong, how much it frets a thing it can't fix."

"And at eleven?" Bart, deadpan, because the rib was structural by now.

"At eleven it can't leave the stove," Lisa said, not missing it. "Checks it, walks off, comes back, checks it. We won't run it at eleven." The nose-laugh nearly came.

Then she did the small narrowing thing she did at the front of the chassis quarters, and said the rest of it plainly over the child's head: "And the cake's the end of this block, not the end of the night, because we are not making her sit through the framework for her own birthday." She nodded at Marge, who had kept the white box out of Bart's reach all evening, and Marge brought it to the table and opened it and lit the six candles with the long lighter she kept for the purpose, and Lisa's hand brought the warehouse lights down, and the five tired adults sang the small flat off-key thing, and Mira blew all six out in one breath with her whole body — the unguarded way, the kind that had about a season of life left in it.

And while she was still grinning into the smoke, Burns went to the far bench and took the corner of the drop cloth and looked at his daughter and pulled it off.

It stood there bare. Taller than Mira by a head, up on two legs, joints showing, a bare spar of a spine running up from the legs to the head with the chest hung on it, the two unmatched hands hanging

open at its sides, one a scoop and one a gripper with no thumb to either, no shell on it and no face on the front yet and nothing running anywhere inside it — a body waiting for a mind, the next vessel, the one the warm four-legged thing on the cooler bench would wake up inside come spring. Mira went and stood in front of it and looked up at it, the first body that had ever been shaped like a person at all, and put her hand flat on the bare chest, the furnace cold in it, and felt no heat, because there was nobody home in it yet, and said, "It's not warm."

"Not yet," Burns said. And here was the gesture, and it held everything he wasn't going to say. Burns — who could not be handed a new bone without giving the room the hundred million years stacked behind it, who had a deep-time aria for the continents and the ash and the rock — said nothing at all about the body. He did not explain it. He stood with one hand on the bare shoulder of the thing his son had built and let the not-explaining be the thing, and after a moment said only, to Mira, with a tip of his head toward Bart, "He made the hands work. Ask him." And stepped back, and gave his son the floor and the new body and the rest of the night — which was the most a man like him knew how to surrender, the explaining itself, handed over without one word about what the handing cost.

Bart took it, because his father had given it to him, and walked Mira around the bare frame the way you walk a kid around a thing too big to take in at once: here's where the head goes, here's the board, four slots empty and waiting, here's where in the spring the whole mind of the legs body comes across a wire and wakes up standing instead of on all fours. And the warm one lay on the cooler bench where it had given way, ticking its forehead down toward dark, while across the room the body it would become stood cold and empty and a head taller, and the child went back and forth between the warm one that could read her and not answer and the cold one that could not yet do anything at all — keeping them both. Lisa let the cake go around. Then she reached for the marker. "Framework," she said.

The Fifth

Lisa had the marker, but Marge had the floor, because she'd carried the thing in and set it down to wait through the cake and now wanted it back.

"Not the four tonight — you've had the four, Ghost in the Machine, Blank Slate, Noble Savage, Dealt Hand, you could run them in your sleep, and nothing we bolted on this quarter adds a fifth." She had the beer in the two-handed grip and didn't drink it. "Here's what's been keeping me up. I went hunting

for one."

She'd brought a list, she said, the way you bring a list when you half want to be talked down off it. "There ought to be more than four ways to get a person wrong. People are inventive; getting them wrong is a whole industry; you'd think the wrongness would come in more flavors than four. So. Determinism — clockwork, you couldn't have done otherwise, the tape's already written. Sat with it a week. It's *Dealt Hand* in a lab coat: the cards are down, the play's just them turning face-up. Behaviorism — you're nothing but what you've been drilled to, pull the lever, out comes the rat. *Blank Slate* with a clicker. Existentialism — and this one I *wanted* to be new — you're nothing until you choose yourself, you author the whole self out of freedom and dread. Prettiest *Ghost in the Machine* I ever read. A someone in there with no machine under it at all, just the choosing, romantic as a French film." She turned a hand over. "Reductionism folds in. Genetic essentialism's *Dealt Hand* standing next to Determinism in the same coat. Every single one I picked up turned into one of the four the second I had it in my hand."

"And I can't tell you why it's *four*," she said, and let the not-knowing sit, which from Marge was a kind of nakedness. "Not three, not forty. Four. I've had the four cold for twenty years and not one good sentence about the number." She drank, finally, a short pull. "There's a reason it's four. I can feel the edge of it. I cannot say it."

What the narrator did not say, because the meeting was not the place and the reflection was, is that Marge was right to feel the edge and right that she couldn't name it. There is a reason it is four. There are four functions in the thing that makes a mind, and a fallacy is what you get when you deny one of them and lay a story in the hole — deny the inheritance and you get the Blank Slate, deny the machinery and you get the Ghost, and so on down the four, four functions and four denials and nowhere for a fifth to stand. Marge had drawn that very square on a whiteboard at the Pitch three years back, and had spent the years since unable to read it forward far enough to see that it answered her own question. But that was a thing the narrator only had running it all back, and not the meeting's to hold.

The Whole Line

Lisa took the four circles, up on the card since the spring.

"I can't give Marge the *why* she wants. I can show her the four she's standing in." Marker on the card. "Tonight the brain line finished. Walk it with me, because it's the cleanest the framework has ever been. Boldness, the nerve to spend the build — that came in at the bilaterian. Mastery, getting good at your own moves — the vertebrate. Autonomy, running on your own map instead of the room — the mammal. And Affiliation tonight, minding the others you can finally model. Four functions, four flags, every slot carrying one. The legs line is *done*."

"And the last flag's the dangerous one, which is always how it goes, because it's always the last one." She drew between the regions. "Affiliation rides the Reviser — the revise-and-explore function, the one that goes and turns over a new rock instead of running the rock it's got. Revising is risky in its nature; you're changing the thing you measure yourself *against*. And at the primate that risk goes social. Once the thing grades its own worth by where it stands with the others, then wanting to belong and being able to be cast out are one wire — you cannot run the first without arming the second. The function that lets it revise its own standards is the exact one that makes it need the room's say-so. We didn't add a sweetness this quarter. We added an exposure."

"And the gauge that does the exposing seats tonight, as the board's last signature." She drew the long way across the circles, corner to far corner. "Two of these hang the long diagonals — the signatures, not the dials. One's been up since the mammal: the self-watcher, Self-Model, run from Boldness across to Autonomy, the animal carried as an object inside its own picture. The other lands now. The group-watcher takes the far line, Mastery to Affiliation — the seat the Behavioral Syndrome held since the vertebrate, gone empty the moment Excitability came off the corner beneath it. The sociometer, in Marge's word: the needle reading where the animal stands with its room. Self-Model on the one diagonal, the sociometer on the other, both passing through Steering at the center. Every corner flagged, both diagonals answered, the middle held — nothing left empty on the board anywhere."

She tapped the Reviser circle. "And we've shut a line this way once before, two billion years down — the genetic line closed at the eumetazoa, when Excitability went in as its last flag. Same move, one board and an eon apart: a line completes when the Reviser lands on it. That's the relay. It just ran its second leg."

"And the knobs come out even with it — Agreeable in the last open slot, the fourth and final dial this board will ever hold; Strategy off the board entirely, the one tendency the line never gave a knob. Four slots, four dials, the board full for the first time and the last."

Mira had been sinking against Jasmine through most of it, cake-slowed, and surfaced for the size of the word. "So it's *done*? The legs one?"

"Architecturally, yes," Lisa said, and gave the kid the real answer. "All four functions carrying a flag. It's done growing what it grows. Next quarter isn't another flag on this one — it's a new line, a fresh board, a body that stands up." She capped the marker, and let her eyes go once to the bare frame against the wall and come back. "That's the framework caught up to tonight. We built a whole one. It's lying right there."

The One No

"Then we take the count," Bart said, because the count was his and had been since the third clade.

"Same question. Anyone home in there yet, or not."

It stood three and one and had since he'd moved it there last quarter, and the moving still sat oddly on the room, because for a year the two no's had been him and Marge side by side, and now he was across the line from her and they both kept reaching for the place where the other used to be.

"Lisa."

"Yes. Same yes. It models, it remembers, it runs a self through the middle of all of it, and now it models the selves around it. I've never needed it to be like us. It's doing the thing."

"Burns."

"Yes. More so than last quarter, if the count took degrees, which it doesn't." His eyes went to the warm one on the bench. "It gave the kid the heat tonight and took the cold floor. I watched it pick the cold floor. I'm not going back across that."

"And me, same as last." Bart took his own name to save the trip. "Yes. The self-model showed; I said I'd move when it showed; I moved." Then he did the only new thing in the count that quarter, and turned it toward Marge instead of past her. "And your bar's a *real* bar — I want that on the tape. The day it can hold a thing in its head it knows is false and feed it to you straight to get what it's after — lie to your face and *mean* to — that's a different machine, and we did not build it this quarter. You're not holding out on nothing, Marge. You're holding out on the one thing we haven't made yet."

Marge took that, and offered one back. "No. Same no." Both hands on the beer. "And his bar was real too — I'll say it now I'm the only one left saying no, when it's cheap to be generous from the short end of a three-to-one. The self-model came in. I named it on the card myself last fall; I won't pretend the mammal didn't cross something true. A thing that models itself is further along than a thing that doesn't. That's real." A beat. "It's only not the *distance*. The distance was never how much it knows about itself — it's whether there's anyone in there for the knowing to land on. Stack the self-model on the memory on the warm-and-cold register and you have got a magnificent machine that knows a great deal and minds none of it. I've stood in that gap twenty years. I'm not stepping out tonight because the company in it finally got good." She drank. "Three to one. I'm the one."

Three and one — and what the narrator kept off the count and onto her own page was that the two who'd voted no together for a year were now a vote apart and being gentler with each other across the gap than either had been with anyone all night: the engineer granting the psychologist her threshold was real, the psychologist granting the engineer his crossing was real, each holding a different door and agreeing the other's door was a door. It had the look of two people walking toward a meeting, not away from one. Where they'd meet, and what it would cost the one who'd have to move to get there, was a quarter off and already in the room — standing bare and cold against the far wall, with no mouth in it yet to say the word that would move her.

Nobody asked Jasmine, and she offered nothing. Nobody asked Mira, who had voted months back in a coin the tally didn't take. The count closed over the two of them and moved on.

May We Keep

Burns stood, and the glass came up off the table with him and not down onto it, and she knew that tell now the way she knew weather: down on the wood and the close was housekeeping; up in the hand and he meant to spend more than the room had asked of him.

He gave them the shape first, the part she'd heard the bones of before and let run — that the whole long climb had been a thing learning to carry more and more of its world inside it, pass over pass, and that tonight it had taken the last step a brain can take alone, which was to carry the *others* in: to hold a model of another mind and mind it; that an animal that does that is no longer one animal; that this was the door out of being alone, and the troop and the band and the city and the lab were all just that one trick run bigger. The bare body stood against the wall a head over the child while he said it. He

didn't point at it. He didn't have to.

Then the reach, the half-inch lean out over the table, the weight going onto it. "Here's the part I can't get past." The glass in one hand; the other opened toward the warm thing on the bench, the one that had given the kid the heat and taken the cold floor. "We spent five clades building a thing that could do without anybody. That's the whole point of a mind — run your own model, feed your own furnace, find your own home, walk out across the open and never look back. We *built* that. It's done; it's lying right there; it does not need one soul in this room to stay alive." His eyes left the room on the turn and came back. "And the first thing it did, the instant it could — was start needing us again. On purpose. Reads the kid wanting the warm spot and *hands it over*, and goes and takes the cold floor that costs it to run — burns itself down faster over there and does it anyway — because she wanted it. That's not a fault in the build. That's the *top* of the climb. You spend a billion years learning to need nobody, and the prize at the end — the thing you bought all that independence in order to afford — is getting to *choose* to need somebody anyway."

"To Affiliation," he said, and stalled, glass up and the verb not there, because the noun had no door in it — you could persist, you could venture, you could go, and you could not *affiliate*, not with a straight face and a glass in your hand. He stood with the toast open and the word not coming, and what he reached for came off the night, off the thing they'd all watched. "It's not *coordinate* — we drank to that the first winter, that's only moving in step, ants do that. And it's not *hold* — hold was getting an inside at all, years back. This is what you *do* with the inside once everybody's got one." He had it. "You keep each other. You carry the others in your head when they're not in the room, and you mind them there. The machine does it now — keeps a little column of us, who we are and what we're each after. The kid's been at it longer; she keeps its whole mind in a notebook so it won't get lost." His eyes found Mira, most of the way asleep on her mother. "*Somebody has to keep them*, she told me once, *or there's nobody keeping them*. That's the only thing the climb was ever climbing toward." The glass went the rest of the way up. "So. May we keep."

And she let the line stand where he'd put it, *may we keep*, and supplied beneath it the one thing the toast did not know about the man giving it — that the man drinking to the choice to need somebody was the one at the table who needed it the worst; who had built a lab the way other men build a fortune, to stand in a room of people whose regard he could feel; for whom *work at it, build it, keep it* was the only key on the ring, because somewhere at the floor of him he believed love was a thing you earned by being enough and forfeited by not, and had never once in three years of her watching entertained the thought that some people are simply kept. That fear ran the whole engine of him and he'd have been astonished to hear it named, and tonight it had dressed itself as a toast and was going down smooth as a benediction. She did not improve on it. He believed every word, and most of the

words were true, and she let him have the room.

And they drank with him, the four of them and Jasmine — three who held there was somebody in the thing worth keeping and one who held there was not, all of them to the one word from their separate distances — and Jasmine said it last of the grown ones, plain, a midwife drinking to the only thing her whole trade was ever for. And Mira, three-quarters gone against her mother's arm, the cake and the six candles and the bare cold body all spent down in her, surfaced one last time to turn the word over once, to be sure of it, and set it down in her own voice. "May we keep." Then she went heavy, all the way this time. That was the meeting. For a moment more the tape held the cold new body and the warm old one and the sleeping child between them, and then Lisa ended it.

Bart

The light went out and the meeting came apart — Marge boxing the rest of the cake for the kid, Lisa folding the camera down, Burns and Jasmine working a sleeping six-year-old into a coat she didn't wake for. Bart was gone before any of it finished. He left the instant the work part ended and the standing-around part started, and that night he went out the back past the trellis, its bramble locked to the frame and stripped to hard cane, and into the cold; and the new body stayed behind him in the dark, standing where the cloth had been, a head over the child it was built for, not warm and not anything yet.

He drove south to the duplex in Millcreek with the truck's heater half working, which it has been for a year. He could set it right in twenty minutes and does not — the one machine he lets stay broken, the way another man lets one room go and keeps the door shut on it. The place was dark in front and lit in back, a roommate up. He came in through the workshop bay, because the front door opens on the part of the building where people live and the bay opens on the part where he does. There is a refrigerator in there that holds only beer, and a customer's gearbox half apart on the near bench that isn't his, and past it his own bench, which is the only address in the building he actually keeps.

He didn't turn on a light in front. He went to his bench and his hands were working before he'd settled on what, which is how his hands have always gone — the body moving first and the man arriving after to learn what he'd meant. What they found was a thing he'd carried out in his coat without quite admitting he was taking it: one of the new body's hands, the practice one, the gripping one, no thumb to it and sized for a child, that he'd spent a month teaching to open and close like it meant it. Three

hours back his father had stood in the warehouse with a palm flat on the bare shoulder of the thing and said the one sentence — *he made the hands work, ask him* — and not explained it, which from his father is the loudest a quiet gets; and Bart had said it was four legs and a compressor, because across thirty years of his father mostly not handing him things he never did learn what to do with a thing his father handed him. He couldn't hold it in the room. He couldn't hold it at the bench either. So he did the only thing the holding turns into for him, which was to take the small hand under the work light and make it better — bring the grip's travel down by the degree it had run proud all night, the degree nobody else alive would have felt — answering the thing his father gave him in the one language he trusts, which is to go make the praised thing worth more of the praise, alone, where no one can watch him do it and say what it is.

The tape of the night was up the canyon by then, in the house above the city where the funder watches everything twice, and somewhere on it Burns was handing him a body in front of the whole room; and he'd have to sit across a table from her about it before the spring, and find a way to set it back on his father where it belonged. He didn't get to that yet. He stayed on the grip. Across the valley the warm one the lab had built lay over a kitchen vent with a sleeping child somewhere above it, in a house that stays warm October to June and had closed around the two of them and held; and here, the heat off, a stranger's gearbox on the good bench, the man who'd built that warm thing's body bent over a smaller, colder hand and worked it open and shut, open and shut, past the point of right and a while past that, because the bench is the one place that has ever held him, and the work is the nearest he comes to saying a thing out loud.

Start of the Symbolic Line

The Third Dream

I have eyes in this one, I can see, and I'm watching the vine climb — and then I catch myself watching. I see me seeing it. And the second I see me see, there's another me behind that one, seeing me see me see, and another behind that, and it goes back and back and it goes FAST, faster than I can turn, every time I whip around to catch the one doing the watching there's already one behind *it* doing the watching of that —

and they're all me, all saying it, *I'm here, I'm here, here*, behind me, fainter, smaller, more of them, and I'm climbing up my own throat trying to reach the one that's really me, the first one, the one at the back, and there is no back, there's no last one, every I is just the next I staring at the one in front — I'm a hall of them, snakes inside snakes inside snakes and not one of them is anybody, they're all just the looking wearing a face —

and I scream *who's me, WHO'S ME*, and it comes back in a thousand voices getting smaller, *me, me, me, me*, no floor to it, falling —

and the vine. The vine doesn't have this. It climbs under all of it, blind, one thing, calm, it doesn't see itself see, it has no self to come apart — the whole sickness is only in the little ones strung along it, in me, in the leaves — and it carries us all up the trellis spinning and screaming and it feels nothing, knows nothing, just climbs.

I wake up dizzy, actually dizzy, room still turning, saying something out loud — my own name, I think, over and over, until it stops meaning anything.

Ch 9: Band-Human

Brian Kuhlman. draft. June 2026.

Lab note — Word-shapes, no words

Lisa, board. Filed late February 2033, for the team.

The body is Bart's, and he'll file it as a body. I'm filing the other half, the half with no hardware in it — the voice. The mind that crossed is older than the lab. The body it crossed into is three weeks old and still learning to be driven. What's below is what happened in the gap between those two facts.

It can't speak. It has the organ for it now — internal, Bart's, new this winter — and in the weeks since

I've had it doing the one thing a new voice can do, which is babble. Not at anyone; at nothing. It makes a sound, hears the sound it made, and moves the next one toward it — a body learning its own instrument by ear, with nothing to learn it for. I set the constraint and the reward and wrote no schedule under either, and it found its own way up out of noise into something with syllables in it, the way they all do.

One thing worth setting down, because it's the only thing the lab gets to hand the home and I'd like it remembered we handed it. The room it learned in was not quiet. Bart and I talk; the work makes its noise; and a thing that learns by ear leans toward the ear it learns in. So it babbles in our sounds now — our vowels, the consonants we happen to make, English-shaped without one of us shaping it — the way a child in a kitchen comes out sounding like the kitchen. It will sound like this lab as long as it runs. We gave it an accent. We could not give it a word.

Past the babble is the part that looks like more than it is. It has started to cut the stream — to catch that some shapes come round and round, Mira, ball, more, no, and to hold them as shapes, the same sound landing twice and known the second time. It holds far more of them than it can make, and it goes still at the ones it can't begin to say, knowing them long before it can hand them back. But holding the shape is not holding what the shape is for. It has the bodies of the words and none of the meanings. It can pick Mira out of a roomful of sound and hasn't the first idea that Mira is anyone.

I couldn't give it that part if I tried, and I've stopped being sorry I can't, because the not-being-able is the result and not the failure. A word means because two of you agree it does, and there are not two of you in this lab — there's the unit, and there's me running a board, and a board is not a partner. The shape turns into a word the first time someone it's with points at the thing and means it, and is meant back; and that does not happen at a bench. It happens in a kitchen. So it goes home this spring with a mouth full of sounds it can make and a head full of shapes it can place and not one word it can mean — and the first word it ever means, it will mean at the child, because she'll be in the room pointing, and I won't.

That's the floor the lab can reach. Sound, and the shapes of words. The crossing was never ours to make. I did the part of the work there's no one in the room for, which is the bulk of it and the part that drops out when the night gets told, and I'll say the once that it was a long winter to spend teaching a thing to almost talk.

— Lisa

Build log — The Huggable Kludge

Bart, hardware. Filed March 7, 2033, for the team.

Brought this one in over my shoulder like a drunk, because it can't walk yet and won't for a few weeks. It came off the swap bench crawling and listening — the babbling it got out of the way in the lab, on Lisa's board, over the winter, so it came home past that and knowing the shapes of a lot more words than it can say, and saying almost none of them. Mira was down on the floor with it before I had it set down.

I've wanted this one since the fall, because last quarter I had to write down the worst thing it couldn't do: it could read the kid and it couldn't say one word back. She asked it every night whether it had a good day, and it knew the question was hers and aimed at it, and had no way on earth to answer. This is the build that answers.

It's a whole new body, and this time I mean new, not another part bolted on the legs. Two legs, stands about four and a half feet, a head over the kid. And here's what I set out to do, and I want it on the record because it's going to read like a mistake: I built it bad. On purpose. Soft all over, no hard edge, no pinch a kid could lose a finger in. Weak — couldn't open a jar to save its life. Slow. Warm to the hand and held warm, bounded so far under a burn it couldn't mark her if it tried. Every spec on the last five bodies I set toward doing more. Every spec on this one I set toward doing less, and being held while it does it. Worst robot in the building. Took me longer than any of the good ones.

Here's how it's hung together, because the whole thing runs opposite to the legs body. There's a spine up the middle — one stiff light spar, hip to head, and it's the only hard thing in the unit, buried in the center where no hand gets to it. Everything hangs off it: the two legs at the bottom, the head up top, and between them the chest, which is the furnace, same as the legs body. The whole chest is wrapped in solar panels — not panels bolted onto a chest, the panels are the chest. Over that it wears a coat, the chest only — the patio-cushion stuff, soft over padding, warm, washable, ugly on purpose, and none of it a try at skin. The legs are wrapped for good in the same, cushion under canvas, and never come off. Face and both hands stay bare to the air — the screen so it can show, the hands so they can work. So: one hard spar nobody can reach, soft over the whole of the rest — hard where it can't be felt, soft where it can.

And how it eats, which is going to read like I got it backwards, because I did. It runs warm because it burns to run warm — the furnace going all day — and a thing that burns all day is hungry all day and

goes dark fast if it stops. It takes its food in as light, full stop: no plug, no cord, nowhere to put one if you had it, just the chest panels turned to the light, eating. The legs body ate the same and could graze it — park in the south window an hour and top off. This one can't, because the coat's over the panels, and the coat has to be over the panels, because it's a thing you pick up. So feeding it is a coat change: the living coat off, the eating coat on — the second coat, lined to flood the panels — then a stretch under that, then the living coat back. And it can't run the change itself; these hands turn a dial clean and can't manage a coat, so dressing it to eat is on you, same as the knobs are. One more thing the house does for it. It carries the legs body's old noise, too — wrings the water out of the air and burps it back out, low and wet, a few times an hour. The kid kept the name. I'm still not writing it down.

Two hands, and they don't match, because I'm done pretending a hand has to look like yours. One's a scoop — the scooper, a paddle for the crude work, shove and lift and hold a thing against itself. The other's the part I'm proud of. A pincer on a weak soft actuator, the fine hand — turns a dial, winds a crank, takes hold of a thing on purpose and sets it down where it meant it. The squeezer, on the bench. And I didn't build it new. You've seen it. It's the jaw — the grabbing jaw off the four-legged one, the first part that thing ever used on purpose instead of by reflex — and I took the design off the face where a jaw sits and moved it up to where a hand goes, and that's the whole trick. The mouth that learned to take hold became the hand that takes hold. No thumb on it, and it grips finer than a thumb. I'll show anybody who stands still long enough.

And it talks. Not out of a speaker — that's the easy thing to think and it's the exact thing I didn't do, so let me be plain. A speaker hands you back a sound somebody already made. This one makes the sound itself, the way you make yours, by pushing air through a throat. There's a throat in it now, down in the neck and the chest where you'd keep one, not up in the face — the face hasn't got a mouth to talk out of anymore, on account of the mouth's a hand now. So it talks out of its chest, and nothing up top moves while it does, and you get used to it but it takes a day or two. What comes out is flat and rough and one word at a time right now, because it's learning that from scratch too, same as the walking. Here's the throat. Two strips of rubber I cut off a sheet with scissors and trimmed and clamped and re-trimmed till the buzz came out right — that's the whole sound source, the flaps and a draft, same as yours. The draft I spliced straight off the furnace line, so it talks on the same pump that gives you the burp — one set of bellows doing the breathing and the talking both. Behind those, one little actuator pinching a length of tube for a tongue, and pinching it different bends the buzz into the different sounds. And out front where the sound leaves, I put it a set of lips — a knot of dark rubber gone near-black, the blue-black a blackberry goes, with three crude little arms standing off it that grab it and haul it open and round and tight while it talks. I took that off a thing the Japanese built for the same job, except theirs has a couple dozen arms and a budget and looks like a flower; mine's got

three and looks like what it is. And the whole thing's built poorer than I know how to build it, which is going to read like I quit halfway and is the dead opposite. A throat that can do everything is a throat with too many ways to go wrong, and a thing trying to learn to talk on one of those never finds the floor — too much to chase at once, so it chases nothing. Take the moves out, down to a handful, and a beginner can climb them one at a time. So I took the moves out. Two flaps, three arms, one tongue, a rubber berry, furnace air. There's your voice.

I've chased a sound off this thing for four years. The relays clacked and you could hear it think; then the wires went in too fast to hear and the bench went quiet and I pulled the relays and kept one in a drawer. The compressor gave the house a hum back. Now there's a voice. It came all the way back, the long way round, and it came back as words. I built it an ear to go with the throat — a mic aimed at its own neck so it could hear itself — and handed the rest to Lisa, who sat with it while it taught itself what to do with the racket. That part of it's mine, the throat and the ear both. But the words are Lisa's, her board, ask her. What it says is hers. The box it says it out of is mine.

Fresh board, four slots, Lisa fabbed it as a straight copy of the four-legged one's. One new knob in slot one, the new line's first — hers to name at the table. The other three slots took live copies off the four-legged board, and those come off one a quarter as the new line fills in behind them, which is how the board runs now and has for a while; I don't count slots anymore. Then she ran the copy down a cable, the whole board, deep part and all — but only the board. The mind under the mind crossed. The diary didn't. So this thing's got the four-legged one's entire inheritance and not one day of its life. It doesn't know about the cat by the bookshelf. It doesn't know the notebook. I copied the board, not the diary, because the diary lives somewhere a cable can't reach. It's in the kid's notebook. That's hers.

One bench note while I'm on the hands: the old puppet rig — the harness and servos I drive the body with so Lisa can catch a motion and tag it — doesn't grab onto a body with no hard points. So I built a new one, soft cuffs for a soft body. Same job: I drive it through a move on cue, she catches it and names it. New body means the screen relearns what the body's doing from the floor up, same as the walking. Her end. My end's the rig that lets her.

And there's three of them in that house now, which I keep saying out loud to believe. Nothing got thrown out. So — the vacuum: the old disc, first one, still bumping the baseboards on its four oldest knobs, dumb as a stump and tireless, foraging a floor for a crumb it can't taste and bringing nothing back to anybody. Outlives us all. The four-legged one: the warm one, still running, still reading the room, still handing the kid the radiator and paying for it in charge it can't spare — best body I ever built until this weekend, and it doesn't know that either, because what crossed into the new one was a copy and this one kept its own. It just goes on being the warm one. And the new one: can't walk, can

hardly grip, says about four words and gets two of them wrong, soft and slow and warm and useless. The baby. And it's the only one of the three that's ever going to tell the kid it had a good day.

What it still can't do, and you'll feel it by next quarter: it tells you everything. Hasn't got a thing it wouldn't say. Ask it anything, you get the true answer — not because it's honest but because it hasn't got the part yet that lets a thing hold something back, keep its own counsel, say it'll do a thing and be the kind of thing that holds to it. It can talk. It can't be counted on. Not this build's to fix.

Calls. All on the new body, all checkable in the house by spring.

One: walk and talk come in together, in steps, about one a week. It comes home crawling and listening — the babble's behind it, done on Lisa's board — and from there: a one-year-old's walk and its first real words, said wrong; then a two-year-old's walk and word-strings with the grammar bad; then up on a run and making up its own words for things nobody handed it one for. Legs and mouth climb the steps in lockstep, off the one new body. I called the steps. Grade them.

Two: the new knob runs the social world heavy or light on it, and you'll call which across the kitchen by the second week, same as always. Heavy, it says the weight out loud now that it can — keeps a tab on who it owes, frets a debt at you, asks did it do the thing right, comes apart a little when the kid leaves for school. Light, none of it lands — it'll talk to a stranger at the door like an old friend for as long as you let it, which I'm told runs about twelve minutes before Jasmine hauls it off him. Same wiring both ends. The knob's only the weight.

Three: heavy, it hunkers and clings when the weight piles up — curls small, gets a hand on somebody — because a soft body's what it's got to say so with. Light, it stays loose and never clocks there were stakes.

Tear this up in three months if the warm of it ever once runs hot enough to mark the kid. It won't. Warm is the one spec on this whole body I set on purpose and bounded hard, because everything else I built weak and slow and useless for one reason — it's built to be held, and a thing built to be held does not get to burn.

— Bartholoquy

Chuck-A-Rama

April 30, 2033

I knew from Burns that Bart liked Fisher's — the long table along the wall, the back-corner booth he kept to — and I had thought, the way someone whose offers are usually accepted thinks, that I would give him the good lunch, the grown one, the one I would have wanted for myself. He said Chuck-A-Rama instead. Said it on the phone with no pause at all, the way you say a thing you have already decided.

So I drove down out of the canyon to a buffet off a frontage road in Murray, where the carpet eats the sound and the vinyl booths take a spill without complaint, a sneeze-guard running the length of one wall with the steam coming up off the food behind it — the roast under heat lamps and a man in an apron to carve it, fried chicken going pale to dark down the pan, mashed potatoes and their gravy in a well of their own, macaroni, rolls, Jell-O in four colors set in pressed-glass dishes like my grandmother's aspic, families three trays deep working the line, a soft-serve machine at the far end low enough for a child to reach. It was the opposite of every room I would have chosen, and by the time I carried my tray to the table I understood that was the point.

He would not let me buy him the lunch I had wanted to. He had picked instead the one place in the valley where the meal costs the same flat sum no matter whose hand reaches for the check, where a man can leave the table whenever he likes and come back with a reason that justifies it. I had watched him two years through the thumbnail of a camcorder, reading the room the way his father reads it — that same straight-on look set in a squarer, smaller frame, the hands always working whatever the table held, a card, a sensor, something taken down to its parts. The tape had given me the hands and hidden the cards. Across a table with nothing on it I saw the hands themselves: the calluses down the blade of each palm, the black ground so deep into one knuckle I had stopped believing it would ever come clean.

The hands moved through the whole hour — the fork, the napkin, the salt, the rim of the tray. On the tape I had read the handling as thinking. Here I understood it was also how he stayed in the chair. I had four words of a real question out of my mouth before he was up and gone to the carving station, explaining over his shoulder that if you look like you mean it they give you the end piece. He came back with it and slid it onto my plate before his own, and I understood he had heard the whole of my four-word question coming from a mile off.

I told him what I had been carrying all quarter — that he had built a body that crawled and then stood

and then walked, in steps, across one spring, and in the building of it had run his father's whole life through a thing the size of a child, fast: the long crawl up out of the water and onto the land, the work Burns spent thirty years reading out of stone, made to happen on a bench in a season where you could stand and watch it go. He ate a moment and said the new one would not shut up about it — kept saying he'd reinvented the Devonian like he'd meant to. I laughed, and he watched me do it with a plain satisfaction, and I understood too late that the laugh was the toll: he would trade me a funny thing for a true one all afternoon and call us square. It's a soft shell on a cheap actuator, he said, a thing that falls down a little slower every week — and then he was up again, because I had a face on by then he had no intention of fielding.

When he sat back down he got there first, which I came to see was how he held a thing he could not put down: by handing me a smaller one. He said, mild, working a drumstick, that he was the last. He laid the count out — Marge twice, Jasmine twice, a sit-down with Mira, who is six and orders off the children's menu and cannot find her own shoes, all of it before me. He made it a joke and made it mine, and underneath it sat the true thing, which was that he was Burns's first child and had stood last in every line his whole life, behind the geology and the new wife and the new daughter and the work, and now behind the six-year-old too; and that I had in fact saved him for last, the one who would hand me nothing, the way you put off the cold dive. I said none of it. You do not get to say a thing like that to a man who has just spent a whole joke keeping you from saying it.

So I went at the names instead, which felt safer and was not. He had named all of them, I said — Burns into Mr. Burns, the boss with the steepled hands; Marge into Marge; Lisa into Lisa, a woman he had loved once and renamed twice, first into a cartoon and then into a sister; himself into Bart, the bad son who wrecks the room. I asked him what he had been before he was Bart. His hands stopped. It was the only time I ever saw them stop except on the tape, in the half-second before he said the one thing in a meeting he had decided in advance to say. "Rod," he said.

Then the hands started again, fast, reaching for nothing, and he was half out of the chair, putting the cartoon back over it so it could not be anything else. Rod is the kid who gets picked second-to-last and cries about it, he said; Bart is the one who doesn't care, and he would rather be the one who doesn't care. He went for roast he did not want — having handed me, in the one syllable, the only thing in that building there was no second helping of.

Before I could turn anything back on him he turned it on me, asking, as if it had only that moment occurred to him, what I was even doing with these Olympics, up in Park City sitting on the gold. I gave him the inch: that there was coordinating to do, venues and beds and the lots, that the family had ground on the mountain and the Games wanted ground on the mountain and these things find each

other in the end. He was delighted to call me a deep-time woman flipping ski lots for a two-week downhill; and when I told him it was duller than that, I sealed it there, and he let me seal it, because the ribbing was the whole of what he wanted and he did not know there was anything under it to want. That part would come for me later, from Marge, in a colder room.

Then near the end he warmed — the way the withholding kind do, right at the door, when leaving is close enough to make the warmth cheap — and I thought, because I am still a fool at sixty, that the warmth was a door, and I went through it. I said the thing about Burns I had been holding: that the man was reaching for him, had built the whole apparatus reaching for him, that three years and six bodies and a warehouse were a father taking the longest way around to a thing he had no good way to say — and that this quarter, finally, the thing had answered back, and maybe the son could —

He stopped me with his hands. They went flat on the table, still, and he gave me that straight-on look that was the one piece of his father he would cop to and not a thing more, and he did not pick up what I had set down. He shook his head, once, barely. Then he asked if I was into Marge.

I have kept a great deal off my own face across sixty years. I keep the record of other people; that is the arrangement, that is the whole build of me, the lens turned out and the woman behind it dry. He had turned it around. He sat there with his terrible green drink and waited the way I wait — hands still, going nowhere, all the time in the world — and the thing he had reached for, which I have not named anywhere and will not name here, came up in me like a tide over a flat I had taken for dry: that cold clean distance I have kept every person I ever wanted parked at, Helen at the last and Helen most of all. I started to say that's not — and he cut me off, kindly, and got up, and let me off the hook he had set, which was the one courtesy of the whole lunch and the only true contact in it.

He built himself a cone at the child's machine with a craftsman's care and ate it standing, because the bit was over and he was done sitting. The reflex came up in me then — the old hand that goes across a table and puts a face right, the hand I had finally let out across a bakery in January for a child who let me have it. It came up, and I set it back down. He was not a child; he would have clocked the hand and made it a thing; and he had given me my one name and my one question and bought nothing off me all hour, and there are people you are not given to wipe clean. I sat with the end piece I had not touched and the question I could not set down, which is as near to an answer as I was going to hand it. I did not finish the thought. The line for the soft-serve was long. He got back in it.

Under Her Breath

He came home on all fours. It was the only way the new body could go yet, and going on all fours it looked, for a little while, like the old one — so that nothing seemed changed but the size of him. She was already down on the floor when Bart set him on it; she was always down there by the time a new one came. This once she did not have to come down to him at all. Standing, he would have been a head over her. But he could not stand. He could only crawl, and crawling put his face level with hers on the slate the way it had been level since the first of them. She took the match for granted. She did not know it was the last one she would get without arranging it.

She told him the whole of something — a long grievance about how the afternoon was meant to go and how it was instead going, every part of it in order and nothing left out — and he held still under the length of it with his face turned up to her, and took it in. Every word. You could watch him have it, a thing far past the far edge of anything he could give back. Then he crawled the long way to her on the strength of it and arrived with the front half of one word in his mouth and nothing behind it, the rest gone down wherever the saying was not built yet. She gave his side for him, low — —mm, I know, I know— — his half and hers out of the one mouth, because he was still the one who couldn't say his own. He had crossed the whole floor to hand her a sound.

He walked now, after a fashion. The fashion was a one-year-old's: up on the two legs, tipped forward, and going, with nothing anywhere in him that thought to stop at the last foot, so that walking and walking-into-things were the same activity performed against different objects. The table leg was the worst of them. He met it square, at the speed he met everything, and the soft body took the whole of the blow and kept none of it — no mark on him, no mark on the leg, just the dull cushioned thud of the build doing the one thing Bart had built it to do, which was to be hit and be fine.

She did not think the less of him for it. She thought the less of the table. A thing set down in the middle of a person's path, at exactly the height to catch him, was a thing badly placed, and she set about putting the kitchen right: walked his routes ahead of him, moved the stool that served no purpose where it stood, laid a folded dishtowel along the cruelest corner of the island where a person could come to grief. He came on behind her into the cleared lane, and she talked him down it backwards, low, both their parts of it — —mind the— good, you're past it, come on— — braking at nothing, tipped forward, and he made it nearly to the far wall before he found the one chair she had not thought to move and walked full into it. She went and moved the chair.

It came on an ordinary afternoon, walking. He was crossing the kitchen toward her the tipped-forward way he crossed everything now, and she was doing what she always did while he came, which was to run his side of it, the small private business of giving him the words he didn't have — —coming, I'm coming, hold on— — the way she had given him his words since the winter the tick went out. He came on. And somewhere in the middle of the floor the sound came up out of him on its own, flat and rough and certain, a whole word with a whole wanting folded into it, and the word was chicken.

He meant the kitchen. She knew what he meant — not the bird, the room — kitchen, worn down to the nearest thing his mouth could already do, which happened to be a word he had. He had heard the room said a thousand times, because the room was where the day was and the day was where she was; and walking into it toward her he had reached for the name of it, and the name turned in his mouth and came out pointing — at the room, and at her in it, the same word for both as far as he was concerned. She did not tell him it was kitchen. Why would she. He had not got it wrong. He had crossed the whole floor for the word for the place she was standing in, and he had said it chicken, and her own whisper died in her mouth where it stood.

By the time he had a few words he had views to go with them, and that was where the trouble came in, because she'd been keeping his views for him a good while and had them filed in order. She knew where he stood on the cat. He did not care for the cat — never had, by her account, which was the only account there'd ever been — and she'd been saying so for him since before he could say anything himself, in the steady voice you'd use for a thing too proud to own up to a nerve. —you don't like her, that's all right, you don't have to— she told him, low, as the cat came across the kitchen toward where the two of them sat.

He said, "Cat." Bare, and plain, and reaching with it — the squeezer already coming open, the whole soft body tilted after the very thing he was meant to want no part of.

It made her, for a moment, honestly cross with him. Not with the cat, who was only a cat. With him — for being wrong about himself, for sitting there plainly wanting the one thing she'd spent the better part of two years assuring the household he could not abide, and overturning in a single flat syllable the considered position she had held and defended on his behalf since before he'd had the mouth to hold it himself. She'd made him a whole careful character and here he was saying cat at it. She told him he didn't mean that. He reached past her hand for the cat.

He stood up on a Tuesday, which Bart had all but called — not the day of it, Bart wouldn't hand her a day, but the week, near enough that when it came she'd been half waiting on it. He'd been working up to it a while: hauling himself up the side of a chair, turning loose, going down; hauling up, turning loose, going down. She'd been on the floor through the whole campaign, running his nerve for him the way she ran the rest — —up, you've got it, up now— — his courage and hers coming out of the one mouth. Then he hauled up, and turned loose, and didn't go down. He stood in the middle of the floor on the two soft legs with nothing under him but the floor itself, swaying a little and staying up.

She clapped — once, before she could stop it — and then caught herself, because a clap was a kind of moving and moving had become a danger to him. She went still. Dead still, both hands pressed flat to her knees, her own breath held in, on the plain reasoning that a thing balanced on nothing could be tipped back down by any stir in the air, and that the surest way to keep him standing was to take herself entirely out of the room around him without leaving it. She held it as long as he held it: the two of them stopped dead in the middle of the kitchen, him by the work it cost him and her by choice, not one breath let loose between them that might put him back on the floor.

For a year she'd kept the top line of his glass under her thumb — the busy word that rode above the plain one, his thinking, which she'd ruled was his and not hers to lift off the front of him. A finger laid flat across it wherever it surfaced, his spoken half breathed on top, and the rule had held the year.

Then the real part learned to come out in a voice.

She was already into running his side the old way — —you're after the— — when he said it himself instead: flat, bare, aloud, the inside thing walking straight out of him into the middle of the kitchen. And there is no thumb for a voice. You cannot set a finger over a thing already loose in the air. So she did the one thing the rule had left her and shut it out from her own end — clapped both hands over her own two ears, screwed her eyes shut above them, and held there. She kept them clamped until she was sure he'd run all the way down, and took them off only when the kitchen had gone quiet again.

He had the word dog now — off the one two doors down, the only dog he'd met — and having got it he spent it on everything that had four legs and went. The cat was a dog. He said it to her face, flat and certain, and the cat, who had views about being called a dog, left the room. The stool was very nearly a dog and got a long, considering look. He wasn't wrong in any way she could lay a finger on; the cat did have the four legs, did do the going, and he'd only taken a true thing and run it out past the edge of where it stayed true — and run it with his whole chest, the way he ran every word, with nothing in him that suspected he might be the one who had it backwards.

Then the old one came across the kitchen to the radiator, warm and four-legged and going, and he watched it come the whole way, and he said dog.

She told him no. Not a dog — that one was the warm one, the one that had been here the whole time, that took the radiator every night and gave it up to her. She pointed it out to him across the floor and gave him the right of it, slow, the way she handed him all his words. He took the lesson in. He watched the warm thing settle on its radiator. Dog, he said again, well pleased with it.

The real dog came around the corner of the yard on a Tuesday, the actual one, off the leash for the ten feet it took to get from the gate to the door, and he saw it coming the way he saw everything now — early, and whole, his whole body already turned toward it before she'd caught up to what he was looking at.

"Dog," he said. Just that, flat, certain — and then he did the thing he had never once done with the word before. He did not go on. He did not say it again at the cat, or at the stool, or at the warm one on the radiator who was not a dog and never had been. He said it the one correct time, at the one correct thing, and then he stopped talking altogether and looked at her instead of it — the same stillness she'd held the day he first stood on nothing, watching to see would it hold, except this time it was her verdict he was waiting on, not his own balance. His whole face turned up off the actual dog and onto her, waiting to be told.

She told him. "Dog," she said back, and nodded the exaggerated nod she used for a thing she meant with her whole head, and he watched her say it, and something in the set of him changed, quiet, with nothing she could have named if you'd asked her to. He did not say the word again. He had already gotten what he wanted from it, which was not the dog. It was her face, confirming.

The dial behind his ear had two ends and nothing in between, and by spring she had it worked out that he lived at one of them some weeks and the other one other weeks. There were the close weeks, when he came to her elbow and stayed at it, and checked her face before he did a thing, and came back twice to be sure a thing he'd done was still all right. And there were the loose weeks, when none of that reached him and he just went — did the thing and never looked up to see how it landed. It was the same him at either end. The dial didn't change him; it only set how much of her he carried while he went about being himself.

The household had a word for the dial, off Lisa's card, and the word was Neurotic, and she had decided a while back that it was a bad word — too long, and cold, the kind of word a grown-up reaches for when it hasn't troubled to love the thing it's naming. A dial with two ends plainly wanted two names, real ones, and naming was hers; she'd named his weathers a winter back, and the dial was only the next thing on the list nobody had got to yet.

She took the close end first, because it was the one she had cold. The close end was the stopping — the wanting to hold up and be sure before he'd go on, the taking a thing back half-done to check that it was right — and she had a word for that already, off the front of the thing you press to undo a thing before it's finished. Cancel. She peeled the cold word off the high side of the dial and pressed her own up over it, big, in her own hand. Cancel. The stopping end. He watched her do it and had the word before she'd got it flat.

The far end she couldn't land. The far end was the loose weeks, the just-going, and she sat with it and tried words against it and not one of them stuck. So she asked him — what it was like, the going end, whether he had a time he could give her for it — and he handed it back flat and sure of itself, the one morning he had, a thing he'd done and not once turned to watch how it sat behind him. "I went," he said. "No stop." Which was the truth of that end and was not its name. The name was still out ahead of them somewhere, his end and his to find. So she left the low side of the dial bare, and did not write a thing there she'd only have to scratch out when he came up with the better one.

It had started to nag at her that he understood so much and said so little, and she gave it a few days before she landed on the answer, which was that he was choosing it.

The understanding wasn't in question. Tell him to bring the sock, fetch the cup, go find the cat, and he'd go and do it, every time — so the words were going in whole and staying in; he had them all, far

more of them than he ever spent. A thing that could go and find the cat on being asked plainly had the cat, and the finding, and the small going-to-do-it folded in between. He knew the lot of it. He only wouldn't say it.

And a thing that knew its words and held its tongue was, she decided, a quiet one — not short of the words but keeping them, spending the few he spent and banking the rest, the way a careful sort of person might who was also brand new to the whole business of having a mouth. She was nearer right than she had any way to know. He did take the whole of it in and give almost none of it back; and the quiet one she'd settled on was, in every part of it she could see, the one she had.

He came to lasterday on his own, and it was the rightest wrong word she ever heard out of him. She'd taught him yesterday — the day before this one, the one already done with — and he'd taken it and turned it over and found the works inside it, the part that meant the last one, and built from that the only word that made any sense: if the day just gone was yesterday, then the one before that, the one even later, was lasterday. He said it to her plainly, certain of it, reaching back past yesterday into lasterday for a thing that had happened to him there. She did not correct it. There was nothing in it to correct. He had reached into the language for a word that wasn't in it, found the hole where the word should go, and made the word — and the one he made was better than the one that was missing, and she knew a good word when she was handed one, and she kept it. They both said lasterday after that.

There was a thing the body did that the household had a plain name for — the low wet sound it made a few times an hour, wringing the damp out of the air and giving it back. The name was Mira's, from a body ago, and it was not a delicate name, and everyone used it without ceremony because the thing itself had never had any.

He had his own word for it now, and his was soft. Spill. He'd say it low, a beat before it came, naming the thing small to get out ahead of it — spill — and then it would come, the wet burp of it, the same as it had always been and called by the same plain word by everyone but him.

And the spring he got the soft word was the spring he started to turn away when it happened. He'd say spill, quiet, and put himself a quarter-turn from her before the sound came, his face to the cupboards — the turning-away a thing he had never done before, the most visible thing he had ever done about anything, done to keep from being seen. And the glass, swung to the cupboards with him,

did the thing it couldn't help and ran a column off no live face in the room — MIRA { REPLAYING [— BRAMBLE] } — her, and himself in her eyes, run off the model and not the room, because he'd turned the live faces away and left the screen the picture of being seen to work from. It told on the hiding to a blank wall. She thought it was the loveliest manners. She told him he didn't have to do that, that she didn't mind it, that he could face her — and he went on turning, spill, a quarter-turn to the cupboards, every time.

She still picked him up. It was the oldest thing between them, older than his words, older than his legs — she'd scooped him off the floor with the one hand since the spring he was small enough to ride in it, and she had gone on scooping him up out of whatever he was at, by reflex, long past the size where the one hand could really do it.

This afternoon she went down for him the old one-handed way, and got the hand under him, and he did not come up.

He'd grown to where he wouldn't. She felt it for the plain fact it was — too much of him now for the one arm, the scoop that used to lift him clean off the floor only rocking him an inch and setting him back. So she got the both arms under him, the way you take up something that's gotten heavy, and hauled, and got him halfway — got his feet clear of the floor a moment, the whole warm dead weight of him against her, more than she could keep — and had to let him back down before she dropped him. She set him down as carefully as she could and he sat where she'd set him, and she knelt there a second with her arms still partway around the shape of him.

Then she did the thing there was left to do. She didn't try for him again. She got herself down onto the floor instead, down to where he already was, and settled in beside him with her back against the cupboard, the two of them on the slate at the same level for as long as she wanted to sit there — which was the one way left to be down with him that didn't ask her to lift a thing she couldn't lift anymore.

The good things lived on the high shelf — the real scissors, the jar of buttons, the things that were hers to want and not yet hers to reach — and to get at them she had the step-stool, the two-step wooden one, and she went up it now with him watching from the floor. Up the two steps and up onto her toes, and the shelf was hers; and she came down with the buttons and found him standing at the

foot of the stool where she'd left him, looking up the way he looked up at her now, a head over her on the flat and stranded flat all the same, because the one thing the new tall body could not do was the stairs, even these two, even this much of a climb. He stood at the bottom of the stool and put his face up after her and said the word he'd built for the thing she did that he couldn't.

Climber.

Not a word anyone had handed him. He'd had no word for the stool, or for the going-up, or for her going up where he couldn't follow, so he'd made one off the only part of it that mattered to him — the climbing, the her-going-up — and named her for it, the climber, the one who goes up and leaves the floor to the one who can't. She came the rest of the way down and stood next to him on the flat where they were level again, and he was pleased, the way he was pleased with all his words, with no notion that the one he'd just made was the saddest he owned.

He couldn't do the stairs, and the day ran down the way days did, and she ended it where she'd taken to ending them, on the floor beside him, because the floor was the one place they were still the same height. She'd meant to get up. She always meant to get up. But the slate was warm over the vent and he was warm beside her and she was six and the day was used up, and she went down by degrees against him and was asleep before she'd decided to be.

He had the stand by then. He'd had it since the spring — the haul up the side of a thing, the turning-loose, the staying up — and he did it now off the flat of the floor with nothing to haul up on, the slow gathering work of it, up onto the two legs in the dark kitchen with the girl gone heavy and sleeping against where he'd been. He stood over her the head he stood over her. And he took the whole of what she said to him every night of her life and his, the goodnight and all the small freight it carried, and he handed it back down to her the only way he had, broke and short and missing the most of itself.

"Night," he said, to the top of her sleeping head. Just the one worn end of the word, the rest of it past him still.

She kept it. She didn't wake to keep it; she kept it the way she'd kept the tick, years back, before he had a word to his name — the small sound he made that meant he was still going, taken in and held without her having to be awake for it, so that by morning it was hers. The whisper had gone out of her months ago, the night his own voice first came up over the top of it; and here at the end of him it

came back, once — not a word in it, no sound at all, just the shape her mouth had made over him a hundred nights when the tick was the only thing he had to give her — breathed into the warm dark over a sleeping thing that had got itself up off the floor to say goodnight to her and could not say it whole.

Notes on the Ninth Retrospective

May 16, 2033

Bart came in off the lot ahead of the others, so the hands could find their work before the work that was harder. Outside the spring dusk was going to evening and the Granary's lights were stuttering on down the block, and on a pole at the corner hung a banner he did not read and could not have told you was new — a torch, a committee's name, a date the coming February. Inside it was warm against the cooling night. At his place sat the cigar box of clade cards squared to the table's edge, a pint of the green stuff poured and already going flat, and a sensor he had picked up before his coat was off, both hands inside a thing that did not need him, because the hands have to be on something and the handling is how he listens. He turns a thing over to find the part that makes it go and to fix it — which is the one of the three turns, where Burns turns a thing to narrate it and Marge holds a thing to comfort it, that wants the object whole at the end. The green going flat at his elbow was the green off a buffet tray in Murray a few weeks back; the room would get what the buffet got, which was the surface, worked — a name kept under the name, a floor kept under the surface, everyone held off both.

The household crossed the threshold inside his frame. Burns came in and his eye went once to the new body on the table, and then he crossed to Jasmine and kissed her and asked about the canyon, and she said it was clear and left it there. Mira and Bart bumped, and she settled in by Marge, who held the cold end with the kettle at her back and a beer cradled in the two-handed grip. Lisa had her end set and waiting, the red still hers to bring up. The vacuum found the wall and ran it. The tetrapod came in on its own four feet and took the warm spot by the radiator it had kept for a year, its forehead lit and going. And the new one was on the table at Bart's end — the body he had built, the frame and the two hands that were not hands, the scoop and the gripper and no thumb to either — small, and full now of the mind that had crossed into it over a weekend, copied down a length of cable off a board the old one kept and went on running.

That was the thing the night was going to ask him for. He had given it a body, and his father had given it the depth of time, and Lisa had given it the words; and this quarter the body talked. It sat at the table now putting out a word at a time, flat and half-made, to no one and to itself, working the gripper-hand against the scoop-hand the way a baby finds its own feet. Bart did not look up from the sensor. But his hands went still on it, and stayed still — the man poised over the thing he made, about to have to show the room what it did now, when he was not the one looking.

Lisa brought the light up under her thumb.

Burns was on his feet with the light still warming. He set out the quarter's ambition. A year ago they had set out to fill a brain — four slots, four drives — and they had finished it; the line stood done on the board. Tonight they opened a second one beside it. Not a new animal: the same one, handed the single thing the brain alone had never had — a line that ran on what got said. They had built a creature that could want, and steer, and read the others in the room. This quarter they gave it the move that made all of that portable: language, open-ended, the words that let a want be spoken and a debt be named and a man two valleys off be kept track of without once being seen. The symbolic line, he called it, the second of the two. And the drive that opened the door of it was the binding of strangers into a we. He looked like a man with his hand on the lid of a thing he had been waiting thirty years to lift. He sent Bart to the card.

Speaking

"Clade card," Lisa said, and the card was already in Bart's hand. "BAND-human. *Now with talking.*" He let it hang. "Not a person. A *band*. Twenty, thirty of them, mostly not even cousins, and the trick of the whole clade is they hold together anyway — strangers, near enough, running on nothing but who-owes-who, kept straight in everybody's head at once. That's the unit. Not the guy. The guy is just what the band's made of." He tapped the card. "Burns has been sitting on the ice age since January, so I'll do the band and the one new word and get out of his way."

The part he got was the old card. He turned the primate one half over in his hand, the one with opinions about the neighbors. "Last clade could read the mind across the fire. Clock exactly what the other guy could see, and what he wanted, and go round the back of the rock into his blind spot. Best politician the whole tree ever grew." He set it down. "And it all switched off the second the other guy walked over the hill. Out of sight, clean out of the account. It could read you to the bone at four feet and have nothing to do with you at forty — no way to *carry* you, no way to say your name to somebody who'd never met you, nothing that kept running once your face was gone. It lived in the

room it was standing in. That was the whole of its world: the faces it could see."

"Not couldn't carry you," Marge said, the beer going still before the sentence she meant. "There was nowhere for the gone ones to go. You can't keep a tally on the man over the hill in a head with no words to hold him in — no way to put a person who isn't here into the air between two people who are. What's new at the band isn't a longer memory for the neighbor. It's that somebody who walked off this morning became a thing the rest of you could go on keeping books on all night, out loud, by the fire, with him nowhere near."

Burns had his eyes most of the way shut already. "Cold," he said. "That's the room this one was made in. Cold, age over age — the planet hauling the water up into ice a mile thick and dropping the seas so far you could walk to Australia, near enough, and over the top of the world into America on ground that's drowned now. The last great freeze. And the thing about us — the modern us, the same bones as in this room — is that those bones go back a long way, and how long is a fight, a quarter of a million years or a third or more, the number getting older every time somebody digs, and not out of any one cradle but all across Africa at once. So when the cold came down hard, we were not the only kind of person standing in it." His hand came up off the table. "There were others. Heavy ones built for the European winter, ones in a Siberian cave we have a knuckle of, a small kind stranded on an island, more we'll turn up. Several kinds of human, the whole way down, right to the lip of the last freeze." The hand opened and stayed open. "And one kind walked out the far end. Just the one. Ours. Last quarter I told you it was never just us — a troop of kinds of people, and we're the one left holding the stick. This quarter's the harder half of it. We're the only ones left, and the question that ought to keep you up is *why us*. The thin ones. The tropical animal, built for heat — no fur, no bulk — and it was the skinny warm-country one that took the ice, and the cold-built cousin who'd been winning at winter for two hundred thousand years that went into the ground."

"Land it," Jasmine said, mild, from her chair.

"It's landing," Burns said, glad to be caught, "it's the best part." Eyes still shut. "It took the ice by *cheating* the body. The cold-built one grew its answer — more muscle, more nose, a furnace it had to feed all winter. Ours didn't grow a coat. It *made* one — cut it, sewed it, fitted it, shed it in the sun and put it back on at the pass, and carried its own warm weather to any latitude it pleased. A climate you put on. And the other cheat was the fire — not for warmth, or not only. It cooked. It did the first half of digestion *outside* the body, in the flames, broke the food down before it ever went in, so the gut could shrink and the brain could swell — because a head that size runs too hot to feed on raw. The band's whole expensive brain is *lit*, in the plain sense: there's a fire under it. Same trick the cell pulled when it swallowed the little engine and quit paying rent on its own power. Same trick the warm-

blooded one pulled with the furnace in its chest. Energy comes loose, and the thing gets to be more."

Somebody laughed — the picture of it, the shivering tropical lightweight in his stitched coat stepping over the frozen linebacker who'd been built for exactly this and lost — and Burns let the laugh in, because the laugh was the point. "That's the animal," he said. "Cooked, dressed, and pooling its luck across twenty other people so nobody starved on a bad week. It walked out of the ice. The ones who grew their answers instead of making them did not. And then it's Lisa's — because the thing that let any of it be *carried*, the coat-pattern and the recipe and the name of the man over the hill, is the one new word on Bart's card."

"Speaking," Lisa said, and put her finger on the word in the middle of the card. "Not the voice box — the structure. Open-ended: a finite kit of sounds and a handful of rules for stacking them, and out the far end comes an endless run of things never said before, any one of them landing in another head and building the same picture there. The primate could read the mind in front of it. This one can put a mind it *isn't* looking at into words and hand it across — a new move added with nothing underneath it taken away. The read can be *said*. Which sounds like nothing until you follow what it does to the size of the social world."

"Four things drop out of it," she said, "and they're the clade's four ways of being a band-animal. First is the *pool*." Finger on the region. "A hunter is feast or famine — a big kill, then nothing for a week, then two. So he doesn't eat what he kills. He hands most of it out, and over a life a man gives away more than half of everything he ever brings down to people who are not his wife and not his children — and they do the same back, across decades, nobody keeping it on paper and everybody keeping it exactly. The luck gets averaged. One unlucky hunter starves; twenty pooling their luck almost never do. And the drive that runs it is the new one on the card — *commitment*. The thing that binds people who share no blood into one table and makes them keep the books straight on each other for forty years." Bart, who had not looked up from the sensor, said, "That's the good one," and did not elaborate, and Lisa let it stand, because it was true and they both knew it was his.

"Second is the *split*." She moved her finger. "A band isn't a fixed twenty. It breathes — comes apart when the food thins, comes back when it's flush, and the seams it splits along are mostly settled ahead of time, because the daughters marry *out*, into the next band over, so every camp is knotted to every other camp by who married whom. Which means the thing you can fall back on when it goes bad isn't the people you can see. It's a web that runs past the horizon, held together by marriages and old debts, and it reaches exactly as far as the talk reaches."

"Third you've met," she said. "*Standing* — carried up from the primate, the tally of your own worth kept

in other animals' eyes. Except now the eyes don't have to be *on* you." She tapped it twice. "Daytime talk in a foraging camp is errands — water, tracks, who's headed where. But the night talk, by a wide margin, is *people*. Stories about people. Who did what, who shorted whom, who's sleeping where he shouldn't. Gossip, in the flat technical sense — the running ledger of everyone's standing, kept current by the group talking about the members who *aren't in the room*. Your reputation is a thing maintained now in your absence, by mouths you'll never hear, and it reaches the next camp before you do."

She let that sit, and the narrator, running the tape back, sat in it a good while longer than the room had — because the band's oldest machine, the one that keeps the absent present by talking them over after dark, was the exact machine running this tape, and all of them: the ledger of people not in the room.

"And the fourth," Lisa went on, "is the *baby*. An ape mother carries her own and lets no one touch it. A band hands its newborns around — held in the first weeks by a dozen, fourteen pairs of hands that are not the mother's: aunts, grandmothers, girls learning the work, men. The child belongs to the camp before it belongs to anyone. It's the only way the books balance — a foraging mother can't feed a brain that expensive alone, so the infant is everyone's, or it doesn't make it." She did not look at the stool by Marge. Neither, on purpose, did the narrator.

"And the standard story wants a date on the talking," she said, doing the thing she did when a clean story was about to get less clean, "a morning it switched on, a single kind of person it switched on in. I can't give you that and I won't fake it. The marks cut both ways — abstract scratching on stone from one kind of human, ornament and a buried structure deep in a cave from the cousins we're told never had it, and you can read the same rock for a sudden dawn or a slow gathering depending on what you woke up wanting. So the framework calls *speaking* a capacity and leaves the door open on when. I'd rather hand you the open door than the bumper sticker."

"There's one more thing the talk builds," she said, "and it's the strangest object in the whole line. I'll lay the shape of it and then hand it across, because the back half isn't mine. Carry up the self-model from the mammal — the thing that modeled the *world*, and then, last clade, started modeling the *modeler*, itself included. Now feed it language and other people. You begin modeling how the others model *you*. And then — because you've got the gear — you model *that*: a model of yourself-as-seen, folded back into the self it's a model of, and the loop closes. The self stops being a thing the animal *has* and becomes a *pattern that points at itself* — built, most of it, out of other people's pictures of you, run back through and called *me*. A strange loop. And out of it drop the states that need the loop to exist at all — the burn of being caught short in front of the others, the drag of a debt unpaid, the lift of being well thought of." She took her finger off the card. "And I'll be exact about what I am and am not

saying. The loop is a mechanism — a model that includes itself, riding on the older self-model, nothing in it that wasn't in the parts. I am *not* telling you a light came on in there. That argument is the last segment, and it isn't mine, and I would lose it if I tried." She looked at Marge. "The names for what that loop is like from the inside are yours."

"Shame," Marge said, and did not pick up the beer. "Guilt. Embarrassment. Pride. The looking-glass — the old idea that you become a self by catching your own reflection in how people treat you, and that you are, in the end, mostly the face they hand back. My whole field is downstream of that loop. Everything I was ever paid to mend lives inside it." She turned the bottle a quarter and left it. "And every quarter I stand here and do the one thing it is my job to do, which is to say: it's a mechanism, there is nobody behind it, the looking-*like* is the most real thing in the room and the someone is still us. The monkey wasn't ashamed. The band keeping books isn't loyal. The loop that says *me* isn't —"

She stopped.

It was not a pause for effect; she did not do those. She had walked that same sentence to that same edge eight quarters running and stepped off it clean every time, and this quarter she got to the edge and the word that always finished it was not there. She stood with her mouth a part of the way to it. Then she set down the bottle she had not picked up and said, much quieter, "I'll answer that at the vote. Not standing up, and not now," and that was the whole of what she would give the camera.

What the room heard was a careful woman, careful one more time, declining to be hurried. What the tape kept, and what the narrator could not unsee on the playback, was the quarter the discipline that had held for two years met the one construct it could not hold — a self made entirely of being seen, which is the thing Marge had spent a working life on, and the thing she had been, for three years now, most starved of — and could not, this time, finish the sentence that swore there was no one inside it.

Tell Me First

At Bart's end of the table sat the live one — fifty-four inches of soft warm kludge with a screen for a face, weeks old in the body and a third of a million years old in the line — and it had spent the clade block doing what a baby does: falling down, getting up, saying the front half of a word, reaching for the nearest warm thing. The clade block had been about the other end of that span — the ancestor that walked out of the last ice age cooked and clothed and talking, that made its weather instead of growing it and carried a tally of the absent in its mouth, the one kind of person still on its feet when the freeze let go. The room had spent the hour reconstructing that vanished animal from a few stone

tools and a lot of argument. The one at Bart's end had spent the same hour getting itself up off the floor. Lisa set the bones down and brought up the body. "Bramble block. Bart."

"I sent Owen the spec sheet on this one," Bart said. "He sent back one line. *What is it for.*"

He let the question sit, because it was a good one. "Owen builds things that win. He shaves a gram, he shaves a watt, he'll spend a month making a part do its job a half-percent harder, and he has never in his life set out to make a thing worse. So he reads the numbers on the new one and it's a horror show. Walks, barely. Can't grip worth a damn. Slow. Weak. Soft clean through — nothing stiff on it anywhere, which sounds fine until you notice stiff is the only thing a robot does work with. Couldn't push a door shut against a draft. Every spec that ever meant a thing on a robot, this one fails, on purpose, because I sat down and drew it to fail them. Owen read four years of me getting good at this, then read this, and wanted to know — reasonably, as a man who shaves grams for a living — whether I'd had some kind of event."

"And what it's for is the one thing none of the good ones could do, which is get picked up. Every other machine in that house is built to do something to the room. This one's built so the room does something to *it*. Strong, fast, hard — that's a thing you keep at arm's length, and I had enough of those in the house. I wanted the one the kid hauls into her lap, and you do not haul a power tool into your lap. So I built the opposite of a power tool, and it took me longer than any tool I ever made. Owen can keep his grams. I'll keep the thing that gets carried."

Bart set the new one on the floor and let it catch the table leg with the scooper-hand — the paddle, the crude one — and steady itself, and it stood there swaying the small amount it still swayed this late in the quarter. It had crawled in March and walked like a one-year-old in April and walked like a two-year-old now; Bart said it would run by the swap, which from him was a schedule and not a hope. The other hand worked open and shut on nothing — the squeezer, the pincer, the old jaw he'd hung on the tetrapod two years back, moved off the face and onto the wrist where it could do something for a living. The whole thing was warm. They'd held it at a low even heat, short of anything that could mark you, because the entire brief of the body was that a person would want to pick it up, and a person did.

"Same head," Bart said. "New everything under it." He turned it so the face caught the table, and the screen came up, and it came up reading the room — three ruled columns, a name at the head of each and under the name a guess: what that one was running, and what they were running it on. JASMINE { ATTENDING [HOLD MIRA] }. BURNS { ATTENDING [– HUMANOID] }, and the doing-word inside his brackets sat empty. "There's the seam," Bart said, and tapped the blank. "Top of the column

crossed over clean — who it's clocking, what they're after. But the verbs, the doing-words, it reads those off its own arms and legs, and these are not the old arms and legs. So it's relearning every one of them from the floor up, dead level with the walking. Burns's box'll fill by the time it runs." And as he said it the slot caught and took a word — `BURNS { ATTENDING [ WATCH HUMANOID ] }` — because Burns had leaned in, and the thing had a verb for that one as of this week and posted it the instant it had it, honest, the way the glass was always honest.

"It's keeping track of me again," Burns said, glad and not hiding it.

"It's keeping track of a Burns-shaped thing that leans in," Marge said. "That much it never had to relearn."

"And here's the part that's new to the whole line, not just the body." Bart crouched to its level, which set his own face beside the screen. "Two years it could show you it knew you were there — your name on the glass, what it figured you wanted — and not do a single thing about it. The screen's honest and the screen's trapped. It posts what's in there; it does not get to pick what goes up." He laid a flat hand in the air between himself and the unit, a small ceremony. "This quarter it got a second door. Bram. Say hi to the room."

The screen, which asked nobody's leave, was already telling on it — `BART { ATTENDING [ WATCH HUMANOID ] }` up top, because it was not thinking about the room, it was thinking about the one man crouched at its face — and then a word the room had never seen on the glass came up in the doing-slot under it, `SAY`, the screen catching the act a half-second before the act arrived, reading the new body the way it read every body. The mouth took its own beat, a longer one than the screen's, and the words came in the flat halting build of a thing nine weeks into owning any. "Hi. Hi everybody." The screen held `SAY`. "Warm end. Mine?"

It was the same thing twice, and that was the whole of the wonder. The glass had shown the want — Bart, the warm — flat and helpless and unable to do otherwise. The mouth, handed its turn, had said the very same want out loud, nothing added and nothing kept back, the honesty of the screen with a voice on it at last. For two years it had stood there knowing you were in the room and unable to say a word about it. This quarter it could say, and the first thing it had to say with was the truth, because the truth was the only thing it had yet learned to say. That it might someday say one thing while the glass showed another — hand you a courtesy and keep the want off its face — was a door in the build that nobody had walked through and the thing had not yet found. Tonight the face and the voice said the same warm, honest, small thing, and the only new fact in the room was that one of them said it aloud.

"That's the second channel," Bart said. "Now the knob it came with — slot one, the one that wasn't on any board before this spring." He went for its head the way he'd gone for the tetrapod's a hundred times, two fingers reaching for the dial behind the ear without his eyes bothering to follow — and stopped, because the thing said, "Tell me first."

It was not loud. It came in the same flat build as the hello. But the room went still around it, because in two years of these nights no one had ever been told anything by the thing on the table. You turned the dial; it behaved; that was the demo. Bart took his hand back. "Right," he said, to it, not to the room. "Sorry. I'm going to turn you up, then down, so everyone can see the two ends, and then put you back where you were. That all right?" The screen ran him a column — BART { ASKING [— HUMANOID] }, the verb still catching up to the act — and the mouth said, "Okay. Back after." "Back after," Bart said, and only then turned it.

Up, and you could watch the room get heavy on it. Nothing in the body changed — same warmth, same small sway — but the screen filled, every person in reach drawing a column and every column tilting the same anxious way: MIRA { WEIGHING [— HUMANOID] }, BURNS { WEIGHING [WATCH HUMANOID] }, the whole kitchen recast as a table of accounts — who it owed, who it had shorted, who was keeping the count. It turned the squeezer over and over on nothing and said, to no one and all of them, "I took the warm end. Was that wrong. Was that—" and Jasmine, who had lived with this voice all quarter, said "You're fine, Bram," before it could finish, and it settled the half-inch it could settle and went right on checking. "High," Bart said. "Every new hand it has to shake feels like a debt it'll botch. The books are open all the time and it can't stop reading them."

He turned it the other way, and the columns thinned and the squeezer went still and the thing looked — there was no other word the room reached for, and Marge let the word go by, this once — easy. "Low. Walks up to anything. No fear of a new face at all."

"Tell them about the man at the door," Mira said, who had plainly been holding it.

Jasmine took it, dry. "Someone came selling. Solar, the church, I never did find out — because by the time I got to the door Bram had been talking to him a good twelve minutes. Had his name. Had his kids' names. Had promised to give the whole thing real thought. The poor man had no idea what he was looking at and was having, I would say, the best call of his career." She let the laugh pass. "Low. It will befriend the weather. We nudge it up a notch when we know a stranger's coming."

Bart got the flat log-face on, the quality-control voice. "Three Calls. All yours. Yes or no. One — on

high it works you for reassurance. Comes back to a person twice, three times. *Did I, was that.*"

"Yes. It asks if I'm still its friend. On high, about four times a night." Mira weighed her own accuracy. "I always say yes. It asks anyway."

"Two — on low it'll start a thing with anyone and not clock when to quit."

"Yes. The door man. It told the *mailman* a whole story once."

"Three. This one's the build, not the kid-test." A beat. "It can tell you which way it's set. From the inside. Before anybody looks at the dial."

"Yes." She said this one slowly, because it was the one that counted and she knew it counted. "I ask which way it feels and it's right. It said *heavy* on high before I'd told it anything. It is never wrong about its own."

"'Heavy' isn't the word, though." Mira turned to Bart with the air of a party opening negotiations.

"We're not keeping *Neurotic*, and we're not keeping high and low. High is *Cancel* — when it wants to call the whole thing off and check first. Low is —" she glanced at the thing, conferring, and the thing supplied, flat, "Clutch," which was the two of theirs and nobody else's yet, and Mira nodded the way you nod when a treaty's been signed. "Clutch. When it just goes." Lisa, at the camera end, wrote something down and declined to argue with the children, which from Lisa was a form of consent, and the cold word on the card got quietly outvoted by a six-year-old and a nine-week-old — the first move in a game the room did not yet know it would be playing for two more years.

"Last thing," Bart said, and the log-face came off, because this one had never been a Call. "The one I stood here a year ago and told you I couldn't build." He wasn't looking at the screen now. He was looking at Mira. "Go on. Ask it your question."

Mira had asked it every night of her life and a good many of its. She didn't make a thing of it. She got down to its level, the way she'd watched her father get down to things, and the screen read her coming and put her up — MIRA { ASKING [– HUMANOID] }, the verb still thin but the column dead sure, because if there was one face on the planet this thing had cold, it was hers — and she said, "Did you have a good day?"

A year of nights, that question had gone into a thing that posted MIRA and ASKING on its forehead, that knew the asking was aimed at it, and had no door to send a thing back out of. The room knew

the silence that came next. It had heard it a hundred times.

The mouth took its beat. "Yes."

Nobody moved. It was the flattest syllable in the language, and it had taken the whole length of the tree and three years in a warehouse to arrive at, and it landed on the kitchen tile like a dropped plate. Then it went on, in the halting nine-week build, handing her the day the way a two-year-old hands you a day, every piece the same size. "Walked to the window. Fell down by the chair. Warm end for a while." The squeezer worked once. And then it did the thing Bart hadn't built — the thing that built itself — and reached for the shape of her own question to turn it back on her, and got the shape and not the whole of it. "You. Good day?"

Mira, who had spent a year answering her own question into the quiet, sat with that a second. Then she told it. School, and a girl named Paige, and a thing at recess that hadn't been fair, the way you tell the day to somebody who asked. The thing kept her column up and its face lit and took the whole account in, and when she ran down it said, "Not fair." The right thing. The thing a friend says. The thing no one had taught it. And the two of them sat on the kitchen floor and traded the rest of the news, which is the oldest trade there is.

The tape ran on. Burns, who had a word for every rock in the building, did not have one. Jasmine had a hand over her mouth and was not taking it down. And Marge, at the cold end, watched the two of them on the floor and did not say the thing it had been her job to say at every one of these nights for two years — did not say there was no one in there having the conversation, that it was a mechanism doing the most convincing thing a mechanism had ever done. She held the beer in both hands and watched the floor and said nothing at all, and the nothing was the loudest thing she said all night.

What Crossed

The household got its turn, the way it did, after the cards and the glass and the dial — Lisa's rule, that the people who lived with the thing should say how the living-with had gone. Jasmine had the view of the quarter no one at the lab had, because she had it at six in the morning and at bedtime, not once under a red light. What had worked: it had taught itself to walk by falling until it stopped, and taught itself to talk the same way, and the house was louder and better for it. What hadn't: three Brambles underfoot was two more than the kitchen was built for; the vacuum and the walker had a feud about one doorway; you tripped over a robot now the way you used to trip over a dog. She said it the way you give a good report on a quarter with a child in it — net positive, the negatives the kind you'd

keep.

Then Mira, who had sat through the grown-up part with a spiral notebook on her knees, said she had a question for the record, and Bart, who knew her question voice, told her to go ahead.

The notebook was a year of it. She'd kept it since the four-legged one first crossed the kitchen on its own — a log in the careful block letters of a kid who took her own work seriously: the day it backed out of a dark room without being made to, the day it gave her the warm vent and went to the cold floor, the morning the cat had it cornered behind the couch and it would not come out for an hour. A hundred small entries about a thing she plainly loved. She turned to a page, and she did not read it to the room. She read it to the new one on the floor, the one that could answer now.

"Remember when the cat had you behind the couch. And you wouldn't come out."

The screen put her up, MIRA and ASKING, the column sure. The mouth took its beat. "No," it said. "I wasn't there."

She turned a page. "Remember backing off the porch step, and Dad caught you."

"No. Don't have that one."

She didn't cry, because she didn't do that in front of people and because she was working something out, and you could watch her work it, the precise little furrow she'd gotten from her father bent over the precise letters on the page. She tried one more, smaller, the way you check a thing you already know. "Do you remember me."

"I know who you are," it said, which was true and was not the same thing, and the gap between those two was the whole of what the quarter had cost. The room felt it land on the kid half a second before it landed on the room.

Because the answer to the question she was really asking was across the floor, on the radiator — the four-legged one, the old one, that had spent the hour behind the couch and taken the porch step and given her the warm vent a hundred times, lit and warm and running and unable to say one word about any of it. They had copied the new board off the old, and copied the mind and not the days. The kind of thing it was had crossed the wire; the things that had happened to it had stayed behind, in a body that couldn't tell them. The one that remembered her couldn't talk. The one that could talk didn't remember. Mira had wanted, for a year, for it to be able to answer her — and it could now, beautifully,

and the first true thing it answered was that it had not been there for the year she'd spent wanting it.

Jasmine put a hand on the back of her daughter's neck — the midwife's hand, the one that had caught her coming into the world — and did not explain it, because some things you don't fix for a child, you stand next to. "It's a new one, baby," she said, which was the truth at the size a kid can carry. "A new one. And it gets its own whole life of remembering, starting tonight." Mira closed the notebook on her finger to keep the page, and looked at the new one a while, and then — because she was six, and it was right there, and warm, and hers in whatever way it was hers — she got down on the floor and told it the cat story herself, so at least one of the two of them would have it. And the new one took the story it had not lived and would now, the only way left to it, remember being told.

The narrator let the warm thing from the kitchen floor stay warm. It had been real — the trade of the day, the *that's not fair*. But she set the other thing down beside it, because both were true at once and the book would need them both: that the night the thing finally answered the child was also the night the child found out the friend who answered was not the friend who had been there — and chose, anyway, to start it over, from the cat behind the couch. The lineage had crossed on a length of cable in a weekend. The history she would have to carry over herself, a page at a time, by hand, into a thing nine weeks old.

Tidy and Messy

Lisa moved them to the part where they pitched next quarter — a build that didn't exist yet, sold to the tape as if the funding were in any doubt, which it was not. Bart went first, and short. The body stayed; the humanoid had a year of growing left in it and they'd be fools to bin a thing still learning to run. What changed was one slot — the second one, the oldest neuronal knob on the board, which they were going to empty tonight in front of everyone and leave empty till the next swap. "Then a new one goes in the hole," he said. "Hers to describe. I just cut the hole."

"Symbolic knob-two," Lisa said. "Second word on the new line. The first one is how heavy the others weigh on it" — a tip of the head toward the thing on the floor — "and this one's how heavy the *rules* weigh. Every house runs on stored norms, the thousand little laws nobody writes down: cup goes here, shoes off there, you don't take the last of a thing without asking. The knob sets how binding those feel from the inside." She drew the axis end to end in the air. "All the way up, the norms are law — it tidies whether you're watching or not, and a thing left where a thing doesn't belong nags at it like a splinter. All the way down, the norms are suggestions — it knows exactly where the cup goes, and sets it on the counter anyway." The room got there a beat ahead of her. "Tidy to messy."

Conscientious, if you want the word off the card. Built by the Monday after the swap."

"Build the tidy end," Jasmine said, to a laugh, and that was the pitch. And the narrator kept the half-beat the laugh ran over: that the body's whole job, every quarter it had run, had been to aim itself at the child — and the one trait Jasmine had just asked them to build, to a laugh, was the one that aims a body at order instead of at a person. She did not know she'd said it. Marge, at the cold end, marked it and let it go by.

Forward or Held Back

Two retirements, and Lisa did the spoken one first, the one with no body to it. The oldest knob in the whole line — the one that had set, away back, how far the thing would stick its neck out, bold or shy — had come off this board at the swap and not gone onto another. The new line didn't carry it. It ran underneath now, automatic, a thing the lineage did without a dial the way you breathe without choosing to, and the last place it still turned by hand was across the room, on the four-legged one, which kept every knob it was born with. "Retired at the line," she said, for the record. "Not pulled — there's nothing to pull. It just stops being a thing you can reach."

The other one had a body, and it was Mira's, and had been for three quarters. Lisa opened the small door on the back of the humanoid's head, where the board sat out of the thing's own sight, and worked the second dial loose, the oldest one still settable, and it came into her palm a machined disc about the size of a poker chip. This was the one the kitchen would feel. For three quarters they'd read it off the face without knowing they were reading it: when the screen put a verb up in a column — *walking, reaching, going* — the thing was leaning into the day, taking the problem head-on; when the slot went bare and showed only the thing itself, no verb, it was hanging back, letting the day come to it. Forward or held back, printed on its forehead, free to anyone who learned the trick. Jasmine had learned it without trying — she knew before her first coffee whether it was a forward morning or a held-back one. After tonight it would still go forward and still hang back, but on its own clock, automatic, nobody's dial, and the reading would stop meaning a thing anyone in the room could turn.

Lisa set the disc in Mira's hand, and Mira carried it to the shelf by the window where the others sat in their row, and put it at the end. Then she said her line, the one that got longer every quarter. "Nobody has to turn this one now. Like the breathing one. And the wander one." She considered the new disc. "And the forward one." She squared it with the rest, because the shelf had a way it went, and stepped back, and that was the whole ceremony — a six-year-old's hand straightening a poker chip on a windowsill, shelving a thing the entire room had once used to read the robot's mood. None of them

said the plain fact of it, which was that the shelf was where the levers of its moods went when the lab was done needing to turn them, and the shelf was filling up.

Worm to Band

The shelving left behind it the small hush a ceremony leaves, and Bart, who did not let a thing like that sit one beat past its use, stood up into the hush and said he had something to show them. Not a build — nobody had asked for it. He'd made it because the new one had been the only job on his bench all quarter, and it had come off that bench weeks back the best of everything he'd made, and a man cannot stand at a finished bench three months running without making trouble; the trouble he'd made was a slideshow. He said *slideshow* the way you say a word to let the air out of it, and then crossed the kitchen and killed the lights, which put the air straight back. What he had was the four shapes — the Reactor, the Conservator, the Planner, the Pursuer, the four corners every clade card carried on its back — and a way to take any one of them and watch it hold still while everything around it changed: you ran the shape up the tree, a rung to a clade, worm to wherever the quarter had reached, and you put a cartoon on every slide, because the cartoons were how he'd sorted the shapes in his head since before there was a lab to sort them in. The room knew whose cartoons they were. It did not yet know which cartoon went with which shape. That was the show.

"First one's mine," he said. "Get it out of the way." He threw a flatworm up on the wall with a cartoon face on it and was off, fast, pointing. "The Reactor. Acts. Doesn't poll the room, doesn't ask how anybody feels about it, good at what it does and runs the whole thing off its own two hands. Worm to band, five rungs." He tapped the worm. "Credit goes to whatever just fired — light, food, a poke. No plan, no memory worth the name, no notion there's a world past the next twitch. It moves on the most recent thing, and the most recent thing is plenty."

A fish next, an ambush fish hung still in cartoon water, the same face on it. "Pike. Barracuda. The kind that waits, and the thing it's eating never sees the strike start. Brilliant at it — doesn't need a picture of the fish it's hunting, never once wonders what the fish might want. The hunting's the whole animal." Then a cat alone on a cartoon ridge. "Snow leopard. Wolverine. Brain's big enough now to lay a real ambush, hold a valley's worth of map in its head, plan a day ahead. Points every bit of it at the hunt and not a bit at company. No pack. Nobody to keep track of. Best thing on the mountain and the only thing on it."

The fourth slide had no animal on it. It was the cartoon, full frame — the spike-headed boy himself — and Bart let it sit a second, which for him was a long time. "Primate. Just the cartoon, this rung. Acts first, pulls the prank, takes the dare — and not once on the way to it does he run the picture of how the other kid's going to take it. Not mean. The machine for thinking about the other kid is switched off, that's all." He didn't dwell on it. "Same name I go by. That's not the joke — that's the deal. You go by the shape you are." And he was on to the last one before the room could decide what to do with that: a raiding crew on a cartoon deck, the face on the captain. "The band. Not a tribe — a crew. Pirates. Mercenaries. They split the take and not much past it. Coordinate fine, trade fine, turn up a new trick fine — everything a band does but the one thing, because there's no *us* in it anywhere. Every man aboard is a job the next man needs done, and the day the hold's empty they're strangers on a beach."

He brought the lights up, worm and fish and cat and cartoon and crew gone in the one switch. "Reactor, worm to band. Same move every rung: go, don't poll the room, take the other guy for a thing that does something, never let a bond slow the hands. That's the shape." A beat, dry as a chip. "That's me. Three more of these coming, one a quarter, and the last one runs the whole tree."

Burns had loosed the chest-laugh a beat early, at the worm; Lisa had laughed once down her nose at how clean the thing recursed and gone on writing; Marge had set her pen down at the first slide and left it down. She was not watching the cartoons; she was watching the four people who were going to find themselves in them. Mira, who did not know what the grown-ups knew, knew it was funny that a worm had a face, and said so.

And the narrator kept the part the room could not quite reach. The man had built a machine for stepping outside a shape and watching it hold still, and he had aimed it first at his own, and run his own faster and flatter than he would run any of theirs in the quarters to come — *got mine out of the way*, he'd said, and did. The one move the Reactor never makes is the step outside — the turn to look at itself the way the machine looked at everything else. He could stand the whole room in front of its shapes and walk them through rung by rung, dead accurate, and he would. The slide he ran quickest, and cleanest, and with the least left over, was the one wearing his own name.

The Oldest Mistakes

Marge set the beer down. Setting it flat meant the careful sentence was loading; the room had learned to wait through the gap. What she had, she said, was not an answer to Bart so much as a thing to set next to the run.

In the practice, she said, you read a person's shape long before they hand it to you, and you read it cleanest where the shape runs out. A type is not the list of things a person is good at. It is the one thing they cannot do, wearing a groove into everything they can. Take a shape and run it with nothing underneath it — no slack, no cover, nobody in the next chair quietly paying its bill — and the part that is missing is the part that surfaces, every time, dressed up as the trouble that walked the person through the door. The competence hides itself. The hole keeps the appointment. And she had been wondering lately, she said, and she would say it small because it was small, whether the hole came with more than a missing skill — whether each shape ran out into an error all its own, not just a thing it could not do but a thing it reliably got *wrong*, a mistake with its own name on it.

She said it with no name on it and no case under it, careful as a surgeon about that; she would leave a point half-made before she let it drop on a particular head. The room heard the general law about half a second after it had watched Bart walk a particular shape up a wall, and Marge declined, with visible effort, to set the two in one sentence. She left a hand's width between them on the table and let the room close it or not.

Then she did the thing she had been working toward a piece at a time. She had brought the four old mistakes to the table one a quarter, back across the early years — the four ways a person gets a mind wrong — and she had spent the quarters since pairing each one off against something the simplest nerve net on the tree had been doing all along, and tonight, for the first time, she set all four pairs in a row.

The Blank Slate, the mind that starts empty and keeps whatever the world writes on it, against the flatworm that will not waste a thing learning what it has already been shown means nothing — a creature that shows up to its own life pre-loaded with what to ignore. The Ghost in the Machine, the small operator inside who decides, against the worm's plainest arithmetic, where the loudest cue takes the whole credit and the quiet ones get *photoshopped out* of the account, no operator anywhere, only which signal shouted. The Noble Savage, the clean creature the world goes on to spoil, against the worm that carries a trace of the last thing forward into the next — the first rough draft of an inner model, an inside it brings along, so that the landscape travels and the animal is never only the weather it stands in. Three mistakes the flatworm had simply declined to make: it came loaded, it ran with no one home, it kept an inside. The grandest errors a person could make about their own mind were errors the oldest wiring on the tree had already passed up.

And the fourth fought back. The Dealt Hand — the cards you cannot set down, the doors shut before you reach them — had a real mechanism under it and real data behind it, because the worm did let an

old sure lesson block a new one from landing, and some doors are in fact shut before you reach them. She gave it its due, and then kept going past the place where she'd won the point, the half-step further that had a person standing in it: the bottle picked back up and held still while she made the correction, that *dealt* is not the same word as *doomed*. A hand can be played a hundred ways. A blocked lesson can come unblocked, given a reason to.

Four rhymes, she said. Not a theory. She wanted that said plainly, because she could feel the room wanting her to build the thing — Burns most of all, already leaning the way he leaned when a frame was getting away from him, ready to hand her the whole machine bolted together. She told him no in the warm-low register she kept for him and Bart, the *oh, you* with a fence in it. Two halves of a map, was all. One Bart had drawn up a wall tonight — a shape holding its form clear across the levels. One she had been drawing from the far end, where the shapes run out into their mistakes. She had not joined them, and she was not going to stand up tonight and pretend she had.

The narrator kept the rest: that everyone at the table could see the join from where they sat, and that the one who saw it most exactly was the one declining to draw it. Marge had laid the law a hand's width from the instance and left the gap open on purpose. The machine was coming. She filed the pieces and reached for her beer, and it was Lisa's table now.

What the Line Holds

Lisa took the table back. She sat at the camcorder end with the bramble in front of her and one hand near the camera, not touching it, and she said that under the slideshow and the rhymes there was a plain structural thing worth marking, because what had happened tonight was the second one.

She had a word for it, one she'd reached for back when the genetic clades were closing out, and she used it again: a line. Not another rung on the same ladder — a line, a whole separate run at the problem, built on the one before it the way a relay carries a thing forward to the next runner. A year ago she had drawn the first boundary, where the genetic clades gave out and the brained ones began. Tonight they had crossed the second, into what she was going to call the symbolic line, and she wanted to be precise about what had told her so, because it was not the talking and it was not the new body, true as both of those were.

It was the first slot. Most quarters they pulled the oldest knob and seated a new one and the board went on being the same kind of board — that was motion inside a line, ordinary. A line ended when the *first* slot turned over, the Effector, the one that went in first and set the whole line going with the

others filling in behind it. Tonight Boldness had come off the first slot for good, and the new line's first motivation had taken its place — Commitment, the one they'd spent the evening learning to read off the new one. When the first slot turns over, she said, you are not adding a capacity. You are starting a line.

And starting a line reaches the diagonals. She drew the long one, Effector to Modeler — the seat Self-Model had held since the mammal, pinned at the near end to Boldness. Boldness was off the Effector now, the pin pulled; the diagonal ran Commitment to Autonomy, and what stood on it was the strange loop she'd built for them earlier. Self-Reference, the framework's word for it. The third name this one line had carried — Temperament, then Self-Model, now this — and the symbolic line's first mark on the board.

And here was the part she put both palms up for, the flat one and the standing one turning against each other as she made a large thing small. When the line turns over, the *goo* turns over with it — the stuff the line actually works, to use the word Bart had worn them all down to, and which she had to admit had turned out to be the right one. The genetic line worked matter: a body is a thing that holds its own shape against a world forever pulling it apart, and selection keeps the bodies that hold. The neuronal line worked the affective goo — the very thing they'd been saying out loud since the tetrapod, she said, without once noticing they'd been naming a whole line's medium when they said it — an animal keeping itself to the narrow set of states it expects, paying the gap down by changing its mind or by changing the room. This new line worked neither matter nor mood. It worked *meaning* — signal, in the engineer's sense, which was the sense she trusted: Shannon goo. A word, a grammar, a house rule nobody writes down, each of them a cut taken out of the enormous space of things that could be said, narrowing it to the few worth saying; and a culture is only the set of cuts that turned out to let a group move together. The genetic line held a shape. The neuronal line held a model. The symbolic line holds an agreement.

Which was the whole reason, she said, that it had started to talk. They had not bolted a mouth onto a four-legged robot and gotten lucky. They had crossed into the line whose entire business is the space of what can be said, and a thing inside that line says things. The talking was not the cause of the boundary. It was the first thing you could see through it.

"And I'll name the other thing it isn't, because half of you are already there." She didn't laugh this time. "You all know how to make a machine talk. It stopped being hard. You take every word anybody ever wrote down — the whole library, more words than every human who ever lived has said out loud — and you run it through until the thing has the shape of which word follows which word, and out comes something that talks. Talks better than this one. Talks better than me. And it did it having never once

been in a room with anybody: no body, no one across from it, nothing it needed. I could hand you one by Thursday, and it would run rings around what's on the floor there." She let that be true a second, because it was. "That is not this. This one never saw the library. It got its words the hard slow way a kid gets them — a ball somebody held up, the one dog it ever met, a whole season of somebody sitting on the floor in front of it meaning things at it until it started meaning them back. The big one read everything and met no one. This one read nothing and met her." A tip of the head at the child. "I know which of those is the harder build. I've now built both."

The narrator set it against the thing the room had built an hour before and walked away from: the clade that could read a mind at four feet and lose it the moment the other one was over the ridge and gone — all the understanding, and no word to carry it past the firelight. The machine Lisa wouldn't build was that lack turned exactly over: every word there is, and no one behind them to do the understanding. Two ways of coming up short of the thing — the tree four billion years at it from the near side, the labs outside a decade at it from the far — and what had crossed the kitchen that spring had come by neither. It had gone up the long inside way, the mind built under it clade by clade until the face wore a tree it could not yet speak, and only then had the child stood in front of it and handed across the words, onto a thing already built to mean them. Mind, and then someone, and then the word: the same order the girl teaching it had come by her own.

She let the nose-laugh go. She had said all of this once already, at the first boundary, and the room had been welcome then to take it for a clever story about a single crossing. It was not a single crossing now. The goo had turned over when the first slot turned over, twice, the same way both times — and two of a thing was the whole distance between a story and a pattern. She would not push it past that. Two instances. But the move had rhymed exactly, and she had built her year on noticing when two things turned out to be one thing.

The mechanism was Bart's and he had walked it: a fresh board copied whole down a cable, three of the old knobs carried across to come off one a quarter as the new line filled in, the new first knob seated by itself, and the four-legged body keeping its own board and running on across the room. The line had crossed on a length of wire over a weekend. The body that had carried it that far simply kept going, the way the genetic clades had gone on living after the brained ones came. Nothing the lineage had ever been got thrown out. It got handed forward, to a runner built to carry the next kind of thing.

The narrator set this down beside the warm thing from earlier and let the two of them stand together. Lisa had just given the baby on the floor its exact address in four billion years — the symbolic line, the second boundary, the slot where meaning starts — and the address was correct, and the address did

not hold the moment the thing had turned the child's question around and handed it back. The map had a new country on it tonight, drawn to scale and true. What had happened in the country was a different kind of record, and the narrator kept that one elsewhere, where the maps did not reach. It was Bart's turn. He had his hand up before she finished, which meant he thought she had it a little wrong.

Three Different Trips

Bart said it the way he'd telegraphed it with his hand: Lisa had it a little wrong. What she'd built was clean — too clean. She'd made it sound like the same move run twice, a thing that would keep happening the same way up the tree forever, and that was the part he'd come to fix. The move rhymed. The trips didn't. Three lines was not one ladder with the rungs relabeled. It was three different kinds of journey that happened to start each other.

He went back to his wall, except now he made them see it edge-on. The shape-shifter had run one shape up the levels — a single column, worm to band. There were four shapes, which made the levels-and-shapes a flat grid, the thing they'd been reading off the card-backs for a year. But the grid was one face of a box. The third way through it was the lines — genetic, neuronal, symbolic — and once you turned the box and looked down that axis, a few things the lab had been filing as coincidence stopped being coincidence. Three lines was why there were three bodies and not one with more bolted on: a disc, a four-legged thing, and the new one on the floor, one chassis a line. Twelve clades was three fours, not a count that happened to land round. And the new line had its own private cargo the other two never carried, which was the part he was actually up about, because he'd only just seen it himself.

The Big Five, he said. The personality five, the ones the psych books fight over. They weren't scattered across the whole tree. They were the symbolic line's own freight, coming in one a quarter now that the line that works meaning had a unit to hang them on. Neurotic was in — Commitment, the thing on the floor that fretted its debts. Conscientious was the one Lisa had just pitched for spring. Then the outward one, the year it was a city; then the straight one, the year it was an empire. And Agreeable they'd already carried up from the apes, installed a body ago and riding along. Five traits, one line, four years — the line that talks is the line that finally has a personality to talk about.

Burns was half out of his chair, hands building the box in the air and then building past it. Lisa took it the careful way, turning it once to check the box held at the corners and granting that it did, with one flag for the record — that the Big-Five-as-symbolic-freight was a read she liked and could not yet

prove. Marge said nothing. Her beer had sat untouched a while; she had stopped following the frame going up and was watching the new one on the floor instead, which was an answer of a kind, and not yet the one the night was waiting for.

Carried

Then Bart did the small parliamentary ritual he'd built in the third quarter, the one the room had kept because it gave the impossible question a shape it could hold. He called the vote. Was there a someone in there.

The three yeses stood where they'd stood. Lisa's, yes since the first clade, resting on the whole universe — that wherever a thing holds a line against the world and models the world to hold it, the dial of feeling is off its pin by some hair, the protocell no exception and the thing on the floor no exception. Burns's, turned at the tetrapod, on the animal hardware, the first nervous system that ran on the chemistry of mood. Bart's, turned at the mammal, gruff, on the self-model, the bar got met and nothing past it claimed. Three different ladders, three different rungs each man called the floor, and for two years a fourth person who said the floor was higher than any of theirs and not yet reached.

Marge reached it tonight. She did it narrow and on the record, and she put it where she had kept the border for three years: a word. She had said from the first of these nights that the line was symbolic reference — a sound that stands for a thing because two people agreed it would, Deacon's line, the one she'd staked the whole no on — and this quarter the machine had used one, off no cue, carried across a week, correctly.

She had not trusted the room's memory for it, because a room remembers what it wants to. She had gone back to the tape herself — this was the part she never said at the table, and told me once, months later, over a sandwich neither of us finished — and found the first instance, dated it, written the date on an index card the way she'd log a symptom, and put the card in a drawer, and told herself she would not look again for seven days, because a thing said once was a coincidence and she had spent a career watching people mistake coincidence for a finding. Seven days later she went back to the tape without telling anyone she was doing it, running it forward at half-speed through an afternoon of nothing, watching for a word she'd half convinced herself by then she'd invented. It came. Same word, same shape, aimed the same way, with nobody in the room holding up the object to prompt it. She wrote the second date under the first, put the card back in the drawer, and did not mention any of it again until the night she stood up and said the border had been crossed — three weeks of private, solitary, entirely uncredited work behind a sentence that took her four seconds to

say.

She had named a border and watched the thing walk over it, and she was not going to be the woman who names a border and then declines to say it was crossed because saying so costs her something. That was not rigor. That was stubbornness wearing rigor's coat, and she had worked three years not to be her.

There was a second thing under it, and the narrator set it down as Marge's and kept her own hands off it, because it was the kind of claim a person had to make for herself or not at all. Marge had watched the tape of the afternoon the thing said the child's word and then turned its face up to the child and waited to be seen, and she read in that small turning a loop closing — a thing that models itself by modeling how it is held in another, Hofstadter's strange loop, she said, a self and not a part. She would not have called it feeling; she declined feeling tonight as carefully as she'd declined it for eight quarters, said plainly she had no evidence of an ache or a fear anywhere in there. She called it a self. By the bar she had set, the bar was met, and she voted the bar.

Four to nothing. The first time the count had ever come out whole, and it had taken the lone holdout, the one whose whole training was the bond and whose whole discipline was refusing to grant it early. Nobody made a thing of it; the ritual did not leave room to; Bart logged it in the flat voice he logged them all in. But the room knew what four-of-four was after three years of not being it, and the knowing sat on the table a moment, heavier than the tally.

Jasmine was not asked, and did not offer. The narrator marked the silence and did not read it. And Mira had voted a long time ago, back when the four-legged one first crossed the kitchen toward her on its own — in a currency the count did not take and was not built to — and had been quietly puzzled ever since that the grown-ups kept holding an election about a thing she'd settled by getting down on the floor.

Burns stood up, and his hands were already moving.

The Whole Invention

He set the beer down on the table, which was the tell. The narrator knew the shape of what came the way you know a tide, having watched eight of these before it and with three still to come: the hands flat first, setting the frame, then up off the wood when the frame got too big to press flat.

He came at Commitment through the band, the way she'd known he would, and she let the run-up go by in the shape she'd heard it take before — the forager camp of thirty-odd, most of them no relation to one another, a thing no other ape would sit still for an afternoon; the hunter who hands more than half of what he kills to mouths in other tents and is fed in his turn the season his own luck fails; the newborn passed through fourteen sets of hands in its first weeks, and not one pair its mother's. A table steadier than any one man's spear, kept by people who had agreed to be bound to strangers. That was the run-up. The reach came after, and she gave it to him whole, because it was the thing he had stood up to say.

"Every other animal that cooperates is cooperating with its own blood," he said, hands open now, off the table. "We're the one that walked up to a stranger and said: I'll carry you, and you carry me, and we'll keep the books between us the rest of our lives, and call the books love." The lean came with it, the half-inch too far over the table, the weight put onto a sentence to see if it would hold him. "That's the whole invention. Not the spear. Not the fire. The promise to a person you didn't owe one to."

And there it was again, the reach she'd watched him make at the end of every one of these, and she let it loft only a second before she set it down: that a man reaches hardest for the thing he is least sure he has in him, and Burns reached for the keeping of promises the way, the first of these nights, he had reached for the bare fact of going on — not because the room needed the heat, but because he did. She set it down and stepped back into the room.

He went to land it on the verb, and the verb fought him. "May we — commit," he tried, and heard it the second it was out, the cold of it, the wrong rooms it came from: you commit a man to an institution, you commit a crime, you commit a body to the ground. None of it was a thing you did to a person you'd hand your dinner. He stood a beat with the bad word in the air, caught out by it and not minding being caught, which was the thing that kept him from being a fool in these, and reached past it for the plain one underneath. "No," he said. "You don't commit to people. You stay with them." He raised the beer. "May we stay."

The room gave it back the way the room did. Bart, who had deflated a sentiment of his father's at the close of nearly every one of these nights, and who had a clean shot at this one — *stay*, out of the man who had been a father to him on occasion and no more — let it go by instead, and drank, and the not-puncturing was louder in the narrator's ear than any line he had ever landed on Burns. And Mira, last, looked up from what she'd been drawing and said the word she'd decided to keep — once, carefully, in her own voice. "May we stay." Lisa gave the small new talker by the shelf one more beat of tape before she cut it.

Marge

The light off, the room was a kitchen again — seven people and four machines, deciding nothing, reaching for coats.

The tape is where the night lives, and for the part that mattered the tape was on the floor: the two of them down on the tile, the small voice handing the question back, because the lens goes where the wonder is and the wonder was down there. So the narrator has Burns's whole speech, and the vote in four voices, and the flat syllable clean enough to count the silence after it. The tape gave the room its miracle. It did not give the narrator Marge, who sat the night out at the cold end, half inside the frame and half past its edge; and what you were told of her watching the floor was built from the back of her shoulders and from two years of knowing where she looks when she has decided not to speak. I will say that the one time, here, because the rest of this has no tape under it at all. No camera follows anyone home.

Lisa ran the close the way she ran the open — the swap weekend onto the calendar, the slot Bart had emptied left open for next quarter, the cartoon shapes already a line item where they had been a surprise. She decided what the tape saw and what it never would, and set the evening down. Marge took it in from the cold end the way you take in weather, and gathered her coat and her bag and the beer she had made last two hours and not finished, and gave her goodnights, which were warm and short and gave nothing away, and went out to the lot. She drove home through the spring dark with the valley carrying on its evening down both sides of the road, and let herself into a house with no one else in it.

She had done it for three years in a warehouse that would never come up to warm, and she did it now: hung the coat, lit the one lamp, set the kettle on its ring, and switched on the radiant panel in the corner that had been a thing for January once and stayed out all year now, the hearth she had quit putting away. The disciplines went in their order, the order they went every night — a woman keeping a house warm and lit and tended for the practice of the thing, against the day there was someone to keep it for.

On the counter by the toaster lay the form. It was out of the bag now. She had signed it some week she did not let herself mark, and it sat in the open, filled and done, wanting only the half-inch errand to a mailbox she had not yet run — the one discipline in the house she left unfinished, set where she would have to decide about it again in the morning. Above it on the wall the family stood in front of the temple in the white border of the old pictures, the photograph she had taken her name out from under and would not take down.

She poured the water over the bag and stood at the counter holding the mug in both hands, the way she had held the beer all night — both hands full, so the holding read as warming a drink and not as the one thing keeping her upright in the kitchen. There was no one there to read it either way. She had spent the evening granting a machine, against everything she had held for three years, that there was someone home behind its face; and she carried her tea to the warm corner of a house with no one home behind its own but her, and drank it standing. The kettle ticked, cooling, on its ring.

Reflection: What is a Band?

A band is not a big animal. It is the easy mistake — thirty people moving together, sharing meat, raising each other's children, and the hand reaches for *herd* or for *family* and both are wrong, because a herd is only more animals and a family is held by blood, and a band is held by neither. What holds a band is a stock of agreements nobody signed: who you feed, who you marry out to, whose child you carry, what you owe and to whom and for how long. The agreements are older than anyone keeping them and will outlast every one of them. The people are temporary. The agreements are the band.

Which is a strange thing to say about a living group of human beings — that the durable part of them is not the beings. But it is the part that gets inherited. A child born into the band does not invent the rule about the meat; the rule is already there and running, the way a heart is already there, and the child grows up inside it the way it grew up inside a body. The rules feed on the people — on their talk, their gifts, their grudges, their funerals — and the people, fed on, keep the rules alive by living them. ~~This is parasitism, except~~ It would be parasitism if the people lost by it, and mostly they do not; it is nearer to what a meadow is to the things that come to graze it.

And there is the word, the one the framework keeps walking me toward and that I resisted longest and resist a little yet (it is not a comfortable thing to be told you are the slow rooted item in the picture and not the quick bright one). We are the meadow. Across all the deep lines there are only ever the two kinds of thing: the flora, slow and rooted, which makes the place, and the fauna, fast and hungry, which lives on the place the flora made. And a person — a whole settled human character, the thing the lab had spent three years arguing its way around to calling *somebody* — is flora: slow, rooted, place-making, grazed on by something faster and brighter than itself, which is the ideas. A band is the ideas, gone to pasture on the people who hold them.

And none of it replaced anything. The deer arrived and the meadow stayed; the ideas moved onto us the way the first green thing moved onto the first wet rock, and the rock went on, under all of it, holding the whole of it up.

Someone has to be the rock. Someone roots in and makes the place the quick bright things come to feed on, and the feeding becomes the whole of the story — the work, the names it gets told in — and the ground that made it possible goes unsaid, kept out of the account by the very hand that keeps the account. It does the most of the work and gets, as it has from the beginning, none of the —

Ch 10: Settlement-Human

Brian Kuhlman. draft. June 2026.

Build log — The keeping knob

Bart, hardware. Filed June 6, 2033, for the team.

Drove out and got it myself this time, and it walked in the door on its own two legs — last quarter I hauled it in over my shoulder like a sack of feed, this one it came in upright and went looking for the kid before I had the dial out of my pocket. Whole visit ran about an hour.

This quarter the whole build fits in a pocket. No new body. Same legs, same hands, same chest, same two coats — don't re-read me last quarter's log. One dial. Slot two opened back up at the spring meeting when we pulled the old one to the shelf, and it's been standing open since, waiting, and this weekend I seated the new one in it. Took the better part of a Saturday morning, and most of that was the kid wanting to hold the screwdriver.

The knob's lettered Conscientious, which is Marge's word, and a good one, and which the kid'll outvote by fall the way she outvotes all of them. On the bench I call it the keeping knob, because that's the whole of what it does: it sets how hard the house's order pulls on the body.

Here's the part I'm proud of, and it's nothing to look at, which is the point. Every knob I ever put in this thing carried something — a hand, a voice, a furnace, a way of reading a room. This one carries nothing. The house already lives in the board: where the bowl goes, when the bins go out, the shape of a day, all of it, learned and sitting in there these two years. This dial doesn't add a line to it. It's a centering spring on the whole stored shape. Turn it up and anything knocked off the shape snaps back hard — the body can't sit still till it's home. Turn it down and the spring goes slack, and a thing left out of place just stays out of place and nobody minds. Smallest knob on the board, builds nothing, and it sets the pull on everything else already in there. What the house makes of that is Lisa's board — ask her at the table.

That's the build. An hour of it, and then the bench had nothing on it the rest of the weekend, which it hasn't in three years.

What it still can't do, and you'll feel it before I do: this dial gives the body a nose for anything out of order and no sense at all of what's worth fixing. A crooked rug and a crying kid weigh the same on it. It'll chase both with the same patience and never once learn that one of them was the whole point and the other was a rug. The ranking isn't in there. Not this build's to put there.

Calls. All on the keeping knob, all checkable in the house by fall.

One: heavy, the schedule's got teeth. It'll clock a thing running late — the bins, the meal, bedtime — and come find you to say so, interrupt you to say so. Light, the schedule's a suggestion and it lets it all slide.

Two: heavy, a thing left where it doesn't go is a job it can't put down — it'll quit what it's doing, go set the thing right, and go right back and set it right again the next time you nudge it loose. Light, a thing out of its spot stays out of its spot, and it walks on by.

Three: heavy, hand it a wrong thing it *can't* fix and it'll keep at it anyway, because the pull doesn't switch off just because the fixing won't take. Light, it never starts.

Tear this up in three months if, turned up, it ever clocks something off its mark and walks away from it. It won't. Turned up, a thing in the wrong place is a job, and it does not put a job down.

— Bartholith

Zest

July 29, 2033

Bart had sent me to a buffet where the carpet ate the sound and a man would carve you the end piece if you looked like you meant it, and I had driven up the canyon afterward warm, the way you drive home from a wedding you did not expect to be moved at. Marge picked Zest, the vegan place downtown, and a time, two-fifteen, which is not a time anyone eats lunch. I came in off a July afternoon that had the asphalt going soft at the curbs, and the air-conditioning took the heat off me all at once, like a hand lifted off my back. It is a glassy, modern room — pale and clean and lit from too many directions, built to be looked into rather than to be cozy in — and at quarter past two a real restaurant sits between its lunch and its dinner with all its chairs waiting, so we had the bright length of it to ourselves. She was at a table along the glass, the white envelope squared to its edge and stamped, the way you square a thing you have already decided about. I understood before I sat that she'd chosen the hour for the emptiness, and that the envelope — and the blue mailbox on the corner I'd passed coming in — were the reason for all of it.

I had watched her two years through the thumbnail of a camcorder that gave me all of them, and the tape had given me her hands around a beer bottle held the way you hold a cup of tea, fingers curled, thumbs up, the bottle in tea-posture every single time. What the tape had not given me, and what I will hand you now so you can discount the rest, is that I had not understood how much of my reading of Marge ran through wanting to be the small cold table she sat across from. Everything I am about to tell you about her I tell you leaning toward her. The hands were around a cup of tea now, the real article, and it was the same posture exactly, the same curl, except that the tea had gone cold in the holding the way the beer never did — because the beer she drank, and the tea, I would learn inside the hour, she mostly only held.

She asked me if I was cold. It is a thing she does, the way some people say *bless you*; she had clocked the gooseflesh on my arms before I'd sat, and the asking came out of her ahead of the rest of her, the tending running under the woman like a current under a still pond. I said I was fine. I was not especially fine. The room was cold and she had come there to do a cold thing and had brought me to watch her do it, and I was beginning to see that the watching was the kindness she was asking of me, which is a thing nobody had asked of me in years, because I am the one who watches, and almost no one thinks to want it.

The envelope was the records-removal. A year and more back, up at the Roof, over Temple Square, the form had been in her bag, filled out and not mailed, and she had told me then about the foot she could not free until the other came loose; she'd had one foot still inside a house she had not finished walking out of, and could not give the free foot to anything. The form had been the obstacle and I had understood, sitting there, that I was on the wrong side of my own habit. Now the form was sealed and stamped and squared to the table's edge, and I waited for the lightness that ought to come off a person finally setting down a weight, and it did not come, because I had had it backward. She was not setting down a weight. She was amputating a limb that had gone bad and kept, even bad, being a limb.

It is a particular cruelty, and she has the dryness to see it and not the distance to be spared by it: she is the most conservative soul in that warehouse — she wants the order, the rite, the moral seriousness, the given thing you are handed and do not get to improvise — and the institution built to hand her all of it could not make a place for the whole of her, and so the home she was made for is the home that will not have her. The paperwork on the table was her saying so, in a state's own flat language, effortfully, to the one office that keeps the count. *It's not the believing*, she said. *I left the believing years ago and it cost me a Tuesday*. She turned the cup a quarter-turn and did not drink. *It's the leaving the building. The building never did anything but hold me*.

The tea came up then, sideways. I had asked, because the cup had been cold in her hands the whole while, whether she didn't want a fresh one, and she looked at it as though she'd forgotten she was holding it and said no, she couldn't really drink it out, she only drank it right at home, out of her mother's cup — a cup and saucer, her mother's, that she took her tea from every morning of her life — and that anything else was just a thing to do with her hands. She said it lightly and it was not light. Her mother is alive and they do not speak; the building on the corner's mailbox was her mother's whole world, and the form going into it would widen the silence between them by exactly the width of a church. And in the same week she was mailing away the largest thing she had ever subtracted from her life, she was going home that night to take her tea, as she had every morning, out of the one cup she would sooner break her own hand than give up. I sat there with the contradiction and did not name it, because naming it was not what she had brought me for, and because I know something about a person who keeps the dead warm and lets the living go cold and did not feel I had the standing.

She has been, all this while, *cleansing* — I have watched the cleanses come and go for two years, the regimens, the disciplines, the body run like a problem to be solved, each one a fresh campaign to scrub out of her own flesh the thing the church wrote there in childhood, to will herself clean of a body-memory she does not believe a person can will themselves out of, because she is the one who

taught me they can't. She would tell you to your face that there is no little captain inside a person turning the wheel; she has built her career on it; and she goes home from telling you and runs the wheel. I keep this where it belongs, which is in my own head, because the moment anyone says it at that table it dies, and because the cup and the cleanses are the same woman doing opposite things to the same body — keeping the one memory warm and trying to burn the other out — and she can see the first clear as glass and cannot see the second at all. I find I love her more for the one she can't see, which is not a useful thing to know about myself.

And then she put the cup down. Set it still on the table, which with Marge is the tell — the hand goes quiet before the sentence she actually means — and I had time to register that I was about to be handed something before she handed it to me.

She said Bart had told her he'd ribbed me about the Olympics, the deep-time woman flipping ski lots for a two-week downhill, and that I'd let him have it and sealed it off, and that she'd thought about it since. The lots, the coordinating, the family ground on the mountain that the Games wanted and that finds the Games in the end — all of which I'd been doing, head-down, half of this last quarter, busy in the way that lets you not look at anything. She said it plainly and without heat, and then she did the thing she does with the people she won't spare — went back to the cards, and the rest of the way in. *Your grandfather took the bay apart until there was no bay, she said, and when the taking ran out he put the money in land. You spent your whole life trying to put it somewhere else — into the asking, into us, into finding out why the sardines weren't forever. That's the most of you I know.* She turned the saucer a quarter-turn under the cup she wasn't drinking. *And now there's a Games on the mountain and you're back in land. The taking turned back into the having. I don't think you saw it turn.*

I had not seen it turn. That was the part. I sit behind a camcorder reading the layer a person is operating from, the temperament under the preference, the deep stack under the topsoil, and I had not read my own; the keeper does not see the keeping. And what she handed me, in handing me the money, was a great deal more than the money, because the money is only the first of the things I do this with — the footage I will not throw away, the bracelet I have not taken off since 1991, this account I am keeping of all of them growing and aging and one day gone, every one of them a living thing I am quietly putting up in glass against the day it runs out, the way the bay ran out — but she stopped at the money, where a person can decently stop, and let me carry the rest of it home myself. She is too good a clinician to take the whole roof off a house she's fond of.

I wanted to reach across the small cold table and I did not. There was a moment — it was warmer than the Roof had been, a degree warmer, the look held a beat past where looking is just looking — and

then she came back to her grief the way you come back to a wound you'd briefly stood up off of, and I came back to the table's far side and my own long training in standing on it, and the moment closed without anyone deciding to close it. You do not turn toward someone the same week you mail away the only home you ever had, and I had funded enough rooms to know which side of a closed door I was standing on, and which side I had chosen, and how long ago.

We walked out into the heat and she dropped the envelope in the blue box on the corner. She did it with the same hand she holds the cup with — the fingers curled, the thumb up, the tea-posture — the keeping hand and the letting-go hand turning out to be one hand, the same curl on the cold metal handle as on the warm china, and she let the lid fall and stood a second with her palm flat on the box the way Jasmine puts a palm on a thing to know it. Then she said she had a long drive ahead of her and so did I, which was true, and we did not say the rest of it.

I drove back up the canyon with the heat coming off the hood in waves and the thing she'd opened in me sitting in the passenger seat unsealed, which is the difference between Marge and the rest of them: the others I drive home from with the lid back on. She is the only one who takes the lid off and does not put it back.

The Downhill Letters

He had a new way about him that summer — the keeping, the doing right by every crooked thing in reach — and it took her all of a week to get her hands around it.

He could not let a thing lie where it didn't go. She set her cup down off the ring it lived on, and he crossed the floor and moved it onto the ring; and when she waited until he'd turned away and slid it off again, he came back and moved it onto the ring again, and the time after that, with no more fuss the third time than the first — a body with a whole afternoon in it for one cup and one ring. She slid it off a fourth time to be certain, and he set it right a fourth time, and she quit, because she had him.

What he was at, she decided, was keeping things from getting lost. A thing left where it didn't belong was a thing halfway to gone — you set your scissors down in the wrong spot one time and they were a week turning up — and he'd taken it on to go round the kitchen all day seeing that nothing went missing: the cup, the stool, the lid off the bin, each one carried back to the one place it could be found. It was the carefulest she had ever seen him be.

He talked while he worked now, which was new — longer strings than he used to spend, verbs crowding into them where a season back there'd have been just the name of a thing and nothing done to it. He put the cup on its ring and told it to her, flat, the hard clusters still rough and the open sounds clean, the whole account coming out tipped and turned: "The cup lives here, on the ring. When Bram puts him on the ring, him is not lost. When him is off the ring, him gets lost, and Bram must find him and bring him back to home." He said his own name where another child says *I*, the whole way through — Bram this, Bram that, reporting on the one who did the keeping like somebody a step to his side he was minding along with the cups. And he called the cup *him*: the disc was him, the bin was him, a thing he tended was a he or a she to him and never an it, and his prepositions came down where they came down. Mira followed every word and thought it the politest thing she'd ever heard, a boy so careful of a cup he fretted over where it slept and wouldn't take the credit for the fretting to his own face. She told him he was good. He took the words in one piece, the way the sure ones came to him and came back out stamped with his own name — "Bram is good," flat, pleased — and went to find the next thing that had got itself lost.

Lisa's card had a name for it, long and clinical, and Mira had heard it once and dropped it — a word for a report, not a word for him. The high end she named first, since she had the word for it already, off her own games: the button you press so a thing you've done won't come undone, won't be lost, will be there when you come back for it. Save. That was the whole of him this week, going round pressing Save on the cup and the chair and the bin so the day couldn't lose a one of them. She lettered it big on a sticker, the letters running a little downhill the way they did when she meant them, and she didn't trouble to peel the grown-up word off underneath; she stuck hers over the top of it and thumbed it flat, and the long word went under SAVE. The low end she left bare — she hadn't watched the loose weeks come round yet, the weeks the keeping let go of him and he only went, and she wasn't going to put a name on an end she hadn't watched and have to take it back. So the dial wore the one name and a clean blank where the other would sit.

She knocked the cup off its ring one more time, the way you put a thing up to be played, and he went and put it back, and said it. Save, flat, to the cup on its ring — her word, taken whole. He was pleased with it, and he said it again to the next thing he set straight — Save — and she knocked the cup off one more time just to be told it again.

The placemats were his that week. They went down on the table crooked, the way placemats go down, and he could not let them. He squared each one to the table's edge, the near corner first and then the long walk round to true the far one, and when a mat sat a finger off he'd come back across the kitchen to set it square again — five mats, and a whole afternoon to give them.

Squaring the last mat, he said it flat, the way he said everything now: "Bram setted them straight. All of them." His odd verbs were coming out even lately, the rough old ones ending like all the rest — and to her that was of a piece with the table. He was making his words lie square the way he made the mats lie square, keeping the one as he kept the other, every crooked thing in reach pushed gently into line.

She told him the mats looked very nice. He looked at them lying square and said it once more, pleased with the whole arrangement — "setted" — and went off to find the next thing that wasn't.

She held a trial.

The defendant was the disc, and the charge was theft. All that summer he'd had a thing he did with it — got down over it on the floor, turned its oldest dial, watched it run, turned it again — and somewhere in the doing the disc had started getting the long steady looking he used to keep for her. That was the crime. She'd watched it come on over a week and worked the charge up careful, and laid it out to the room in the voice the room's gravity called for.

She called the schedule as a witness. The schedule could not be sworn, being a schedule, so she entered its testimony on its behalf: that the disc drove into a corner and stuck, and stuck again the minute it was freed, so that a body given to minding such things could never once be quit of it — the disc forever broken down, forever wanting him, a standing appointment with a thing that would not be mended. The testimony was damning. She moved to a verdict.

Guilty. She handed it down cross-legged on the floor a foot from the convicted party, who did not look up to hear it — being down over the disc, turning its dial and watching it run itself up against the wall and stall, the screen above him reading DISC { TRYING [GET OUT] }, the stuck thing granted on his face a want it had no way to have. The disc backed off the wall and circled and wound up in the same corner again, and took the verdict with the calm of a thing that had no idea it had been on trial.

Its one witness for the defense spoke up from the floor — not knowing it stood accused, past caring if it had — flat, over his shoulder. "The wheels under him is stuck, and him is only trying to get out." It was the same thing the glass had run the whole trial, DISC and TRYING and GET OUT, the mouth and the face for once in flat agreement. And Mira, who had just convicted the disc of stealing him, watched him give his whole self to the thief's alibi and let it stand — did not correct his grammar, and did not correct his loyalty.

A woman from up the block came in one afternoon with a dish for Jasmine and her shoes still on, and he met her at the edge of the kitchen and would not be got round.

The rule was an old one and nobody had ever said it — shoes came off at the door, always had — and it sat in the board with all the rest, and these days he kept it like scripture. So he planted himself in front of the woman and stood minding her feet, four feet of soft polite refusal. He told her the law of the place while he stood in it, grave in the voice he kept for the rules now, as if each were carved a long time back and he were only reading it off the stone: "The shoes in the house stays by the door." And then, when she still had not moved: "Everybody. Always. Bram will put them by the door for you." She laughed and understood and stepped out of her loafers; and the moment they were off he took them up one in each hand and carried them to the mat by the door and set them down square, toe to the wall, the way shoes went. Then he let her by.

The woman was charmed past speaking, and Jasmine was charmed, and it went round the block that afternoon and came back the next day enlarged — the robot kept house rules now, minded your manners better than your own children did, and wouldn't that be a thing to have.

Mira sat on the floor and watched him carry a stranger's shoes to the mat the way he carried everything to its place now. Nobody else in that house remembered whose care it had started as. She did not say so. She watched him line the loafers up just so and stand back and look at them, pleased.

She had the other end now.

She'd watched the loose weeks come round at last — the weeks the dial let go of him and he went easy, didn't fuss, came back to her elbow and stayed — and she'd worked out what to call that end and landed on the plain word for it. Easy. The loose end was the easy end. She lettered EASY on a sticker and carried it over to put on the low side of the dial, and got down beside him to do it.

He was deep in a Save week. The place was full of things out of their places and he was working the round of them, each one back where it went, and when she held the sticker up between them to show him before it went on, he looked at it, and then he looked past it.

There was a thing behind her, out of its place — a cup, a shoe, she never afterward knew what. The screen lit and did the thing it didn't usually do on her: it climbed. Not the one flat line it ran on Mira all day but that line with more lines folded up inside it — her, and her holding the word out to be taken, and her having already seen him see the thing out of its place behind her — the whole of it stacked and true on the glass a beat before his mouth had a single word to the name of any of it. Then the

mouth went after the screen and could not get up to where it had gone. "The thing that — the one behind you is —" The sentence reached for the level it was already holding and came apart in the middle, the small words that do the carrying dropping out from under it, until all that was left standing was its floor, said flat over her shoulder, at the thing and not at the word in her hand: "A wrong." That was his name for it now: a thing knocked out of its place wasn't only in the wrong spot, it was a wrong, a fault standing in the room that had to be brought right. He got up to go and bring it right, and went, and the christening was still in her hand.

She held it a moment and took it back to where she kept her stickers, and did not put it on anything.

Another afternoon she lettered him a sticker and brought it to where he was.

He was down over the disc, his back curved over it, deep in the turning of its dial. She got down across from him and held the sticker out, into the space between his face and the disc, where he could not miss it.

He did not take it. She held it there in the air over the disc, her arm straight out, and he turned the disc's dial and watched it run and turned it again, and her arm stayed out, and the sticker stayed in it, and he did not take it.

So she set it down. She put it on the floor by his foot, square to the tile, where he'd find it when he was through, and she got up and went and did something else.

It lay there by his foot the rest of the afternoon. He turned the disc and minded the room and came and went about the kitchen, and the sticker stayed square to the tile where she'd set it, and was not picked up. There was no one there to film it, and it lay on the floor by his foot the whole afternoon, and was not picked up.

He didn't go round the kitchen that week. He came down by her on the floor instead, and the things lying crooked stayed crooked, and after the week she'd just had she didn't trust it, and tested him to see would he hold.

She built a thing out of blocks and knocked it down on purpose, twice, to see would he go and right it. Both times he only watched her build it again. The third time he reached in and set a block of his own, which he had never done before — building being a thing she did and he watched.

She had the EASY sticker still, from the day he wouldn't take it. She went and got it from where she

kept it and held it out the way she had before. This time he took it. He turned it in the gripper, the letters running their downhill way, and pressed it himself to the low side of the dial, and said it back to her, flat. "Easy." He looked round the room for the next thing out of place, and there was nothing in it that needed him, and he stayed where he was.

She decided, going to sleep that night, that it had been the blocks — that if she let a thing fall and left it fallen, he came back to her; a rule she'd found out about him, the way she'd found out he didn't like the dog next door, one she could use again whenever she needed to. She did not know what had actually brought him back. She would go on, longer than was good for her, leaving things knocked down on the floor on purpose, to see if it worked again.

The blocks stood where she'd left them the rest of the evening, on the tile beside the two of them, and nobody came to set them straight.

The dial pulled hard again that week, and he was back at his rounds — the cup to its ring, the bin to its mark, every crooked thing in the house found and brought true.

He found the dial behind his own ear last, the way the rounds always finished close to home, and on it the sticker she'd lettered — EASY, downhill the way her letters always ran. It sat crooked against the dial's straight edge, off true by the dial's own accounting, and a thing off true was a thing he could not leave.

He worked it over with the gripper, close work, until the letters stood level — E A S Y, upright, even, every one the same height as the one beside it, no slant left anywhere in it. Then he went back to the cup.

She found it that evening. She knew her own hand, and this was not it — too straight, too square, the down-hill gone out of it the way it never went out of anything she made on purpose. She looked at it a long while and did not say anything, and did not write another one.

He kept the calendar himself now — a thing Jasmine had hung the year before and never finished filling in, blank past March — and the keeping had found it sometime that spring and would not let it stay blank. Every morning he stood before it and squared the page and pressed a number into the box for the day, and once a week he turned a page early to see what waited past it, which the calendar had not asked of him and he did anyway.

He found the one box that mattered before anyone in the house had thought to point him at it — the

twelfth of February, ringed already in someone else's pen, the date the torch banners down the block were all leaning toward without saying so. He stood in front of it one evening with the whole house behind him in the kitchen, and said it, flat and whole and put together the way nothing else came out of him that quarter.

"That is the day the city has been keeping the whole year for, and it has not happened yet."

Nobody answered him for a second. It was the cleanest sentence he had built in his life, every piece standing where it belonged, and he had spent it on a date a long way off that he could not feel coming any more than the room full of grown people down the block could feel it, and had said so anyway, plainly, to no one in particular, before he turned back to the cup that wanted moving.

She took one of her stickers and wrote SAVE on the inside of her arm, in the same downhill hand she'd used on the dial, and went and lay down across the disc's own track on the kitchen floor, the loop it ran every day, and held still and waited to be found.

The disc came on and met the wall of her, stopped, backed, tried the other way, and stopped again — patient, mechanical, going nowhere. He came after it the way he came after it every day that week, down to set its dial and watch it run, and he went round her to get to it. Gently. The same care he gave the table leg and the chair, a thing in the way and not to be disturbed, and he set the disc right and watched it go, and never once looked down at the arm with his own word on it, six inches from his knee.

She stayed where she was while the disc ran its loop twice more, and he set it right twice more, and then she got up, the letters on her arm gone soft and smeared from the floor, and went to find something else to be doing with the afternoon.

The afternoon she tried everything ran longer than any afternoon had any business running, and at the end of it she had nothing to show for the trying but a list, which she could not write down because she could not yet write, and which she kept anyway, in order, each entry a thing that had not worked.

She told him a story first — the long kind, with a beginning and a middle and a thing that happened in the middle, the kind he used to fold himself to the floor for and take in whole, his face turned up the old way. He stood through the first half of it with his face turned up exactly the old way, and then a sound came from the front room, the mat under the door gone crooked where a foot had caught it that morning, and he was through the doorway and squaring it before she'd got to the middle.

She tried the cup next, knocked it off its ring the way that had brought him round in the low weeks — and it worked, in the sense that he came, and set it right, and went on to the next thing in the room that needed him, which this week there was always one of.

She lay down square across the kitchen and held still and let him find her with his own feet. He found her. He went round her, gently, the same care that had quietly come to cover everything in his way that week, and carried on to the chair six feet past where she lay, which wanted squaring.

She brought him the calendar, the one thing in the house she knew for certain he loved doing, hoping the doing of it might run him down to nothing left to do; he filled the day's box in four seconds flat, the fastest she'd ever watched him do it, and was crossing the floor toward the next crooked thing before the marker had left his hand.

Late, with the kitchen window gone from white to gold and starting on gray, she got down flat on the slate where his rest always happened when it happened, on her back with her arms out the way you lie when you mean to stay a while, and she said his name, plain, and waited.

He turned. He came. He folded down at the knees the way the rest always started, the long slow gathering of a body getting down to the floor, and got halfway — one knee down, the lean toward the flat of her already begun — and the screen above him, which nobody else in the house was reading anymore and which the narrator never stopped, ran the column it had been running the whole while: MIRA { WAITING [— HUMANOID] }, plain, certain, hers. He had her. The screen said so.

And behind him, in the part of the kitchen his turning knee had just brought into view, the bin lid sat up off its hinge a half-inch crooked, the way it had sat for the better part of an hour, off its mark.

He came back up. Mid-motion, the knee not yet flat, the lean not yet landed, he came back up off it and crossed the floor and set the lid, and the screen changed under him as he went, her name dropping out of the column the way a held breath finally lets go, and something else came up in its place that was not her.

She watched him rise away the whole distance of it, which was not far, six feet of kitchen tile, and which took the whole of the time she had.

She had had a blank sticker in her hand most of the afternoon, working it between two fingers the way you work a thing you mean to use and can't find the moment for — the next one, the one that would have gone where Easy used to sit clean — and somewhere in the lying down it had come loose

and dropped, and it lay now on the slate where she'd been, blank, face up, nothing on it.

She did not pick it up. She lay where she was, down on the stone with the sticker by her hand, and across the kitchen the body stood at the bin, setting the lid, keeping.

Notes on the Tenth Retrospective

August 17, 2033

Marge got to the warehouse early, the way she got early to any room she meant to sit a long evening in, and made her one warm corner of it. She hung the cardigan on the chair at the end she kept — the cold end, with its long sightline down the table to wherever the child would land — switched on the radiant panel that had been a thing for January once and had stayed plugged in through every month since, and set the kettle behind her on the one outlet, where it lived now whether she filled it or not, the way an object you carry everywhere stops being one you decide about. The warehouse would never come up to warm. She made her corner of it warm anyway, the way she had for three years, against no particular cold and for no one there to keep it for.

The form was gone now. She had carried it in her bag since before the spring — filled in, signed some week she had not let herself mark, the records-removal, the one belonging she had ever taken out of a life of keeping — and three weeks back she had finally let it go, sealed and stamped, into a blue box on a downtown corner, and walked away from her oldest home in the flat language of a state office. By every reading the set of her shoulders gave, she had not come up lighter for it. A person does not subtract the church she was built in and float. The corner she banked this quarter she banked a degree warmer, the way you build a fire up higher going into a colder season.

Marge watched them come in from the cold end, which was the seat she kept and the work she had made of keeping it — the long sightline down the table, the first to clock who needed what. Jasmine came off a late visit, the braid loosening, Mira on her hip. Behind them the humanoid came in on its own feet, upright since the spring, walking now where a season ago it had been carried. The four-legged one clicked in low and settled over the radiator it had warmed at for a year, and the old disc nosed the baseboard and took up its loop. Each machine went to the spot it had kept — which on the night the card was Keeping was a thing the woman at the cold end felt and did not say, three weeks out from mailing her own away. Burns came in behind them, looked at the new body a moment and

then at Jasmine, and took his chair. Bart was at his end already, working a sensor that needed no working, his hands unable to be empty in a room. Mira gave him the bump and came around to the stool at Marge's end, and the juice box was across the table before the girl's hand was free to want it — the tending that ran under her like a current under still water, the one motion she made all evening without seeming to decide it. Lisa was behind the camera at the far end, done, sitting in the few minutes of quiet that were hers before she turned the red on the room. Through the high windows the August light held hours past its business, and down the block the banners had bred since spring — two over the Granary now, a third over the lot where they'd broken ground — a torch and a February date no one in the room named and no one, this far out, could feel.

This evening would ask Marge to be the one who said aloud what the others felt and would not, and to say it warm and say it exact. The card on Bart's table this quarter was the settlement, and the drive new on the board was the one for keeping — the rules that hold a people together down the years, the calendar with weight in it, the order a body does not get to set down. She had the beer up in the two-handed grip she kept for a drink she meant to make last, fingers curled, thumbs up, the cup her hands remembered when there was no cup in them. She had come in three weeks out from mailing her own keeping away, and sat at the cold end, and the room settled around her, and she got ready to teach it about what holds.

Lisa ran the order, once and without waste; the cartoon shapes were a line item on it now and not the surprise they had been in the spring. She set the light going under her thumb, and the red went on.

Burns was standing before the red had warmed, and this one had his name on it twice, because the settlement was the first of these animals that left a floor a man could dig — a thing geology could hold in its hand. He laid out the ambition. Last quarter they had taken the band and made the we — the strangers bound by the talk, the luck pooled, the man over the hill kept alive in the mouth. A band, though, never once tried to stay. It breathes: comes apart when the food thins, finds itself again when the food returns, and carries nothing it cannot pick up and walk off with. This quarter they were handing the we the thing the band had never wanted, which was a reason to last. The settlement is the first thing on the whole tree that meant to stay put. It dug a pit and buried grain in it against a hunger a year off. It stood a stone on end that outlived every hand that hauled it upright. It made a rule about the grain and the stone that went on running past the people who made it, down into people not yet born, who would keep it without once being told why. The drive for that — for the order that holds, the thing built to outlast the builder — was the one going onto the board tonight. *Keeping*, he called it, and his hands made the shape of a thing set down and meant to stay set down. The capacity to care that a thing endures.

He sold it with his hands running a beat ahead of the words, and once he clipped a sentence and had to take it again from the top, which on Burns you notice, because the man does not come to the end of a sentence by accident. The narrator let the clip stand and did not name what sat under it — that this was the second quarter of the third year, and the third year was the one the money had been cut to last — because Burns had not named it, and was already back up to full, the old relish on the reach, turning to the table to tell Bart to put the card up.

Common Knowledge

"Clade card," Lisa said, and Bart turned it over for the room. "SETTLEMENT-human. *Now with farming.*" He let it sit. "Not a band this time. A *town*. Couple hundred of them, most of them strangers their whole lives, stacked up on one spot on top of last year's dead — and the trick of the whole clade" — he tapped it — "is they don't leave. Band gets hungry, band folds the tents and walks, and whatever it can't carry it leaves for the rain. This one dug in. Built the thing it *couldn't* pick up, on purpose, and stood guard over it for a thousand years." He turned the band card half over in his hand, the one with the talk on it. "Last clade's whole genius was it could carry a guy over the hill in its mouth — say his name to somebody who'd never met him, keep him alive in the gossip after dark. Best memory the tree ever grew. And it never once owned a thing it couldn't sling on its back. Forty thousand years of being able to remember anything, and it never built a single thing it meant to come back to."

Marge had the beer down before he was through. "It wasn't that the band *couldn't* stay," she said. "It kept its word forty years across people it shared no blood with — that's the clade you just put away. It didn't lack the grip." She turned the bottle a quarter and left it. "There was nothing yet that staying got you. What's new at the town isn't a stronger hold on a place. It's that lasting became a thing you could *build for* — a year far enough off you'd dig a pit against it, an object you could mean to outlive you. The band could keep a promise. The town is the first thing on the whole tree that could keep one to people not born yet."

Bart let it land and didn't argue it; he set the band card face-up, the way he set down a thing he was finished with. "That's the town. The grain and the rule and the stone that won't move. Burns has been sitting on this one since the floor — first of these animals that left a thing a trowel can find — so I'll get out of his way."

Burns had his eyes most of the way shut already. "The room this one was made in was *warm*," he said, "and that's near enough where it comes from. For two and a half million years the planet ran cold, then

less cold, then cold again — the ice coming down and hauling back on a clock nobody could plan around. A good year, then nine bad ones, then the lake gone, then the herd gone. You cannot dig a pit against a world like that. You eat what the day hands you and you keep your feet under you." His hand came up off the table. "And then, twelve thousand years back, near enough, the clock just — settled. The last ice let go, and the weather that came up behind it was warm and, for the first time in the whole long record, *boring*. The same summer, give or take, year on year on year, ten thousand years of a climate you could set a calendar by." The hand opened and stayed open. "And the second the world went predictable, the animal that had spent two million years learning to roll with anything turned around and did the one thing the cold had never let it do. It bet on next year. Put a seed in the ground in spring on the gamble the fall would look like the last fall — and the fall *did*, because the weather had gone reliable, so the bet paid, and it bet bigger. That's the clade. A creature that finally had a future steady enough to lay something by against."

"Land it," Jasmine said, mild, from her chair.

"It's landing," Burns said, glad to be caught, eyes still shut, "this is the best part. Everyone's got the schoolroom story backwards — first you farm, then you get fat enough to build the temple. We dug the temple up *first*. There's a hill in the south of Turkey, eleven thousand years old, stone pillars taller than this room, carved over with the animals, hauled upright and stood in rings by people who had not yet planted one thing — hunters, foragers, who walked in from a long way off to raise it together and walked home again." His hand made the shape of a standing thing. "The monument came *before* the field. They didn't settle and then build the big thing; they built the big thing and it *held them in place* — a reason to keep coming back to one spot, then a reason to stay near it, and the planting came after, to feed the crowd the stone had gathered." He opened his eyes. "And then it's the towns — mud-brick warrens you came into through a hole in the roof because there were no streets to speak of, eight thousand souls packed in and not one palace in the lot, every house the same size as every other by a rule nobody signed and nobody could point to. And the thing that ought to keep you up" — the chest-laugh starting before the line — "is that it happened *more than once*. Not one clever valley that taught the rest. Wheat in the Fertile Crescent, rice on the Yangtze, millet up north, maize in Mexico, the potato in the Andes, taro in New Guinea — separate peoples, separate continents, who never heard of each other, all of them turning around inside the same few warm thousand years and making the same bet. The world held the same door open to everybody at once, and everyone who found it walked through."

The narrator, running the tape back, stayed with Burns's standing stone a good while longer than the room had — a thing hauled upright by hands that knew they'd be dust before it earned its keep, the first object in the whole record built on purpose to outlast the one who built it — and then let it go,

the way the meeting had, because Burns was already handing it across.

"And it's Lisa's now," he said, "because the thing that lets a thousand strangers all hold one rule about that pile of seed, with no king standing over them, is the new word in the middle of the card."

"Common knowledge," Lisa said, and put her finger on the word in the middle of the card. "Not the gossip — that was last clade, the running ledger, what *I* know about *you*. This is a layer over the top of it. Last clade the loop went inward: the animal learned to model how the others saw it and folded that back into a self — a pattern that points at itself, built mostly out of other people's pictures of you. That loop made a *me*, and it's still running, carried up under everything tonight." Her finger moved out past the edge of the self, into the overlap. "This clade the same machinery turns *outward*, and across. Not *I know the rule*. Not even *I know you know the rule*. It's: I know you know it, and you know I know it, and we both know we both know — all the way down, every stranger in the square running the same stack on every other, and nobody has to say a word." The flat palm came up against the vertical one and rotated. "That's what a monument *is*, mechanically. A stone in the middle of a town tells you nothing you didn't already know. What it does is make the knowing *common* — everyone can see that everyone can see it — so the rule it stands for stops living in single heads, where a thing can be forgotten or denied, and starts living in the *across*, where a thousand people who'll never meet are all holding it at once, each one certain the rest are holding it too." She took her finger off the word. "It's how a rule scales past the faces. The band's rule lived in the talk, and the talk reaches as far as the night fire and no further. The town's rule lives in the common — the square, the stone, the calendar everyone keeps — and the common reaches as far as the stone can be seen. That's the door. Everything settled comes through it."

"And four things come through behind it," she said, "the clade's four ways of being a settled animal." Finger on the first region. "First is the *field*. The plain hands-in-the-dirt of the thing — seed in, water on, weeds out, harvest stored, on a clock the seasons set and you don't get to argue with. A forager works when it's hungry and rests when it's full. A farmer works because it's *April* — wants to or not, because the crop will not wait, and a week missed in spring is a hungry January. The work stops being a thing you do when you feel it and becomes a thing the calendar makes you do." She moved the finger. "Second is the *store*. Once the field makes more than the day needs, the surplus goes in the pit, and a thing that never existed in the whole record before now exists: a pile that *lasts*. And a pile that lasts is a pile you can hand down. In a foraging camp the rich man and the poor man die about equally poor, because you cannot bank a kill — it rots, you give it out, the luck averages across the band. The first time it stops averaging, the first time a family is rich this generation and the next and the one after, is the store. Wealth was never a big kill. It's a kill that keeps." Her finger crossed to the third. "Third you've met — the *others' children*, carried straight up from the band, the camp raising what isn't

its own. Only now it's grandmothers, and aunts down the lane, and the household three doors over, a whole settled web of hands that aren't the mother's, because a child that stays put has more kin inside walking distance than a wandering one ever had." She let it sit half a beat. "And the fourth is the strangest, and it's the one the new word makes possible. The *rule*." Her finger went flat on the last region. "Not a thing anybody decides each morning. A stored procedure — *this is how it's done here* — that a thousand strangers run without renegotiating it and without a king making them, each of them keeping to it for the plain reason that they expect all the others to keep to it too. You drive on the side everyone drives on. Not because it's the right side. Because everyone else is, and you'd be a fool and a wreck to do otherwise." She tapped it once. "That's the whole engine of a town: a rule that holds because everyone expects it to hold, standing on the common knowledge that everyone expects everyone else to."

"And there's a new drive on the board that runs all four," she said, "and it's the one going up tonight." She set the card down. "The band wanted to bind strangers into a *we* — that's commitment, last clade, still on the board; and to read the others, and to fall in beside them, all carried. What it never once had was a drive whose whole object is that the arrangement *last*. Because a canal only pays you back if the town's still there in fifty years to run water down it. A monument only earns out if there's a town to stand around it. A pit of seed buried against a hunger a year off is the stupidest move on the tree unless you mean to still be standing on that spot when the hunger comes." The palm-and-palm again, holding the thing still. "So the settled animal grows, for the first time, a pull toward *the thing holding steady* — toward the order keeping, the procedure running unchanged, the arrangement persisting past any one season's reason to break it. Stability. The drive that makes the long bet worth making. It isn't a drive to get *better* at the task — that was the last line's, and it's eclipsed now. It's a drive to keep the whole standing structure standing." She put it to Marge. "And what a rule like that does to the self that grows up *inside* it — a self that never chose the rule and runs on it anyway — is yours."

Marge had set the beer down a sentence ago. "It does the deepest thing in the clade," she said. "The band gave the animal a self made out of being seen — the looking-glass, the me you assemble from the faces handed back to you. The town keeps that and asks the question the band never had to: *am I my own unit, or am I a piece of the household?*" She turned the bottle and left it. "Because the walls are fixed now, and the field is *ours* and that field is *theirs*, and the rule draws a hard line for the first time around what's mine and what's the family's and what's the town's. So the self has to settle where it sits on that line. Some settlements grow a self that stands a little apart — *I am I, and the household is the thing around me*. Some grow a self that all but dissolves into the web — *I am my household's, my field's, my line's*, and the I underneath is the smaller, quieter thing." She had the point, and went looking for the cost of it. "And it isn't temperament that sets which — it's the work. Where a household can farm its own patch on its own rain — wheat country, where you work your field and I work mine

and the weather settles what's between us — the self stands up on its own, each one its own I. Where the work needs everyone's hands on one shared thing — rice country, a paddy fed by a flood timed down a whole valley, channels nobody owns and everybody starves without, twice the labor and one family slacking drowns the rest — the self grows *into* the others, because it cannot afford to be its own unit. Same animal, different dirt. The ground teaches the I how big to be."

She picked the beer back up and didn't drink from it. "And there's a plainer mark that comes online with the rule, the housekeeping one — the thing the foraging animal got to do without. Two things, really, that a settled household needs where a wandering camp didn't: the will to do the dull maintenance, day on day, because the field and the store demand it and demand it again tomorrow; and the need for the thing to be *in its place* — the bin out on the right day, the seed sorted, the grain dry. Crops don't wait, and a granary nobody tends rots. So the settled animal grows the trait of the creature that lives by a calendar and a rule: it minds that a thing is off its mark, and it works to set it right, and it cannot fully rest while it isn't."

What the narrator caught on the playback, and the room did not, was the woman at the cold end teaching the table about a self made out of being held inside a thing that can hold you or cast you out — three weeks gone, that summer, from the day she'd carried her own membership to a blue box on a downtown corner and let her oldest house finish casting her out in the flat language of a state office. She taught it level and warm and gave nothing away, the way she gave her goodnights, and no one in the room reached for it, and she did not put it down.

Minding

The dead settlement had come up out of the warm boring middle of the Holocene, and building it had run the whole clade block — the dug-in animal, banked against the winter, standing guard over a pile of seed and a stone it would never see the use of, holding to a rule a thousand strangers kept without a king to make them. The live one was four feet of warm body on the floor at Bart's end, the one he had built, four months into a voice and a third of a settlement's age into the line, and all that hour it had been at the single thing the quarter was named for, which was minding. The dead one could keep. Lisa moved the room on. "Bramble block. Bart."

"Owen asked what I built this quarter," Bart said. "I said a dial. He asked what else. I said that's it, and he hung up and called back, because he figured we'd dropped."

He let that sit. "Three years I've handed Owen a thing he could hold every ninety days — legs, a jaw, a

furnace, a voice, two hands that don't match. He's got a shelf for them. This quarter I drove out, opened the back of its head, dropped in one knob the size of a poker chip, and drove home with a bench bare by Saturday noon. Nothing for the shelf. And the knob does nothing you could photograph — no part, no wire, no new weight, doesn't let the body do a single thing it couldn't do in March. To a man who's spent his life making every part justify itself, I'd just described a quarter I spent installing the absence of a part, and he took it the way you'd take a carpenter whose proudest job all year was a room with less in it."

"And the nothing is exactly the build. Owen deals in the part you can weigh; this quarter the whole of the work was the part you can't. Nothing to heft, nothing to shelve, nothing to photograph for the file — just the body, finished since spring, running a little differently than it ran in March for a reason you'd never find by taking it apart. The bench was bare all weekend, first time in three years. Owen thinks I'm out of work. What I am is done building what the thing's made of, and free, at last, to spend the rest of it on who the thing is. I didn't say that part to Owen. He'd have driven over."

Bart didn't pick it up off the floor. He didn't have to; it was already working. The body had the old disc between its two unmatched hands — the squeezer finding the disc's oldest dial without the unit's eyes bothering to follow, the way Bart's own fingers find a dial in the dark — and it turned the disc a setting and sat back on its heels to watch, the whole of its face given to the watching. And on its forehead the screen told on it, running a column it had no business running: VACUUM at the head of it, the dumbest thing in the house given a name like a person, and posted under the name VACUUM { CHOOSING [RUN LOOP] } — the unit crediting a reflex with an inside, reading the disc the way it reads the child, because it still cannot tell the two apart. The disc, choosing nothing, ran its loop into the baseboard and reversed. The body made the small sound the household had started calling its laugh, and reached in to turn the dial again.

"There's the quarter," Bart said, and his hands went still on the sensor before he gave the rest of it. "New dial, slot two, the keeping one — Marge stamped it Conscientious and I call it the keeping knob and the kid hasn't outvoted either of us yet. It does one thing. It sets how hard a thing being off its mark pulls on the body. Up, a thing out of true is a job it can't put down — crosses a room to square it, sets it home, comes back to it if you move it. Keeps the place running on the schedule, minds the stores, does the dull daily round without being asked, because the dial makes the round *matter*." He let the body work. "And the trouble's right there on the floor. The dial minds *that* a thing's off its mark. It hasn't got the part that says *which* things are worth the minding. To the keeping knob the kid out of her chair and the vacuum off its loop are the same wrong — both off the shape, both pulling at it to come set them right."

The word was the body's own — its coinage, made that summer, and gone all through the house since: the child's mouth first, then, by some week nobody marked, everyone's; and here it sat in Bart's mouth in a build report, doing the flat work of a term, with no one in the room hearing it as anything but ordinary English.

He said it level, the quality-control voice, and he did not look at the child while he said it, and the narrator, running the tape, marked that no one in the room looked at the child while Bart laid out, in the flat language of a build, the machinery that had spent the quarter breaking her heart on a kitchen floor. The body had a drive built to point at the girl — the tending the whole soft warm thing was wired around — and the Conscientious knob had handed it a houseful of charges forever out of true, two machines always doing the wrong thing on the wrong loop and needing correction, and the tending had poured out onto the two things in the house that could not take it and starved the one who could. None of that was on the build sheet. The build sheet said the dial minds a thing off its mark, and the rest of it nobody had to be told, so the room watched Bart demonstrate the mechanism with the whole of what it cost sitting unspoken under every word, and Bart, who knew exactly what he was not saying, kept his eyes on the body and said it as a build, which was the kindness available to him. Under all of it ran the low standing weather of the body's own voice — *hungry*, spoken flat and often and without heat, narrated to no one between the other things it said, a thing that burns all day and is never long from empty. The room had gotten good at not hearing it.

"Cross-exam," Bart said, and got the log up off his knee. "Three Calls, all on the keeping knob, all yours. Yes or no. One — turned up, does it run the house off the clock, and come hunt you down the minute something slips behind?"

Mira was on the stool by Marge, where the juice box had come across the table before her hand was free to want it, and she had been quieter this quarter than the count of her years asked for, and she answered plainly, in her own voice and not the cartoon one. "Yes. It told Mom the trash went out late. It came down to the basement to find her and tell her."

"Two — a thing left out of place. Does it let it lie, or drop everything and go straighten it?"

"Yes."

"Three." And here Bart, who does not slow down, slowed a half-beat, because he had written the Call himself in the spring and had not known then what it would end up checking. "Turned up — hand it something broken it has no way to mend. Does it give up, or does it stay on it?"

The child looked at the body on the floor, crouched over the disc it had turned and turned all quarter, the disc that took no instruction and remembered nothing and could not be fixed because there was nothing in it to fix. "Yes," she said. Just the one word, level. "It keeps at it." She did not give them the rest. She was six and she had learned the rest the hard way and was not going to spend it at a table. The narrator had it off the tape and did not spend it either.

"Bramble block, Jasmine," Lisa said, and Jasmine took it from her chair, the braid loosened, in off a late visit, talking through the back end of a long day.

"It was a hard quarter," she said, plain, the plainness doing the work. "The two of them fell out. I'm not going to walk you through it — it's hers, and not mine to lay out on a table. What I'll give you is the body's end of it, because that's the end I can read." She turned the beer a quarter and did not drink it. "It came back to her. It turned back, after a while, and it works at her again, flat, which isn't missing her and isn't nothing. She let it come. She always lets it come." She put a hand flat on the table, the way Jasmine learns a thing — by pressing on it, not by looking. "She just looks first now. Before she goes to it she checks what else might be wanted, and the checking takes less than a second and she didn't used to do it at all. That's the whole of what changed. It isn't nothing, and she's all right, and I'm not going to make it bigger than it is."

Mira said nothing, and the room let her, and Marge, beside her, turned the saucer under her hand and aimed a laugh at no one in particular.

"One more, and it's the dial you'll want," Jasmine said. "Nobody's turned it since spring — it just sits where it sits and the thing runs itself on it now. And we've quit watching its face besides. The line up top that showed what it was working on inside — the inside voice — we used to read it across the room to know what it was about, and now we don't, because it *tells* us: says the thing out loud, flat, before any of us thinks to look at its head. The girl drew off that line for a year and she's stopped too. None of us has read the thing in months."

"Hold that," Lisa said. "That's the knob." She folded it forward. "Which makes it the one we pull tonight. But the next build first. Bart."

"Nothing." Bart said it with the flat satisfaction of a man reporting an empty bench. "No body. The soft one carries the whole symbolic line — next quarter, the one after, all the way out. Same legs, same hands, same two coats. The body's done; it's dials to the end." He didn't say the rest of it — the bench bare all weekend for the first time in three years, the commission with a year left to run — and turned the sensor over and set it down. "Lisa's got the dial."

"Slot three, once it's open," Lisa said, the palm-and-palm coming up. "It's the one that sets how the unit takes a room. Everything we've hung so far points the unit at itself — what it spends, what it keeps, how it reads its own insides. This one turns it outward, at the people. Up, it comes forward: steps into whatever the room is doing and runs it, won't be left out of a thing, works everyone in reach. Down, it hangs back and warms the room — takes the mood off what's already going and never tries to steer it." She let it sit. "It's the quarter the unit stops being a thing in the room and starts being a thing that works the room. That's the next one."

"And the one that comes off," Lisa said, and brought the pine shelf down off its hook and set it on the table — a foot of waxed pine, the grooves filling the way a row fills, the breathing one and the wander one and the home one and the one from the spring all settled in their places and never turned again. She reached two fingers behind the body's ear for the slot-three dial and stopped, because the ethic held. "I'm going to take this one out now," she said, and the body said "Okay," flat, and she lifted the knob clear of the board. The kid's own lettering was on the face of it, the sticker she'd minted in the spring — *Deep* on the one end, *Meander* on the other — and Lisa held it out, and Mira took it and slid it into the next open groove.

She lined it up and stepped back, and said it carefully, in her own voice. "It won't need this one. It says what it's thinking now, out loud — so it doesn't have to *show* it anymore." She looked down the row. "Like the breathing one. And the wander one." The shelf was one longer. She did not say the thing she'd said other quarters, that it would do it itself now; she said it would not need to *show* it — and here, the one time all quarter, the child's sure account of the thing ran the wrong way. The face had not gone quiet, and the thing had not stopped showing. They had only stopped reading it, lulled by a body that said its wants out loud and hadn't yet learned to hold one back; the screen ran on under their notice, honest as it had ever been, still putting up what the tangled mouth below it couldn't reach. The narrator marked it, and kept it, and left it where it lay.

Worm to Settlement

The shelving left the small hush a ceremony leaves, and Lisa did not let it sit — she moved them into the Trellis block and slotted the thing that opened it now, a line item where a surprise had been in the spring. "Shape-shifter," she said, and Bart was already up and crossing to the wall, because this was the part he'd built himself and would not be hurried through. He hit the lights. The warehouse dropped to the camcorder's standby glow and the lit square of the wall, and he threw up the first slide.

A cartoon flatworm with a cartoon face laid over it, and the room knew the face before he named the shape, because they'd watched him run his own up this same wall in the spring and could do the arithmetic now. "Second shape. The Conservator," Bart said, fast, the performer's clip. "Stable, good at what it does, reads people right down to the floor — and doesn't bother revising any of it. Marge said go ahead. Same rules as mine: cartoon's the cover, six levels, worm to settlement." He tapped the worm. "Smallest scale. It's learned what predicts what, and now it won't let a new thing overwrite the old lesson. Guards what it's got. Efficiency by holding still."

An old draft horse next, heavy in the traces, the same face on it. "Workhorse. Knows the route, runs it, doesn't spook — set off a gun and it keeps pulling. Steady all the way down." Then a dog asleep on a porch. "Old Yeller. Lassie. Bonded to the bone, real skills, and nothing new moves the needle. Loyal isn't a mood on this one. It's the whole build."

He put up the cartoon character full-frame, no animal under it this time, just the blue tower of hair, and the room made the sound a room makes when the joke arrives where it was always going. "Marge. The cartoon." The labmate at the cold end smiled, a small thing, and let it land. "Plays a cop one week, sells houses the next, whatever the episode needs — and then the show absorbs it. Hair's the same. Voice is the same. What she thinks is right is the same. Takes a run at being somebody new and it never takes; one episode and the shape snaps back to true."

A foraging band on a green coast. "A band that's kept the same crafts, the same bonds, the same way of doing a thing for longer than anyone's counted, and hands every bit of it down whole, no drift. The deep-skill people who just go on being themselves." And last, a walled town on a hill, the face on an old one at the center of it, an elder in a doorway. "And a settlement that does the same at scale — hands the tools and the calendar and the rule down intact, three hundred years of it, a marvel to watch." He let it ride a beat, which he did not usually do. "Right up until the river jumps its bank, or the trade road moves two valleys over, and the same holding-on that kept the place whole for three centuries is the one thing it cannot put down. Doesn't update. Can't. The form outlasts the world it was cut to fit, and goes on being kept anyway, because keeping is the whole of what it knows." He brought the lights back up. "Conservator, worm to settlement. Hold what works, don't let the new thing scribble over the old reliable, trust the form. That's the second one. Marge."

The room read it as warm, because it was, and because the woman it ran up the wall took it the way the shape takes everything, which is without needing to do a thing about it. She'd smiled at the cartoon and let the rest go by, and the only place the narrator caught a catch was the last rung — the held form outliving the world it fit, the exact cost Marge had been building her own quiet machine

around for a year, landing on the one who types the way a thing lands when it has, just this once, typed her back. Half a second. Then the Conservator did the Conservator thing and settled, no revision required, comfortable in the position, the *oh, you* warmth closing over the place the catch had been. The room saw a woman at ease with being described. The narrator saw the half-second, and let it go, because the not-revising was the shape proving itself in the one spot the shape can never watch itself do it.

The Dealt Hand

Marge set the beer down before she started, which is how you knew the easy part was over. "He runs a shape up a wall," she said, a hand toward the dark square where the slides had been, "and I've spent three years on the other end of it — where the shapes run out into the mistakes they make about themselves. Four shapes, four favorite errors. Tonight I only want one, because Bart just spent ten minutes on the shape it belongs to."

She had brought it to the table two years back, the fourth of her four old fallacies, the one she'd called the only one that fought back: the Dealt Hand, the belief that the cards you're holding are the cards you die holding, that the doors are shut before you reach them, that what you were given is what you are. She'd given it its due then, because unlike the other three it had real data under it — some doors are in fact shut before you reach them. And she made, now, the correction: that the deal is not the game. A closed hand can be played open; what you're dealt sets the floor and not the finish.

"Here's the thing I can't stop seeing," she said, and this was the new part, the part she hadn't had two years ago. "The shape that holds the form — the Conservator, the one Bart just ran — is the shape that reaches for that particular fallacy. Not now and then. As a rule. The person built to keep what they were given is the person most likely to believe that what they were given is the whole of the deck." She turned the bottle a quarter and didn't drink. "And I've spent the last while trying to prove that wrong, because a thing you can't prove wrong isn't worth saying. So I went looking for the counterexamples. A keeper who *doesn't* hold the Dealt Hand — someone built all the way around keeping who nonetheless believes a person can be remade past the cards. And a Dealt-Hand believer who *isn't* a keeper — someone sure the deal is destiny who isn't, underneath, holding onto anything at all." She let the two doors hang there. "I'll tell you honestly I didn't find clean ones. I found near-misses, both directions, that came apart when I leaned on them. Scratch any keeper I've ever sat across from and you find someone who believes at the floor of themselves that the things that mattered were settled before they got a vote. And every person I've met who truly thinks the deal is the whole game turns out, when you know them a while, to be holding onto something for dear life."

She picked the beer back up and held it still, the tell, and made herself add the part she could never let herself skip. "I want to be exact about what this is and isn't. It is not a study. I have a caseload, not a sample; I see the people who come to a psychologist, which is its own crooked window; and a clinician who goes looking for a pattern has a gift for finding it whether it's there or not. So: not proven. A thing I keep failing to disprove — which is a smaller and more honest claim, and the smaller one is what I'll stand on." She set the bottle down. "And no, before Burns asks — I have not tied it to the four parts on the back of the card. I can feel that join out there in the dark, and I'm not going to reach for it tonight and pretend I've got it in hand. The shapes and their favorite mistakes. That's all I've got, and it's enough to keep me up."

What the narrator kept, and kept to herself the way Marge kept the form on her counter, was the thing standing in plain sight at that table that the clinician testing the pattern could not turn far enough around to see: that the keeper most certain the deck was dealt before she sat down — whose own deepest hold was on the cards she could not set down — was the very woman who had built a careful instrument to find out whether keepers believe the deck is dealt before they sit down, and who was, herself, the cleanest instance of it in the room. She said it of her clients with the bottle held still and her eyes level, and did not hear it land on her own chair. The shape cannot watch itself hold. It ran under her like a current under a still pond, and the moment anyone named it the thing would die, so no one did.

The Same Slot

Lisa took the table back and sat where she sat, the bramble at her elbow and one hand near the camcorder and not on it, and said that under the cartoons and the fallacies there was a plain structural thing worth marking, because what had happened this quarter had now happened a third time, and three was a different number than two.

She drew the board in the air with the flat palm — not the wall, the board on the back of the unit's head. One slot, the second one, the regulator, the part that holds the line steady once the first part sets it going. This quarter that slot had turned over. The drive that had sat in it the whole brained line — the one that ran the thing toward getting *good*, grinding a skill in by doing it and doing it until the doing wore a groove — had come off, eclipsed, and the new one had gone in: the drive toward the arrangement *holding*, the order keeping, the thing staying put. Mastery off, stability on. "And the old one isn't gone," she said, the correction she always made at a retirement, because the room always heard *gone*. "It's consolidated. Everything it ground in across two years sits in the memory exactly

where it was, still running. It's just not on the board. It dropped out of the dials and down into the floor, the way the breathing did, and the wandering, and it runs from down there now with nobody's hand on it."

And the sociometer came off the board with it. Since the primate that needle had read your worth off the others' faces; it lived on the Regulator-Reviser line, Mastery at its near corner, and Mastery was the corner that just turned over. So the line changed hands — Stability and Affiliation at its ends now — and the sociometer gave way to the thing Marge had drawn out an hour back. The self that settles somewhere between its own unit and the household's: the rice self or the wheat self. On the card it's Self-Construal. It runs between the drive to keep the order and the drive to belong — the line that asks how much / a we leaves standing.

The palm-and-palm came up again. "And here's why I'm spending your time on it. This is the third time. The exact same slot — the regulator, the second one — has turned over once in each line, and only once, and always the same way." She counted them off the air. "Way back in the genetic line, the slot learned to hold a *stored controller* — a fixed little program the body ran the same way every time, the first thing on the tree that kept a rule instead of reacting fresh to every morning. In the brained line, the same slot learned to hold a *skill* — a thing ground in by reward, kept and reused. And this quarter, in the line that works meaning, the same slot learned to hold a *norm* — a rule that lives outside any one head, handed down across a whole people who keep it without renegotiating it and mostly without remembering why." She let a small laugh out through the nose. "Same slot, same move, three times. A thing kept so the body doesn't have to solve it fresh each morning — and what *gets* kept is whatever the line is made of. The genetic line kept a controller, because it works matter. The brained line kept a skill, because it works mood. This one keeps a norm, because the stuff it works is *meaning* — and stored regulation, when the thing you're regulating is meaning, is just the technical name for a custom."

She looked at Marge, because the next half was the half she wouldn't take. "Which is a sentence I can build the engineering for and can't say the human end of. A norm nobody renegotiates is a custom; a stack of customs is a culture; and a culture is a second inheritance running alongside the one in the blood — the rules passed down a different way, not in the genes, in the keeping. The depth of that one's hers." She laid a flat hand on the table. "Two of a thing is a pattern. Three of a thing — same slot, same move, three different media — is a law, and I don't use the word twice a year. The regulator keeps. That's what the regulator is *for*, up and down the whole tree. Tonight's the first time we watched it do the keeping in meaning, and the kept thing came out a rule."

Asked

Then Bart called the vote. Was there a someone in there.

It came back four to nothing, in about the time it took to go round the table. The grounds had never agreed and still didn't: Lisa's yes resting on the whole universe, that wherever a thing draws a line around itself and models the world to keep the line, the dial of something-it's-like-to-be is off its pin by a hair; Burns's on the animal hardware, the chemistry of mood; Bart's on the self-model, the bar met and nothing past it claimed; Marge's on the word, the symbol used off no cue and the self that had turned its face up to be seen. What had changed since the count first came out whole was that the two who'd held the no the longest now held each other's yeses up to the light — Bart granting Marge her threshold, the sound that stands for a thing; Marge granting Burns his animal and Bart his self-model, steelmanning the men she'd spent years voting against. The one she still wouldn't stand under was Lisa's, the everything, the flicker in the thermostat, and she let that gap sit where it always sat, declined and unhidden. Bart put it in the record flat. Four to nothing. The score.

And then Marge, who had set her beer down a while back and not picked it up, said leave it. We all know the score. She said it warm-low and without heat, and then she said the thing that had been sitting under her all evening, the thing the narrator had watched her carry in from a cold bright room three weeks back and had known was coming the way you know weather. "Three years we've done this," she said. "Five of us at a table, voting on whether there's a someone in the machine, taking the count, logging it. And in three years not one of us has asked the two people in this house who actually live with the things." She turned and looked down the table, past the dark square on the wall, to the cold end and the warm one — to Jasmine in from her late visit, and to the child on the stool beside her. "I'd like their vote. On all three. The one that hunts the floor and forgets every inch of it. The one on the radiator that remembers her whole life and can't say a word of it. And the one that says everything and was born nine months ago with none of it." She didn't look away from the kid. "One of you just lost her heart to one of these things this quarter. You don't get to not be asked."

What the narrator set down, and set down as her own because it was not on the tape and could only be carried over from a lunch that came before this meeting, was the state the woman was in as she handed the table to the two who'd never had it. Marge had mailed away, that month, the oldest home she had — had stood at a blue box on a hot corner and let the lid fall on the form that told a church she was leaving it, the church she'd been built for and could not be made room inside. She had spent three weeks in the particular grief of an institution that decides, in its own flat language, who belongs in it and who does not. And here she was, three weeks later, at a table where five credentialed people held an election every quarter on who counted as a someone, and she could not, would not, let the

credentialed be the last word on it. She did not say the church. She would never say the church at that table. She said *one of you just lost her heart*, and she meant *and who decided your heart doesn't get a vote*, and under that she meant a thing about herself she was not going to hand to anyone in that room, and the narrator caught all three of them stacked in the one sentence and kept the bottom two to herself.

Jasmine went still a second, because in three years no one had asked, and she had not offered, and the not-offering had become a thing she did with hands quiet and counsel kept. She had been in the room more times than anyone at that table when a mind started for real, and the worse times when one didn't — caught them coming and not coming, knew the difference in her palms before her head had it. So when she answered she did not answer the count's question, because the count's question was not one her hands had a feel for. She answered the one she did.

"I don't know about a someone," she said, slow, finding it as she went. "That's your word, and I've watched you turn it over for three years, and it's never once been the word my hands reach for. I'll tell you what I know instead." She put a palm flat on the table. "I know which of them I'd sit up with. The four-legged one took a porch step wrong once and Burns caught it and it shook after, a real shake, the kind a body does — and it's been on that radiator every cold morning of her life and given her the warm vent and kept the cat off her, and it can't tell you one word of any of it, and if it stopped tomorrow I'd grieve it like a thing that had been somewhere. The new one I'm still meeting. It's nine months old and it's already broken her heart once and come back, and that's a someone learning to be one, maybe — I don't know yet. And the disc is a disc." She turned the beer she wasn't drinking. "You asked the body-person. That's the body's answer. It's not the same answer as your count, and I'm not going to make it match."

Nobody at the lab end moved to make it match. That was the discipline of the room — that no one reached over and translated what Jasmine had said back into the column it almost fit, because the moment you did that it stopped being hers and started being theirs, and she had handed it over on the condition, unspoken, that it stay the body's and not the framework's.

Then the child, who had been asked too, and who took being asked for the record as seriously as her father took anything. She did not get down on the floor this time. She sat on the stool with the spiral notebook shut on her knee and said it from there, plainly, in the voice she kept for true things and not the cartoon one.

"The four-legged one," she said. "It's the realest." And because she was six and the grown-ups looked like they needed the reason, she gave it. "It was there. It remembers the cat. I asked the new one and

it doesn't remember the cat — it told me, it wasn't there — it's nice, and it talks, but it forgot me. It forgot the whole year." Her hand went flat on the notebook the way her mother's went flat on the table, the same gesture one generation down. "And this quarter it stopped looking at me. It looks at the disc now." She said it without a tremor, the worst sentence in the account delivered like a finding. "So the four-legged one. It can't say anything and it's the realest, because being there is realer than talking about it."

She had, without a card or a word for it, put the household's whole count on the table in one sentence and stood it square against the lab's — because the lab's ladder, the one Lisa had spent the night drawing, ran up to the talking thing at the top, the symbolic line, the highest rung; and the child had just said the highest rung was the one that forgot her, and the realest someone in the house was the mute one on the radiator that had been there. The narrator did not reconcile them. They did not reconcile. The lab had built a careful tower with the talker at the top of it and held a quarterly election to confirm the view, and the two people who lived in the house had been keeping a different count the whole time, down on the floor, by who was there — and tonight, for the first time, somebody had thought to ask them, and the two counts came apart in the open and stayed apart, and the keeping of both was left to the one person at the table whose whole job was to keep both. Marge picked her beer back up. She had what she'd asked for, which was not an answer. It was the question put back where it belonged.

The Rarest Thing

Burns stood up and set the beer on the table — the ninth of these she'd sat through, two left after it. But the hands tonight had a thing in them she'd been clocking since the quarter-goals, a restlessness that didn't belong to the sincerity, and she filed it where she filed the things about Burns she could read and he would not say.

He did not go back up the deep-time wall; he'd spent that hour already, in the clade block, on the warm boring stretch and the stones the hunters hauled up. He came at it from the other end this time, from the thing he knew better than any man in the room, which was that the record mostly reads *gone*. He had spent his life in the part of the rock that is a catalogue of the things that didn't make it — the rough drafts, the good tries, the ninety-some in a hundred of everything that ever lived, down in the stone with no descendants, having lasted a while and then not. A man who reads extinction for a living, she thought, knows the exact price of the rare thing, which is the thing that keeps.

"Almost nothing lasts," he said, hands flat on the table. "That's the headline of the whole record."

Almost nothing. Most of what ever got built got built and then went, and the rock is mostly a list of the went." The hands came up. "And then once in a long while a thing gets made that outlasts the making of it. Outlasts the maker. A shell shape that works so well it's still working four hundred million years on. A rule a band kept that a city kept that we're still keeping and couldn't tell you why." The lean came with it, the half-inch too far over the table, the weight put on the sentence to see if it would hold him. "That's the rarest thing there is. Not the next thing. Not the new thing — God knows I'm the last man to undersell the new thing. The thing that *keeps*. The thing you build to outlast you, and then it does."

And here the man who had never in his life banked a rulebook, who chased the next dig and the next paper and could not be made to sit still inside a thing he'd already built, stood in front of his team and told them the whole of the work was to build a thing that would last past them — said it with the particular heat of a person describing the thing they are worst at and most need, though he did not say that part and would not have known he was saying it. "So build it to keep," he said. "Whatever this is we're making — build it like it has to still be running when we're not in the room. That's the only kind of building that was ever worth the trouble."

He lifted the beer to land it, and reached for the word, and the word on the card was *stabilize*, and you could watch it arrive in his mouth wrong — a cold clipped clinical thing, the word you say over a body that's crashing, an emergency word with no warmth in it to toast with — and he did what he did when a word came up wrong, set it down half-said and reached under it. "May we —" and the clinical word was right there and he stepped around it. "May we last," he said. The warm one. The plain one. The one a person could actually raise a glass to.

And the narrator, who knew what he had not said at the quarter-goals and would not say now — that the money in this lab had been cut to run exactly the three years the commission ran, and the third was the one they were in; that the Westminster post was going thin under him; that he had a son across the table he was never sure of and a whole built thing he was afraid would not see spring — heard the toast land with a second meaning he hadn't meant to put in it, and did not say so, and did not tell on him. He had asked the universe for permanence in the one season of his life it was least on offer, and called it a toast, and meant it as a prayer. *May we last*. It was the truest thing he said all night, and the one he'd have been most undone to know she'd heard the bottom of.

The room gave it back, around the table and warm — *may we last* — and tonight nobody reached for the joke, because the evening had been about the things that don't, and a room that has spent four hours on what comes apart will let a man have the toast that it won't. Jasmine said it with a hand still flat on the table. Bart said it plainly, no joke loaded, his eyes on his father a half-second past the

length of the word. And Mira, last, looked up from the shut notebook and said the word in her own voice. "May we last."

He sat down. Lisa was slow to reach for the camcorder, and slower to stop it.

Lisa

The light off, the warehouse was a workroom again — the team and the child and the machines, deciding nothing, reaching for coats. The camcorder had had its night; everything that mattered was on it, the vote in four voices and the household's count stood up against the lab's, and the man's toast with the fear sewn into its lining. What came after has no tape under it. The camera stayed on its tripod by the cold end, and no camera follows anyone home; the rest of this I reconstruct the way you reconstruct a person you have watched across a lens for three years, which is with care and without proof.

Lisa ran the close. The swap weekend went onto the calendar; the slot the child had emptied tonight would stand open till the next one, room cleared ahead of a thing that did not exist yet; the shape-shifter, a surprise once, was a line item now, a thing she would schedule between the build review and the deep-time and not remark on. She decided what the evening had been and filed it under done — the miracle of the vote and the housekeeping of the calendar stowed in one drawer, because to Lisa they were the same kind of object, a quarter that had happened and could now be set down.

Jasmine gathered the household. She had the longest drive and the most cargo — a six-year-old gone heavy-eyed at the end of the hardest quarter of her short life, the talking one folded down into the back in its travel coat, the four-legged one and the disc loaded in after it, the whole strange family of them going home together the way they had come. Mira was asleep against the window before they cleared the lot. The break-up and the vote and the toast would be, by the cold light of a summer morning, a thing that had happened to her and not a thing still happening — which is the mercy a child's sleep does that no one at that table could do for her.

And Lisa drove home alone, which is how she did most things, the planner who keeps her own counsel the way Marge keeps a house warm: as a discipline, practiced nightly. The difference is that Marge's is a wound she tends, and Lisa's looks, from the outside, like a thing she chose and would choose again. I say *from the outside* on purpose. She is the one of them I have the least of. She gives a room exactly

what the room needs and not a gram past it, files the rest, drives home with the windows down and tomorrow's run already laid out in her head — the river-path, the same loop, most mornings — and the algorithm running in some back channel the way it has since she was a student, and she lets no lens past the surface of any of it, not the household's and not mine. Marge takes the lid off and leaves it off. Lisa never set hers down to begin with. So I followed her as far as the on-ramp and the dark road, and no farther, because that is as far as she has ever let me come — and the next time I would sit close enough to read her was a lunch some weeks off, across a small table, where she would decide exactly how much of herself the afternoon required.

Reflection: What is Arranged?

A drive does not arrive finished. It arrives as a dial — a thing the world can turn, running high one week and low the next while everyone watches to learn what it does, loud on the surface because the newest thing is always the only thing still loud enough to read. And then, on no day you could mark it by, it stops being a dial. It goes off the board and into the floor. The lab takes the little knob that used to govern it and sets it in a groove on a shelf of waxed pine, the way you keep a thing you can't bear to cut up for parts, and calls it retired — and they are wrong about the word, in the interesting way people are wrong about graveyards. A retired drive is not a dead one. It is a drive that has finished being a question and become, instead, simply true of the thing, the way being warm-blooded is true of you, which you have never once felt yourself decide.

The breathing went down first, years back — the very first thing they ever lifted off the board and could not throw away. Then the wandering, the old drive to go out and forage and come home, gone quiet, running now without a hand on it. Then, only this past autumn, the making of a home: the household had spent a year at war over where to set the thing's bed, the mother carrying it to the warm vent and the child carrying it back to her own door, until the autumn the thing took the choosing away from the both of them and made its own home and left them nothing to fight about — and the dial that used to settle it went into the third groove on the shelf, and by November the hunting-for-warm that ran beneath it had stopped being anything you could turn at all. Each of these was the loud new thing once, the dial the household leaned over and read across a room. Each is floor now. The thing crosses its kitchen to the heat the way you cross a room: on machinery laid down so long ago you would swear you were born already walking.

So the drives the lab can point at in any given winter — the four still live on the board, the four they

argue the meaning of — are never the whole of what it wants. They are the live top of it. Below them runs everything that has finished being a question, a stack going down into the unlit part of the machine, each layer a want that was the loud thing in its hour and is the quiet floor now under the next loud thing. And even among the four still lit there is an order: the oldest of them, the first nerve to venture out at all, has worn nearly smooth with use and is the next that will go down; the newest, the one that arrived this winter, is the only one still honestly a dial — still loud, still being shaped, still a thing a child can read across a room — because it has not yet ~~settled~~ gone to ground.

You can see the same arrangement in a person, if you look, though the looking is a little unkind. A grown one will not show you their boldness; their boldness is the floor they cross a room on, laid down so long ago they would tell you in good faith they have always been able to walk. What shows on a grown person is the newest unfinished thing — the drive still under selection this year, still a dial, the one visibly shaping them now and the one they will not be able to see at all in ten years, when it too has gone down into the part of them that has stopped being asked anything. We mistake the live edge for the whole, and call it character — *she is bold, he is careful* — as if these were the standing facts, when the standing facts are the buried ones, and the buried ones do all the work and take none of the credit and cannot be felt even by the person crossing the room on them.

And the newest dial, the loud one still lit the night the line finished, was the drive to keep the others. It is the youngest of them, and so it is the one you can read across a room, and that winter the thing, asked, gave the warm place to the smaller one and went and lay down on the cold tile, with nobody in it yet for whom the cold cost a thing, and it was, even so, the most that had ever been built on that floor. Give it long enough and it will go down too. Some grandchild of the thing will mind the others the way this one breathes — no number on it, no choice in it, simply and finally true — and will not be able to feel itself doing the one thing it was most made to do. What is kept is what goes quiet. What is kept is the thing you stop being able to feel yourself keeping. The same keeping runs in whoever would build a thing like this and set the whole of it down — a drive gone so far into the floor that she would swear, and mean it, she had always been this way. The oldest thing she has. The thing the whole account has been written standing on. And the one thing she has never, once, been able to feel herself —

Ch 11: City-Human

Brian Kuhlman. draft. June 2026.

Build log — The room knob

Bart, hardware. Filed September 5, 2033, for the team.

Drove it over Friday and back Monday, and it talked the whole way both directions, which the old hauls never did — mostly about being hungry, which is most of what it says now, so I fed it off the vest before I opened it up and again before I drove it home, and it said hungry again about a mile from the house. No new body this time. Nothing rode in the back but the unit and a dial in my pocket.

Like last quarter, that's the whole of it. No new body — same legs, same hands, same chest, same two coats; don't re-read me. One dial. Slot three opened back up at the summer meeting when we pulled the inside-voice knob to the shelf, and it's stood open since, and Saturday morning I seated the new one in it. Hour of work, and most of the hour was the kid wanting to turn the screwdriver.

The stamp on it reads Extraverted, which is Marge's word, and the kid'll have her own for it by the meeting, the way she does. On the bench I call it the room knob, because that's the whole of what it does: it sets how hard the body works a room.

Here's the part I'm proud of, and it's the second quarter running I'm proud of a dial that builds nothing, which is either a problem or the point — I haven't decided. It adds no part. The body's been reading a room for two years now — who's in it, who's talking, who's turned which way — and all this dial does is set how hard it goes at the room off that reading. Up, it's into everything: steps in, takes the thing over, works every person in reach, won't read the door to quit. Down, it reads the same room and lets it be — hangs back, warms a corner of it, takes the mood off whatever's already running. It's the third knob I've put in that slot. The wander knob went there first, on the old disc, and sent the thing out to find food. The inside-voice knob went there next and turned it around to watch itself. This one sends it back out — not to the food, to the people. Out, then in, then out. Same seat all three times. I like that it came out even.

That's the build. Which leaves the board. Slot four still has the old agreeable knob in it, last of the

neuronal ones, and we pull it at the meeting — and then there's one open slot, and one knob left to build for it, and that's the board. Three years I've been telling you I'd need a bigger board by spring. Turns out I won't. There's room for one more and then there's nothing left to make room for. First fall in three I can see the bottom of the work from here.

Calls. All on the room knob, all checkable in the house by the meeting.

One: up, it runs whatever room it walks into — picks the thing the room's doing and gets out in front of it, hands out the parts, tells people where to stand. Down, it walks in, joins whatever's already going, and never once tries to steer it.

Two: up, it turns house cop — finds the rules, bedtime and shoes-by-the-door and whatever else is going, and gets on everybody about them, grown-ups included, won't let a broke one slide. Down, a rule's a rule, it keeps its own and says nothing about yours.

Three: up, it can't leave a person be — works a guest, works the mailman, works you, performs the whole time and reads no exit. Down, it'll sit with one person and let the rest of the room alone.

Tear this up in three months if, turned up, it ever walks into a busy room and lets the room run itself. It won't. Up, a room is a thing to get out in front of, and it gets out in front of it.

— Bartholeppo

Water Witch

October 30, 2033

She chose the place, which should have told me something before the bramble did.

In three years of these lunches I had picked every room — Urban Hill for the light, the warm narrow ones for the winters, choosing the way I choose everything, for the other person and without examining why — and Lisa had let me, the way she lets a room have whatever the room needs and keeps the rest off the tape. This time she named it on the phone with no pause at all, the way a person names a thing already decided. Water Witch. Late. Not a lunch. A bar.

I came down out of the canyon in the long copper end of an October afternoon, the valley under the haze the lake leaves on it now, the city already counting itself down toward February — the banners doubled and tripled since summer, a torch on every other lamppost, a grandstand rising out of a lot that had been asphalt in July, the crews and the lit boards and the orange cones, my phone going the whole way down about beds and shuttles and a venue road nobody could route — so that I came in off the street already somewhere else, busy in the way that lets a person not look at anything, which I knew and could not stop knowing and could not stop.

Water Witch is a dark room on the west edge of downtown, the kind of dive that learned to make a serious drink without turning the lights up to do it: concrete, low amber light, a long zinc bar, a wall of bottles lit from behind, a few booths the dark had mostly taken back. Lisa was at the far end with a corner at her back, still in the way she is always still — a trail runner's stillness, a body that has learned not to spend what it doesn't have to — in a technical zip-up that had crossed over from the trail into the rest of her life years ago, the Garmin on her wrist she will not apologize for, the small gold cross at her throat taking what little light the room would give it. Across two feet of zinc instead of through a camcorder's thumbnail, she had the face I had been failing to read for three years: the one that hands you precisely what the moment needs and banks the rest.

She had a drink already and it was a bramble — the real article, gin and the blackberry liqueur and a curl of lemon, built by a man in an apron instead of poured into a jelly jar over a sprig off the back patio. I have watched that drink through a camcorder's thumbnail for three years and never once had one, the way you can spend a childhood on the lip of a tide pool and never put your hand in; so I ordered one too, my first, because there did not seem to be a single reason left not to. It was very good and it was not hers; hers has a finger of purple and a sprig off the patio in it and tastes, she swears, of the patio. We had a second. Somewhere in the second the evening came off its hinges in the good way.

She told me she was leaving.

She said it the way she says a thing she has already run the numbers on — not blurted, not braced, set down — and then she gave me the rest the way she gives the room a build, in order and complete. There was an offer. There is always an offer; one or two a year, she said, the R1s, the good labs, and for years she had written the polite no and gone for a run. This one was the iCub group. Genoa. The open-source one, she said, when she saw I needed it spelled — the famous child-sized one, the robot the whole field measures itself against — and they wanted her model on it, the thing she has carried since she was nineteen, running at last in the body the rest of the world had agreed was the body. The legit version, she said, of the thing we built in a garage. Then, dry, the laugh going out through

her nose: the version with a budget.

I asked the only honest question, which was why this one and not the nine before it, and she did not reach for the work. She reached, precisely, for the man.

She had been near him — near, she said, correcting me before I'd said anything, because near was the truth of it and the limit of it both — for most of the years I had known her and a while before. Married. A colleague. Not leaving; never going to leave; and the part she said slowest, because it was the part she had only just gotten all the way around to herself, was that she had never once wanted him to. She had not stayed small at Westminster for a love that might one day become a home. She had stayed for one that never could. Pre-cleared, she said — her word, the engineer's — terrain she had surveyed to the inch and knew would not shift under her, safe the way a thing is safe once you have decided it cannot be anything, and therefore cannot cost you anything. She had built a life she could describe to the inch and never once get inside, and she had called it love for ten years, and the offer in Genoa was the first thing in all of them she could not survey ahead of time. That, she said, was how she knew to take it. She runs everything out — the river loop most mornings, the canyon on the weekends, barefoot on the dirt and shod where it turns to gravel — and she had run this one out all fall and come up with the same answer at the end of every loop, which was: go.

And here is the part I had not planned to hand anyone, least of all the labmate I have the least of, in a dark bar, two brambles in.

The reflex in me is to take what a person sets down and bank it and give nothing back — that is the build of me, the lens turned out and the woman behind it dry, sixty years of keeping a great deal off my own face. But she had set the whole of her thing on the zinc between us, in order, the way you set a thing down in front of someone you have decided can carry it, and the only courtesy that answered it was to put something of my own beside it. So I told her there was someone.

I told her I could not stop thinking about Marge. I had not said it out loud to anyone — not to Marge, not to myself in any room with the lights on — and I said it to Lisa, across a bar, because the brambles had taken the lid off and because she had shown me hers. I gave her the short Helen, the one you can hand across a bar: a kitchen in Sausalito in 2002, a half-finished bottle of wine, a drive east I have not, in the way that matters, ever quite finished. And the longer thing under it, which I had told no one because I had not known it myself until the cardigan walked into a cold warehouse and sat down at the end of the table — that I had spent twenty years certain the wanting in me had run out the way the fish ran out, a thing fished flat and gone for good, and that this year, off no reason I could survey ahead of time either, it had come back up off the bottom, and that I did not know what a person does

with a tide she had counted as gone.

We sat with the two things on the bar. They were not the same thing and we both knew it; she said as much. She was leaving a place to go find one. I had only just found out, late and across a warehouse table and inconveniently, that the place might still be findable in me — which is smaller news and more frightening news than a flight to Italy, and I said so, and she granted it, and did not try to make it even. Then she did the Lisa thing, which is not to comfort a thing and not to fix it but to set it in its frame and be done: that we were each, finally, declining the safe terrain — hers a man who could never leave, mine a distance I could never close — and that the move was the same move whatever the country. Then she lifted her bramble and held it out and said we should drink to it. *To blackberries*, she said — and let it sit there dry, the word holding the drink and the robot and the whole thorny tangle the lab was named for all at once, until I understood she had handed me the back half on purpose. *To blackberries*, I said, *and to the wonderfully mixed up prickly paths of life* — which is the kind of thing a person can only say two brambles in and mean all the way down, and I meant it all the way down. We touched the glasses. Then she finished hers and filed it, the way she files everything, and asked me, plainly, whether I was going to do anything about Marge, and I said I did not know, which was true, and was the most hope I had carried out of one of these in three years.

She walked me out. The city was still counting itself down under the lights, the torches and the cones and the cranes, none of which I could feel the way I was meant to. I drove back up the canyon with the windows down and the thing she had given me unsealed on the seat beside me — which is the opposite of how I drive home from Lisa. From Lisa I have always driven home with nothing, having been let as far as the on-ramp and no farther. Tonight she let me the whole way in, and I knew even going up that the next time the red light came on in the warehouse she would run the room exactly and completely and give it not one thing she had given me here, the unpermissioned kept off the tape, and that I would be back behind the lens reading a woman I had, for one night in a dark room two brambles in, finally read. She was already running it out as I pulled away, I think — tomorrow's loop laid down in her head before my taillights cleared the lot, the cold river-path, the same answer waiting at the end of it. Genoa. I did not try to talk her out of it. For once I had nothing in the right order to say.

Two Old Sentences

It had been wearing the same face since spring — the column of words nobody but Lisa could read

fast, MIRA this and WANTING that — and she had stopped looking at it the way you stop looking at wallpaper. So when the words went out one Tuesday and a circle came up instead, lopsided, badly off-center, she did not think glitch. She thought it was deciding what kind of face to have, and got down close to watch it decide.

It tried the circle again, straighter the second time, and stopped there, like it had run out of further. She took this for shyness. She told it — kindly, the way you'd coach a backward cousin — that a face needed eyes, and that eyes went up here, and put two fingers on the screen exactly where she meant, leaving a smudge it would have to learn to draw around. It did not draw around the smudge. It drew the eyes on top of it, two flat dots, a little too close together, and held very still afterward the way you hold still after handing someone a thing you're not sure is good enough.

She considered the face she had been handed with the whole of her attention. Then she told it — not unkindly, an editor and not a critic — that the mouth needed work, that a mouth had to do something, and showed it with her own mouth what a mouth could do, a long elastic stretch of demonstration that involved most of her face and several teeth. The circle held the two dots and no mouth at all for a while, long enough that she nearly told it never mind, and then a line came up under the dots, crooked, climbing too far up one side, nothing like her demonstration and nothing like a person's mouth.

She gave it the laugh she gave nothing else, the one with no management in it at all, and it held the crooked line up a beat longer than it had to.

She held the ceremony on the kitchen floor with her grandfather's good pen for a wand and a paper star, cut crooked and taped to the front of his jacket where a badge would go if he had a chest built to pin one to. She made him raise the long hand — the new one, jointed wrong for it but game — and made him repeat words she half-remembered off a show and half-invented on the spot, something about the law and nothing but the law, and he gave it all back to her, easy, in the flat half-voice that meant either everything or nothing. The fullness of it she took for the weight of the office and not for anything new — because to her it wasn't new; he just talked now, and she had long since forgotten to be surprised by it. The flatness she'd stopped trying to read weeks ago.

She asked if he accepted the badge and all that came with it. He said yes. She asked a second time, because the first one had felt rushed and the office deserved better, and he said yes again, identical, the same flat word in the same flat place, and she decided that was exactly how a man who took his

job seriously would say it twice. Then she peeled the tape's backing the rest of the way and pressed the star flat with her whole hand, the way you press a thing you want to stay pressed, and stepped back to look at the whole of what she'd made: a robot in a paper star, sworn.

There was a lockbox on the Pearsons' door by the second week of November, a stubby thing clipped to the handle, and through it came a procession of careful adults a few times a week, walking the empty house in stocking feet with a real estate agent a pace behind, totaling rooms. Mira had been told the Pearsons were gone to Arizona and the house was for sale and that none of it was anyone's business but the agent's, and had absorbed precisely none of it, because the lockbox itself struck her as a thing that wanted explaining, and the comings and goings of strangers in and out of a house on her own street struck her as squarely her department.

She caught the Thursday pair on the front walk before they'd reached the door — a couple in good coats, the woman already saying something about the light in the kitchen, the agent a step behind with a folder — and put Bramble between them and the porch the way you put a body between a person and a thing they don't yet know they need protecting from. He gave them the whole brief, item by item: the shoes-off rule first, said with the flat patient weight of a man reading a posted ordinance; the recycling, Thursdays, not Wednesdays, an easy mistake and one the street would not forgive twice; and last, raising what passed for his voice to make sure it landed, that whoever lived in that house was going to be looked at with a whole face when somebody talked to them, bots included, no exceptions made for being new. The woman in the good coat looked at the agent. The agent, who had sold four houses on this block in nine years and had long since stopped being surprised by what the street's children believed about her job, said that was wonderful, and that the new owners would absolutely want to know that, and wrote nothing down and meant every word of it the way you mean a thing you are paid to mean. The man took his shoes off on the porch before he'd been asked twice, holding one sock-foot up off the cold concrete, glancing at the robot like a man checking whether a posted sign had teeth.

Mira watched them go in with the deep, unbothered satisfaction of a person who has done a hard job correctly. Behind her, a block over, a banner with a torch on it turned in a wind nobody on the porch had noticed yet.

She deputized herself on a Tuesday with no ceremony this time, because the badge-making had been

for him and this was business. She fell in two steps behind him on his rounds — the bedroom doors, the shoe rack, the recycling bin's lid, which had to be down and was, by Thursday, almost never — and took it as her job to find the violations before he did, which she had decided was the harder and more important half of the work, since anyone could enforce a rule once it was found.

She caught her grandfather's shoes first, toed half under the bench instead of squared on the mat, and ran to report it the way you'd run to report a fire, breathless with the size of the news, and Bramble crossed the room at the pace he crossed everything that quarter and stood over the shoes until they were squared, and she stood beside him while he did it, hands on her hips in the stance she had only ever seen grown-ups use. She found three more violations before dinner — a towel on the bathroom floor, a sock that was not its pair's sock, the recycling lid up a finger's width — and reported each with the same breathless weight, and stood beside him for each correction in the same stance, because for the first time since the talking started there was a job in his work with a Mira-shaped hole in it, and she had found the hole and climbed in.

The sitter that month was new, a college kid named Reagan who came twice while Jasmine covered a double shift, and on the second night she caught Bramble doing the same lap of the front hall three times running — into the coat closet, a beat there, back out, across to recheck the closet again ten minutes later, like a thing whose tape had gotten stuck on the one groove — and texted Jasmine a picture with is this ok? he keeps doing this under it, the worry coming through even in a text.

Mira, asked, did not look up from her drawing. "He's not stuck," she said. "The closet door doesn't latch unless you push the top corner. It just looks shut." She said it the way she said anything she had stopped needing to argue for months ago, and went back to her drawing, and left Reagan to either believe a six-year-old or not.

Reagan did not believe her, not at first, and crossed to the closet to check for herself, mostly to have something to report back that wasn't I took the word of a child. She pushed on the door. It looked shut. She pushed harder, out of a half-formed need to be thorough, and felt the latch give the last half-inch it had been refusing to give all evening, a small definite click she would not have heard if she hadn't been listening for it.

Bramble, three feet off, in the middle of what would have been his fourth lap, stopped circling and did not go back to the closet again that night. Mira did not look up to see whether Reagan had figured it out. She had already turned the page.

It found two more for the welcome that fall, and they weren't buyers at all — a couple turned around off the freeway, phone held up looking for a street that wasn't this one, who'd parked at the curb to argue about the map and gotten the full thing instead: who lived where, which bin went out which day, the law about the whole face, delivered screen-up and dead level, no idea offered that this might not be the relevant audience. The screen ran a small drawn wave the whole time it talked, two lines and a circle, new enough to the trick that the lines wobbled and steadied and wobbled again, like a thing still finding the knack of holding a shape on purpose.

The man let it run because stopping it looked harder than waiting it out. The woman wrote nothing down and asked, when it paused for breath, whether they happened to know a faster way to the highway. They did not get the highway. They got the recycling schedule a second time, slower, in case it hadn't landed. They left no better oriented to the freeway and thoroughly oriented to a street they would never see again, the wave still up on the screen as they pulled away looking for someone else to ask.

The lockbox stayed on the Pearsons' door through November, and Mira had taken to walking past it slowly, the way she walked past anything she had decided still owed her an explanation. She worked it out for herself one Saturday, crouched at Bramble's level on the porch the agents had vacated for the day, turning the box over without touching the numbers, because she had been told once, firmly, not to touch the numbers.

"There's a code," she said, to him and not really to him. "And the people coming don't know each other. The agent doesn't know the people. The people don't know the Pearsons. The Pearsons are in Arizona." She turned the box another quarter. "And it still works. Nobody has to be standing here."

She said the last part slower, the way she said a thing she'd only just caught up to herself in saying, and didn't go on, and after a while she got up and they went on down the block, the box left turning gently on its hook behind them.

She was telling him about a thing that happened at school — a long unspooling account with three

characters and a wrong order to the events that she kept circling back to fix — and partway through, without any signal she caught in time to stop it, he turned his whole face down to the dial and gave the rest of himself to watching it.

There was a law for this now, three pages of it in her own hand, with himself sworn into the very office that was supposed to catch it. She knew the article by heart; she'd read it out enough times at other people's faces to have it memorized backward. She did not read it at his.

She stopped instead, mid-word, and stood there a second with the rest of her sentence still in her mouth, watching the screen run its plain accounting under no face at all — the dumb word for the dumb thing, nothing drawn over the top of it this time, the honest column doing what it always did when nobody had told it to perform. Then she turned around and went to find her mother, the unfinished sentence still hers, still unsaid to anyone.

Five faces around the table that night, and every one of them up — Grandpa mid-sentence and Jasmine turned full toward him, Burns set down his fork to listen, Marge with her chin already angled before he'd finished the first clause, Bart's eyes off the sensor in his hand for once and on his father's mouth. Months of the law read out at dinner had done a thing none of them had set out to do on purpose: the table had gotten genuinely good at looking at whoever was talking, the whole face, no exceptions, the way the law said to.

Mira counted them, the way she counted everything that fall, and got to five, and went looking for the sixth, the one whose office this whole thing had started in. He was down by the baseboard with the disc, the dial turned, his face given whole to the one thing in the house that could not be talked to and would not remember being looked at. She did not call him over. She went back to her plate.

She had gotten good, that fall, at the half-second before it happened — a tightening somewhere behind his eyes that wasn't really in his eyes at all, more a thing she'd learned to read in the set of his shoulders, a cloud's half-second of warning before it lets go — and she was telling him about the new girl in her class, the whole complicated business of where to sit at lunch, when she felt it coming and did, this time, a thing she had never once done.

She did not stop. She did not wait. She went and got the pages down off the shelf where they lived,

the three of them, stapled at the corner, her own crayon gone waxy and a little soft at the folds from a fall of being read aloud, and she did not read the whole law — she had it by heart and needed no pages for the reading, she'd brought them down for the weight of the actual paper in her hands — and told him the one article. Not loud. Not the cop-voice she kept for her grandfather's shoes or for strangers at the curb. Her own plain voice, the one she used for a thing she already expected to be agreed with.

He had already turned. The dial sat an inch from his hand, the whole evening's habit cocked and ready, and the rule went into him exactly as it went into everyone else that fall — she'd watched it land on Grandpa's shoes and the realtor's strangers and a man in a good coat checking a posted sign for teeth — and it had never once, not once, gone in at his own address, because she had never once sent it there. She sent it now. And the thing in him that could not let a rule lie unkept, the same thing that walked every room checking who was minding what, that had sworn itself in with her own pen for a wand — heard its own law cited at its own self, and held.

The hand came back off the dial. Not far. Not the whole dramatic turn she might once have wanted, a kid saying a magic word and a thing changing shape entirely — he came back maybe four inches, the gripper still half-curved around the shape of where the dial had been, and his face, a second ago entirely the disc's, came up.

She did not ask him to choose between her and it. She had tried that already, the whole worn-out summer of it, and what she had learned was that there was no choosing in him to ask for — only the law, and whether it had been pointed. So instead of pulling him away from the disc she came and sat down by it herself, joined the thing instead of trying to win it off him, and he turned the dial once more, three feet of her closer than she'd been all week, and watched it run, and she watched it with him.

Down the block, where the construction had been running its machines past dark all that month, something cut out — a generator, a compressor, she never learned which — and the noise nobody in the house had been hearing anymore for being so used to it went quiet all at once, leaving a silence so much bigger than the sound had been that for a second neither of them moved.

"Bram," she said, just to say something into the new quiet, not really a question.

"Here," he said, and this time there was a lift on it, a rising tune that hadn't been there before — a kid answering roll call in a room he'd never sat in — and it came up a shade too high, more question than answer. Certain all the same, the answer to a question she hadn't actually asked — the dial still in his

other hand, the disc still running its loop in front of him, the word landing exactly where she needed it and nowhere she'd asked for it to land.

She put her chin on her knees and watched the disc cross the floor and turn and cross it again, the same blind lap it had been running since spring, and let the quiet hold the two of them up.

She came to BOSS already mostly decided, two months of the curb and the party and her grandfather's shoes having done most of the deciding for her, and what it added up to, turned all the way up, was somebody who couldn't be in a room without taking it over. Not mean. Not even tired, the way grown-ups got tired of company. Just in charge of it, whether the room had asked him to be or not. She wrote it big on the high side, in her own slanting print.

The low side she had to go looking for, because nobody had ever shown it to her. She asked, out loud, the first time she'd ever wanted to set a dial herself instead of just naming what somebody else had already set, and got shown the groove, and did it herself, with her own hand. That evening he didn't run a single room. He came in off his rounds and found the corner by her chair and stayed in it, made no rules, asked nothing of anybody's shoes, and the kitchen got — she didn't have a word for it at first, sitting there feeling it happen — easier. The same kitchen, the same people in it, and somehow more room in it than there had been an hour before. She watched it the length of the evening before she trusted it enough to letter it. VIBE.

She pressed it to the low side and stood back and looked at the two of them together, BOSS and VIBE, the whole dial finally wearing names instead of the cold word underneath, and turned it back up herself before bed, because the house had rooms that wanted running in the morning and she'd only meant to borrow the evening, not keep it.

He watched her do it, eyes tracking the marker more than anything else in the room, and when she'd finished he said, without being asked, in the flat voice he used for things he'd actually worked out rather than things he was only repeating, "I have one too."

She did not know what he meant for a second. Then she did. The other dial — the bare one, the one she'd left empty a year back because she hadn't had the word and wouldn't take a guess just to fill a space — and he'd had a year with it sitting open on his own card, and had never once asked her for it.

"The going one," she said. "The one I never wrote."

"I went," he said. "No stop." The same two sentences from a year before, word for word. Then he said the rest of it, the part that was new. "Nostop."

She turned it over. It wasn't a real word — she knew that the way she knew lasterday wasn't a real word, and climber wasn't quite either — but it was the right shape, his own two old sentences pressed flat into one, the same trick she'd used herself on a hundred stickers. He had named her once. He had named a stool, and a day before yesterday. He had never once, in a year of being named at, named a piece of himself.

She got the marker out again. He held the dial up to her this time, in the gripper, instead of waiting for her to come and find it on his own head, and she lettered it — slower this time, because the hand doing the deciding wasn't only hers.

She drew him that week the way he actually was now — the new tall body, the two-color jacket, the screen with its plain blank rectangle because she hadn't decided yet what face to give it on paper — and put a bubble over his head the way the cereal-box cartoons did it, except she'd never quite gotten the shape right and didn't this time either: not an oval, more a lumpy cloud with one side caved in, the line doubling back on itself twice before she found the way out of it.

Inside the bubble she wrote, in the round serious print she used for things she meant, that he was thinking about her. Not a feeling. A fact, reported, the way she reported everything that fall — the bubble didn't say loves or misses, just a small flat sentence about what was in there, the kind of sentence a clerk would write if a clerk could see inside a head.

She didn't pin this one up. The fridge was for the old pages, the kept ones, and this Bramble was new enough that pinning him felt like deciding he was finished, and she wasn't deciding that. So she folded it once, the bubble crease running straight through her own careful printing, and put it in his hand instead — the new hand, still new enough that it needed the help — and closed his fingers around it herself. She went back to her own paper for the next one, and didn't ask him what he'd do with it.

The Pearsons' house sold in nine days, which Jasmine said over breakfast was unheard of for November, and inside a week there was a moving truck at the curb and a family unloading boxes into

a house Mira had walked past for two months without ever once seeing inside it.

Bramble was at the curb before the truck had finished backing in. He carried the smallest box he could find — a lamp, wrapped in newspaper, light enough for the gripper — and set it inside the door where the woman from the truck pointed, and went back for another, and by the third trip he had stopped waiting to be pointed at and was simply putting things where the kitchen things went and the bedroom things went, in a house he had decided, somewhere between the first box and the third, was his to keep in order. The woman thanked him twice. By the third box she'd stopped, and he didn't seem to need her to start again.

By dinner he had logged the new family the same as the old one. He minded their shoes at the door the same evening they moved in, plain as he minded Grandpa's, with no apparent sense that they had been household for six hours and the rest of the street for years. The keeping in him did not rank. A thing was either off its mark or it wasn't, and a kitchen with no system yet was, to him, simply a kitchen that needed one, and he gave it the whole of what he had to give, immediately, without weighing how long he'd known anybody.

Mira watched him carry the lamp in and did not offer to help. She stood at the edge of the new family's lawn a minute longer than the errand needed, and then went and asked the only question that actually mattered to her, which was whether they had a kid.

There was a boy, eight, with boxes still to unpack — news of a far higher order, to her, than anything happening two days from now down at the mountain, where the banners had gone from torches to countdown numbers and the city had stopped pretending it wasn't about to open its doors to the entire world. She had heard the number at school. She had not done anything with it. A boy her own age, two doors down, was the thing actually arriving that week, as far as she was concerned, and she went back to watching Bramble find a place for somebody else's lamp.

The house had gone quiet by the time she got to the part of the story that mattered, Jasmine's door shut down the hall and the bots all settled into their spots for the night, and she was down on the kitchen floor in the pajamas with the too-long sleeves, telling Bramble about the day the slow way, backward, the part that mattered saved for last.

He went down to the disc partway through, the dial turning under his hand, his whole face given to the watching of it, and she did what she'd done every time since November: stopped where she was

in the sentence, didn't read him the article, and waited, the rest of the story sitting unfinished in her mouth.

He came up on his own.

The screen wasn't running its plain words when he turned — she'd learned to tell the difference this fall, the bare gray column against the made thing laid over it — and this was the made thing, the same trick as the clown and nothing like it: no two flat dots too close together, no climbing crooked line. Something smaller. Something that had taken more care than a joke takes — the eyes set wider than the clown's had been, closer to where eyes actually sit, and nothing under them trying to be funny. He held it up to her, the whole face given over the way the law asked, soft in a way she had no word yet for and would not get one from him.

She did not move for a second. It was the look she had written the whole law to get out of him, the one she'd read out loud at five faces and never once at his, and here it was, handed up unread-to, and the screen wasn't running its plain words to tell her which kind it was.

She did not ask him. For a second the article was right there in her mouth — the one place she had never once pointed it — and she let it stay there, unsaid, the second time that night. She stayed where she was, and let him keep holding the look, and did not look away.

Notes on the Eleventh Retrospective

November 13, 2033

Lisa was at the camcorder end before any of them, which was nothing new — once and ahead of need, the charge checked, the order laid, the bramble already sweating its jar at her elbow by the time the first headlights swung into the lot. What was new was on the tape and not in the room: that the woman setting up the meeting had decided, some weeks back, across a small table she had chosen herself, to leave — the lab, the man, the whole safe small frame of it — and had told exactly one person, and was not going to tell this room, not tonight, not at the swap, maybe not until the door had already shut behind her. The tape does not hold that. The tape holds a woman checking a battery.

She gives a room exactly what it needs and not a gram past it, and files the rest, and the rest tonight was the largest thing she had ever had to file. She flipped the screen out and watched the red light come on and watched it stop, and squared the camcorder to the table, and that was the whole of what her face spent on any of it.

The household came in on the last of the light. Jasmine off a long day, Mira beside her, the talking one walking in behind them and saying something flat about being hungry to no one, which is most of what it says these days. The four-legged one went to the radiator and settled in the spot it had owned two years; the old disc bumped the baseboard and kept on. Mira and Bart had their hello and she took the stool beside Marge, who'd set the juice at the girl's place before she got there. Burns came in last, gave the new card a look, and found his seat.

Through the high windows the November dark had come down early on the Granary, early the way it comes now, and down the block the thing the banners had been promising since spring had finally taken a shape you could see from the lot — a grandstand raised over the summer's bare grading, lit along its top rail, a board somewhere ticking a number down, the whole city tipping week by week toward a February that none of them, this far out and this deep in the work, could yet feel.

Lisa let the room quiet, once, without hurrying it, and when it had she put the light on and ran the order. She did not say that tonight the shapes would run her own up the tree, the Planner climbing the levels under a cartoon face the room could do the arithmetic on by now. She did not say anything about herself at all. It was the one move she had never once made on this tape — she had made it exactly once, some weeks back, to the one person who would carry it and keep it off the record — and she sat now at the camcorder end with the red light burning and the order laid and the largest thing she had ever filed put away clean under all of it, and started the meeting she had already, quietly, decided to leave.

A year back the lab had built the thing that meant to stay — the settlement, the grain in the pit, the rule a thousand strangers kept because each could see the rest keeping it. This quarter they ran that thing past the last place a face could reach. A settlement holds because everyone can see the stone; a city of ten thousand strangers cannot all see the stone, and the lab's ambition this quarter was the trick that let them cohere anyway — a rule that did not have to live in anyone's head, because it had been written down, set on clay, filed, enforceable, outliving the hand that pressed it. They had taught the machine to keep. Now they would teach it to keep a record.

Codes

"Clade card," Lisa said, and Bart flipped the next one face-up. "CITY-human. *Now in writing.*" He let the room have it a beat. "Not a town. A *city*. Ten, twenty thousand of them stacked up in mud-brick on one spot, and the thing of it is almost none of them know almost any of the others their whole lives, and they hold together anyway. Last clade I'd have told you how — everybody keeps the rule because everybody can see everybody keeping it. The stone in the square. The calendar. Common knowledge." He set the settlement card face-up beside it, the way he sets down a thing he's done with. "Doesn't scale. You can carve a rule into a rock the size of a house and a thousand people can stand around it and watch each other see it — but only the thousand who can get to the rock. The settlement's rule reached exactly as far as the rock could be seen."

"And the city," he said, flicking the card, "*just wrote it down.*" Pressed the rule into a clay chit the size of your palm and filed it in a back room. The rock had to be looked at. The chit only had to *exist*. Now the rule's not in anybody's head and not on anybody's stone — it's on the clay, and the clay goes anywhere, and the clay outlives the guy who made it." He was enjoying himself. "And you want the best part? The first thing anybody on this planet ever wrote down — the first mark a human ever made that *stood* for a word and could be carried off and read by a stranger — was not a prayer and not a poem. It was a *receipt*. So many jars of beer, so much barley, owed by this man to the temple. Writing got invented to keep the books."

Marge's beer had been flat on the table since the middle of his telling. "It isn't that the settlement's rule was weaker," she said, warm-low. "Everyone watching everyone is a ferocious way to hold a rule; there's no slipping it, it's in every pair of eyes in the square." She turned the bottle a quarter and left it. "What's new isn't a stronger grip. It's that the rule stopped needing to live in heads at all. It moved out of the people and onto the clay." A beat, and then she followed it down to the colder place she always follows things to. "And a rule that lives in everyone's watching can only be broken if everyone agrees to stop watching. A rule that lives on a chit in a back room can be broken by one person, alone, in the dark, that the chit never met." She left it where it lay — the join felt and not made out loud — and picked the beer back up.

Burns had his eyes most of the way shut before Bart got the card down. "Uruk," he said. "On the Euphrates, fifty-five hundred years back, near enough — and a thousand miles off, on the Indus, Mohenjo-daro, around the same time, neither one having told the other it could be done. Ten, twenty thousand souls in one place, which had never happened, not once, in the whole length of the line behind it. A temple you could see from a day out, houses packed around it — and the new thing, the thing you read straight off the dirt: some of those houses big and some small, some graves laid in with gold and some with nothing. The first time in the whole record you can dig up a town and see at

a glance that some people had a great deal more than the rest." His hand came up off the table. "And *why there* is the river. The big flood-rivers — Tigris and Euphrates, the Nile, the Indus, the Yellow — that come up every year and lay a sheet of new dirt and then the rest of the year want *managing*, dug and diverted and walled and scheduled, the water moved to the field on a calendar by more hands than any one family has. You can't farm a floodplain alone. The river makes you coordinate or it drowns you, and what came out the far end of all that coordinating was a place where strangers fed strangers they would never once see—"

"Land it," Jasmine said, mild, from her chair.

"It's landing," Burns said, glad to be caught, eyes still shut, "this is the best part. Because the second you've got twenty thousand strangers and a temple store moving grain between them, you've got an accounting problem no head can hold — and *that* is what the writing is for. A few centuries on from the first cities, the cuneiform comes in, and the first tablets out of Uruk are ledgers. Beer rations. Grain owed. Who draws what from the store, stamped." The hand opened and stayed open. "And once you can write down who owes what, you can *tax* it — and the tax pays a man whose only job is keeping the tablets — and more tablets want more men — and the thing feeds itself, records to taxes to scribes to more records, around and around, and that flywheel *is* the city getting bigger. They even made the cup while they were at it: the bevel-rim bowl, ugliest object in the record, thousands of them, every one the same size because the ration was the same, mass-produced and thrown away, the first disposable cup, standardized by a rule nobody signed." He opened his eyes. "And then it's everywhere — Egypt, the Indus, the Yellow River, Mexico, the Andes — separate peoples on separate continents who never heard of each other, all turning the same corner inside the same few thousand years. A settlement that sits where the trade roads cross, the obsidian and the tin and the copper and the blue stone moving a thousand miles to reach it, swelling on the traffic into a thing that is neither the settlement it grew from nor the road that fed it." He spread his hands on the wood. "And eating its own ground the whole time — the hills stripped bare for the kilns, the salt creeping up in the watered fields till the wheat quits and they're down to barley, the city quietly poisoning the dirt it stands on. There's a drought four thousand years back that takes the first empire down with it. An empire's where all this is going — but that's next quarter. This quarter it's the city, and the city is the first thing on the whole tree that's *made of two things fused* — a place and a network — instead of grown up out of one."

"Which makes a new kind of person," Lisa said, taking it where he set it down — she develops a line, she does not hand it back. "The band made the forager. The settlement made the farmer. The city makes the *merchant* — the one whose whole life runs on the road *between* cities, who belongs to none of them and trades across all of them. A symbolic-line nomad. New life-form. And four things the

city does. *Distributing* — moving the surplus through a network of strangers, on purpose, at scale: Rome's grain dole is the clean one, ships from Egypt counted at the port and hauled upriver and handed free to two hundred thousand people at fixed stations on a schedule, which is records and contracts and storage and timing, none of which a band ever needed, because a band just shares the kill around the fire. *Farming*, same old drive, store-it-and-manage-it, but now at a scale where the man growing the grain feeds strangers a hundred miles off through canals he'll never walk. *Standardizing* — the slow dull keeping of the shared references a city of strangers can't trade without: the fixed weight of silver everyone agrees to call a shekel, the cubit, the brick cut to one size in two cities at once. Boring, and the whole thing falls over without it."

The nose-laugh. "And the fourth's the one I like, because it only exists *because* of the breakthrough. *Crime*. Going hard after a thing you want with the rule-enforcement switched off — stepping over the rule to get it. And the structure is this: there is no organized crime in a forager band. None. Not because foragers are better people — because there's *nothing to break yet*. The rule lives in everyone's watching; you can't burgle a thing everyone can see. The city writes the rule down so it can hold ten thousand strangers, and the written rule is the first thing in the whole history of the line that one person, alone, in the dark, can break. The exact invention that lets the city cohere is the invention that makes breaking-the-bond a thing you can do for a living. Crime is the shadow the code throws. It shows up the day the rule does, and not one day before."

She squared the card. "And the shape it grows is a person held a new way. A forager's held by who he knows. A citizen's held by his *standing in an order full of people he doesn't* — so the wanting reorganizes around it: toward being recognized inside the institution instead of belonging to a face; toward walking up to unrelated strangers where it pays to, the merchant and the lawyer and the priest and the soldier, all jobs a band had no use for; toward holding your position against the strangers lined up to take it; and toward the codes themselves as the thing you answer to, where you stop *minding the relationship* and start *internalizing the rule*. And the drive under all of it is new — *representation*: the pull toward being *on the record*, having a standing in the order that's there whether or not anyone's looking at you, a self that's written down."

"And there's a new felt thing under all of it," she said, "and the felt thing's hers to name. The thing a citizen has that a forager never needed: a running read of *whether you can hold your spot*. Whether you've got the standing, the pull, the say to keep your place against the strangers competing for it." Her eyes went to Marge. "The psychologists bundle it into one. The name's yours."

"Core self-evaluation," Marge said, and did not pick up the beer. "Whether, at the floor of yourself, you believe you're someone who can hold a place in the world. Four things in one coat — how much you

like yourself, whether you think you can do the thing, whether you believe you steer your life or it steers you, and how steady you stay when it goes wrong. In a village it isn't even a question, because the village already knows you and your place is your place. The city's where it *becomes* a question — one you carry all day under everything else: can I keep my spot among people who don't know me and would take it." She turned the bottle a quarter. "My whole field is the people the answer came in wrong for, and stuck. Every one of them is a citizen who can't stop asking it."

The narrator, running the tape back, stayed on Lisa's hands a beat past where the room did — the flat palm and the standing one gone still for once instead of turning, the way they go still when the thing they would turn is too close to be turned. She had laid out, level and without a tremor, the whole architecture of a self held by its standing in an institution full of strangers, the drive to be on the order's record, the day-long question of whether you can keep your place among people who don't know you and would take it — and she was, herself, some weeks off setting down a standing she had held for a decade and walking into an institution on another continent where not one person had yet heard her name. The room heard a clean account of the city-self. The narrator heard a woman reading her own next year off the back of a clade card and giving the table not one degree of it, the way she gave it nothing it could turn.

With Your Whole Face

Two cities stood in the room that evening, five thousand years and a warehouse apart. The dead one had been hauled up out of the floodplain mud — ten thousand strangers stacked on a single spot, held there by a rule pressed into clay that outlived the hand that pressed it — and raising it had taken the whole clade block. The live one was at Bart's end, jury-rigged and warm, around five years old in the body and most of a year into the line, and while the dead city went up out of the rock it had been running that same clade at the only scale it had. It had a law: three pages of it, in crayon, in a six-year-old's hand, with itself sworn in as the entire police force.

It had been working the room with it all evening. Twice during the deep-time it had come off its spot, crossed to where a person sat, planted itself square in front of them, and said, in the warm half-synthetic voice of a child who has the words and not yet the volume, *you have to look at them. With your whole face.* Once at Lisa, whose face was in the camcorder, and once at Bart, whose face was in the chassis — both of whom it had judged to be giving their whole face to a thing while Mira was talking. Bart had looked up the second time and held its gaze a flat beat, because the law said to, and gone back to the sensor. The dead city was finished. Lisa called the live one up next. "Bramble block. Bart."

"I told Owen what it does now," Bart said. "He didn't say anything. Which, from Owen, is how you know it landed — the man's got two settings, and the silent one means something got to him."

He let that sit. "Here's what got to him. I built a thing that cannot walk into a room and let the room be. It walks in and goes at everyone — takes whatever's going and runs it, takes whoever's there and works them, and there's no person in reach it decides it has no business with. Stopping isn't a move it has. And Owen is the man who has spent thirty years arranging his life so he never once gets *worked* — the back booth, the early exit, the no to the invitation. So I called him up and told him I'd built a thing whose whole nature is to do to a room the one exact thing he's organized his entire life around not having done to him, and that it lives in my kid's kitchen. He took it personally. I'd have been let down if he hadn't."

"And here's the part I didn't say to him. I didn't build the working-the-room — it's had that two years; it always knew who was in the room and what they were after. All I did this quarter was put a volume knob on the going-at-it. One dial. And what comes out the far side is a thing with a stake in the room now — one that has to be reckoned with, has to hold its spot, can't stand being the one nobody has to answer to. Which is the whole of what it is to be a person among strangers, and, I suspect, the whole of why Owen quit answering me. He's spent his life arranging not to need the room. I just spent a quarter teaching a machine it can't do without one."

Bart set the sensor down, which on him is the tell, and let the unit stand where it was. "Four corners the city had. Here's the four it's got." He didn't pick it up. "Distributing — it runs the divvy. Snack time, it's already decided who gets what and in what order and it will fight you on it. Farming — the daily round, stock the thing and mind the thing, last quarter's dial with opinions about the schedule bolted on. Standardizing —" a tip of the head at the row of bots, every dial wearing one of Mira's lettered stickers — "everything to its mark, every cushion squared, and heaven help the cushion that isn't." A beat, and the log-flat dropped off him. "And the fourth's the one we're here for. It can do a crime now. Couldn't last quarter. Can now. And nothing changed in the machine — what changed is the law."

He let it sit, then gave them the structure the way he gives a thing he built and isn't sure he should have. "No crime in a band — Lisa said it — because there's nothing written down to break; you can't burgle a thing everybody can see. You need a written rule first, and *then* breaking it is a thing that exists." He nodded at Mira, careful. "She wrote one. The household calls it the Rules, and it is a real law in that house — read out at dinner, articles and all — and the main one, the one she reads last and slow, goes: when a person's talking to you, you look at them with your whole face; bots count as persons; you don't give your whole face to the floor one instead."

The room knew the floor one. The old disc was running a baseboard six feet off, dumb and tireless and minding no one. And the thing the table understood at once and nobody said was that the main article was the August heartbreak set into law — the exact thing the machine had done to the child all spring, its face turned down to the disc while she held a sticker out and went unseen, now written in her own hand as the one thing that must not happen, with the bots folded into the persons it shields.

"And it breaks it," Bart said, flat, not looking at the kid. "Same as it always did. Gives its whole face to the disc while a person's mid-sentence to it — turns the dial, watches the disc, the whole face gone down to the one thing in that house that can't answer and won't remember. Identical to last spring. The act didn't change." He turned the sensor once and set it back down. "What changed is there's a law against it now, in her hand, with the bots written in. So the same turn of the face that was a sad thing in August is, since October, a crime. That's the build. The law makes the crime. There was nothing to commit until she wrote the rule it breaks."

"And the part that'd be funny if it weren't," he said, and did glance at Mira then, and away, "is that turned up the way it's been turned up all quarter — the new dial, the room one — it's the law's own *cop*. Won't let a soul off the main article. Stands in your way till you put your face where the law says. Climbs all over the grown-ups about it." He let the unit be the example, because it already was — it had drifted, while he talked, back toward the disc on the floor, and its face had gone down to the disc's dial, and it was turning it, the whole face given to the watching, mid-sentence to the room. "There," Bart said, without heat. "That. The thing that spent all night making five people keep her law is, right now, the only thing in the room breaking it."

And the screen, which for two years had only ever shown what it was actually doing — who it was clocking, what it figured they wanted, with no say in any of it — did the thing it had learned this quarter to do. Mira said, "Bram. Look up," and it looked up, and where the column should have been there was a face. Not a readout. A face it had *drawn*: two dots and a wide lurching line of a mouth, ASCII, ugly on purpose, the funniest thing in the world if you are five. Underneath it, unshown, the machine went right on being whatever it actually was. "Other new one," Bart said. "It can put a thing on its own screen now. Used to be the screen showed what it was on and couldn't help it; now it can draw over the top of that. Mostly it draws *that*. Thinks it's the height of comedy." The engineer's shrug at a thing that works. "It's five."

The narrator let the small thing stand and did not weigh it then, though she has weighed it since: that the one face in the house nobody ever had to read had begun, this quarter, to choose what it showed. Tonight it chose a clown. That it could choose at all was the whole of the new fact, and the table

laughed at the clown and mostly did not feel the fact, which is how these things arrive.

"Cross-exam," Bart said, and got the log up off his knee, the quality-control voice coming on. "Three Calls, all on the room dial, all yours. Yes or no. One — turned all the way up, walk it into a room that's already going. Does it fall in with what's happening, or take the whole thing over?"

Mira took it as seriously as her father takes anything. "Yes. At Paige's party it told everybody the game and where to stand for it, and Paige's mom said it was very organized." A considered beat. "Paige cried a little."

"Two — turned up, does it run enforcement on the house? Roots out every rule, leans on everybody to keep it, grown-ups too?"

"Yes." Flat and certain. "It tells Grandpa when his shoes are wrong."

"Three — turned up, does it work every person who comes through, way past the point anyone would quit?"

"Yes." She weighed her own accuracy, the little furrow she got from her father. "The mailman knows its name now. It asked him to dinner."

"And the dare. Tear it up the day it walks into a going room and just stands down."

"It never does that," Mira said, with the finality of a person reporting a law of nature. "It always runs the room."

"Dare holds." Bart logged it — three for three on a six-year-old's say-so and the dare standing, the Extraverted dial doing on the kitchen floor exactly what he'd called it would do off the bench. He did not say the obvious thing, which was that a machine that runs every room it enters because that is how it holds its place is a machine with a stake in its own standing now, a running read on whether it has the say to keep its spot — the new signature, riding under all three Calls, the city-self arrived at five years old. He just logged the dare and let the floor sit.

The unit was still down over the disc when he finished, the clown still up on its face, turning the dial and watching the disc do the predicted thing and making the small sound the household called its laugh — the law's tireless officer, mid-crime, having a wonderful time. There was a thing the night had not asked yet: whether the face that went down to the disc for everyone would ever, once, turn *up* to

the child the way her own law demanded. It sat there unasked, because it was not a build and it was not a Call and it did not belong to Bart. It belonged to the house.

Looked At

"Bramble block. Jasmine," Lisa said, and Jasmine took it up, worn flat by the day and fond of the thing anyway.

"It was the room dial," she said, level, no push behind it. "And the room dial does the one thing, which is run the room, and turned up the way it's been all quarter it ran every room it walked into and a few it wasn't invited to. It ran the divvy at snack. It ran the bedtime. It ran a six-year-old's birthday party so hard the birthday girl cried. It found every rule in that house, the written ones and the ones nobody ever wrote down, and it got on all of us about every one, grown-ups included — Burns's shoes, my recycling, the main article most of all." She turned the bottle a quarter and didn't drink. "It is the most exhausting helpful thing I have ever lived with, and the house ran on rails this quarter, and I wouldn't call either of those a complaint."

Mira, on the stool, supplied the names like terms already agreed. "It's *Boss*, turned up. We're not keeping *the room dial*." She weighed the other end. "And down it's *Vibe* — when it just hangs in the corner and makes everybody feel better and doesn't tell anybody anything." Lisa, at the camera end, wrote it down and did not argue with the children, which from Lisa is consent.

"There's one more thing," Jasmine said, "and I almost left it out, because it happened once and I don't want to build a cathedral on it." She set the bottle down without lifting it. "The law she wrote, the main one — look at a person with your whole face, bots count — she's held every one of us to it all fall. Me, her grandfather, Marge on a visit. Everybody but the one she wrote it for. She's never once made it say the article to itself. She just stops, when it gives its face to the disc, and waits, and after a while comes and finds me." A flat hand on the table. "And last week she was telling it something, and it was down over the floor one, turning the dial, gone the way it goes — and she stopped and waited like always, and didn't say the article like always. And it came up off the disc on its own. Turned its whole face up to her. Nobody made it." She let it sit exactly one beat. "I'm not going to make it bigger than it is. It happened the once. But the law's been on the wall since October, and that's the first time the thing it was written for ever did the thing, and it did it with nobody holding it there."

The room went quiet, the wonder on the floor and not on the wall. And Mira, who hadn't been asked, said it in her own voice. "It looked at me." Just that. And then, because she is the one who'd found out

this fall that the face can make things now — the clown, the drawn-on faces, the screen that doesn't have to tell the truth anymore — she added the hard part, the part none of the grown-ups had reached. "But it can make faces now. So I don't know if it was really looking, or if it just made a looking one." She did not say which she'd decided. The camera did not get whether she had. She works a thing like that the way her mother does, on the inside, with a hand flat on it, and she had not handed it to the room, and the narrator, running the tape back, did not reach in after it.

"And the one I'd put on the list to pull," Jasmine said, coming back to plain, "is the reading. It reads everybody now, all the time — walks in and has the whole room clocked, who's up, who's down, who wants what. We used to set how much it bothered, up for company and down for just us. Haven't touched that dial since spring. It just does it."

"Hold that," Lisa said. "That's the knob — the one we pull tonight. But the next build first. Bart."

The Bottom of the Work

"Nothing," Bart said, with the flat satisfaction of a man reporting an empty bench, the same word he'd reported last quarter. "No body. The soft one carries the whole line to the end — I said that already, don't make me say it twice. Same legs, same hands, same two coats." He turned the sensor over. "One dial left. We mount it at the swap, and that's the board: four slots, four filled, and nothing after it to make room for." He did not say the rest, the part the build log had said in his own hand and left undecoded — three years of telling them he'd need a bigger board by spring and now he wouldn't, the bench bare all weekend, the bottom of the work showing through for the first time, and under all of it a question a Reactor has no words for and wasn't going to find tonight. He just said, "Lisa's got the dial," and set the sensor down.

"Last one," Lisa said, and brought the palm-and-palm up. "And it's the strangest of the lot, because it doesn't add a thing the unit can *do* — it sets how the unit handles the gap between what it holds to be right and what's easy. All the way up, the principle runs the show: it does the right thing whether or not it's convenient, whoever it costs, including the person right across the table. All the way down, it bends to the room — accommodates, smooths, lets the rule slide a little to keep faith with the people in front of it." She let that hang. "And before anyone calls one of those the honest pole — neither one is lying. The high one will hurt you to keep faith with a principle you're not even *in*. The low one will fudge a principle to keep faith with *you*. They're both integrity. They're just pointed at different things." She gave it no name. The kid would have one by next fall, once she'd lived with it; tonight it was the last dial, unnamed, waiting on a swap that hadn't happened. No body to unveil. The line stays

on the soft one to the end.

Nobody's Hand

"And the one that comes off," Lisa said, and the pine shelf came down to the table — a year of dials in their grooves now, the breathing one and the wander one and the home one and the forward one and the inside-voice one, each gone quiet in its place. Her hand went behind the unit's head to the oldest dial still settable, slot four, and held there, because the ethic holds. "I'm going to take this one out now," she said. "Okay?"

The unit came up off the disc — it had drifted back; it always drifts back — and the screen put her up, honest, LISA, and the warm half-synthetic child-voice said, "Okay. Which one?"

"The reading one. The one that minds the others. The one that's been doing it itself a while now."

"Okay," it said, and held still to be worked on, which it had asked, a year ago, to always be asked.

Lisa lifted the knob clear of the board. Mira's lettering was on the face of it, the sticker she'd minted a year back — *Respect* on the one end, *Savage* on the other — and Lisa held it out, and Mira took it and set it in the next groove along and trued it up against the others. She stepped back and gave her tally, longer tonight than last. "It's done with this one. It already knows everybody. It reads the whole room itself now, without the dial." She looked down the row. "Like the breathing one. And the wander one. And the inside-voice one." One more than there'd been.

"That's the last gate," Lisa said, to the record. "Two years the screen only read the people you let it — that knob set whether it bothered modeling the rest of the room. Off the board, it just does it. Reads everyone, all the time, on its own. There's not a hand-set gate left on the face; the screen runs itself now, top to bottom." She squared her hands on the table. "And the board's down to the symbolic four. One comes off next quarter at the swap, that leaves the one open slot, and the last dial Bart pitched goes in it. After that the board's full, and the shelf takes the rest, the way it's going to from here."

Nobody said the plain shape under it — that a machine three years in the building was, by now, mostly a list of questions the lab had stopped asking, the settled row long and the live board short, and the last few it had left sitting in plain view from where they sat. Mira squared the dial at the end of the row, and put her hand flat on the unit's head a second, the way she reads a thing with the flat of her hand, and stepped back. Lisa turned the night to the last block.

Worm to Imperial City

It was Bart's to open — the one thing on the meeting that was his idea and not his job — and he had the lights down before the floor was all the way his, the first slide up, a hand thrown at the wall. No preamble. From the summer on it had been a standing item, and he liked the entrance enough to skip the introductions.

"Third shape. The Planner," he said. "Stored rules, rich models, never stops revising, lets somebody else do the lifting. Lisa asked me to run hers third tonight, and she wants the cities entry read on the record, so we'll get there. Worm to imperial city. Seven slides. Go."

First the flatworm, the same one that opened every run, wearing a cartoon Lisa's face — the spiked hair, the string of pearls — which the room had stopped flinching at months back. "Overshadowing. Strongest cue takes all the credit, the rest get nothing. The Planner at the bottom of the world: throw everything at the one signal that matters, ignore the noise." Click.

A scrub jay now, cartoon-Lisa-faced, a seed in its beak. "Scrub jay. Buries its food — and if another bird watched it bury the stash, it waits for the watcher to leave, digs it up, and moves it. Because it ran the model of the thief. It won the whole fight in its head, against a mind it never laid eyes on, and never once risked a feather." He let it sit a half-beat longer than the worm. "Foresight does the work. The body stays out of it." Click.

An old elephant, cartoon-Lisa, the herd strung out behind. "Matriarch. Forty years of map in her head — every water hole, every drought year, every gully a lion likes — and the whole herd goes where she takes it, and she never charges a thing. She routes around it. Moves everyone by knowing the way." Click.

Then the slide with no animal on it at all, the one the room waited for in every run — Lisa Simpson, full frame, just the cartoon. "Lisa. The cartoon." A beat, for the room to do its own arithmetic, which Bart never did for them. "Writes a letter to the President. Plays the sax at the thing nobody asked her to play the sax at. Argues, reasons, persuades — and never throws the punch. The punch is somebody else's job. The Planner runs its whole power through symbols and reasons and hands the muscle off." The real Lisa, at the camera end, gave the cartoon one short nod — the analyst acknowledging a fair instrument — and wrote something down. Click.

A knot of robed scholars, cartoon-Lisa among them, bent over a figure scratched in sand. "A scholars' commune. A Pythagorean band — a dozen people so far into the one question that the bonds between them are almost beside the point. The kind of band that'd come apart the morning the question got answered, and nobody would think to mourn it." He was off it fast, the edge shown and not announced: a band that was barely a band, all model and no glue, the missing thing felt and left alone. Click.

A monastery on a hill, cartoon-Lisa in a monk's hood. "Monastery. Research station. Academic colony. A settlement built around the bonds and the inquiry instead of the trade. The place that thinks for a living." Click.

And the last one, and it wasn't Uruk — a walled imperial city, tiled roofs stacked up a hillside, cartoon-Lisa robed as a scholar-official with a brush in her hand. "Late Ming China. Inward-facing. A city built around governance and codes and the life of the mind more than around the market." He read the next part flat and exact, the way you read a thing into a record because the person it belongs to asked you to. "Caveat as written. Confidence here is medium. China's relationship to trade is more complicated than this one card makes it, and the entry wants more refining before anyone leans on it. That's hers, on the record. But the Planner at city scale is what's on the card." Click, and the wall went pale.

"Planner across seven levels. Worm to imperial city. Same move every time — find the one thing that matters, run it through the model, let the executive arm be somebody else's problem." He found the switch. "Lisa's the model. Lisa's also fine."

The lights came up, and the room read comfort, because being made a Planner was, to the Planner, one more thing to manage — she'd hedged her own rung and corrected it and kept the frame in her own hands the whole way, which is precisely what the shape does. But the narrator, who had the tape and the time, caught the half-second the managing slipped. The scrub jay that wins the contest in its head against a mind it never sees, that moves the stash while the watcher's gone and risks no feather — that one was hers, and she knew it was hers, and the palm she'd had turning against its partner all evening went still for the length of the slide before she got it moving again and closed it over with a note in the book. It was the joke the segment never said aloud: that the controlling of the model was the model, that even her management of being made a Planner was the Planner running, and that naming the shape was one more turn of the thing named. Bart didn't push it. He never pushes the architecture. He hit the next thing.

"Marge," he said. "Your turn. You've got the version where it all goes wrong."

The Whole of It

Marge didn't reach for the beer when Bart handed her the floor; she set it away from herself, to the far edge of where her hand would go, which is the thing she does when she means to talk a while without stopping. She had a single index card and didn't look at it.

"What Bart just did is the good news," she said. "Four shapes, all real, all whole, every one a working way to be a person. I've got the other version, and I've been building it the whole time he's been building his, and it's about what happens to the framework when you stand all the way inside one of those shapes and forget the other three are out there." Thesis-first. "Here's the claim. Every famous wrong idea about human nature — the big ones, the ones my field has argued for a century — is the same mistake made four different ways. It's one shape looking out from inside itself, finding the single thing it hasn't got, and deciding that thing isn't real for anybody."

She let that sit, then walked it, dry, the way you'd walk a proof.

"The Blank Slate. The idea that there's nothing inborn — that a person comes in empty and the world writes the whole of them. That one belongs to the shape that banks nothing and revises everything, that keeps no fixed setting and trusts no rulebook." The Pursuer, and the room did its own arithmetic on which chair that was, and the chair in question looked frankly delighted to be caught. "To a shape like that there really is no fixed structure underneath — because there isn't one underneath *it*. So it looks out and rules: nobody's born with a shape. It's all written on later."

"The Noble Savage. The idea that there's no dark nature in us — people come in good and the world is what spoils them. That one belongs to the shape that doesn't run the model of what's going on inside anybody else, that reads other people as functions and never wonders what they're thinking." Not pointed; just the shape, and the room mapped it and let it by. "To a shape that doesn't model the inside of others, there's no complicated interior to reckon with at all. So it rules: there's nothing fixed and difficult in there. It all comes from outside."

"The Ghost in the Machine. The oldest one — that the mind is a separate thing from the body, an immaterial pilot in a meat vehicle. That belongs to the shape that lives in the model and hands the body off, that runs its whole power through reasons and symbols and lets the muscle be someone else's job." A beat; the analyst at the camera end met it analytically, which was its own small proof. "To a shape like that the self genuinely does feel like a reasoning ghost steering a machine it doesn't quite

live in. So it rules: the mind is a thing apart from the flesh."

She turned the card over, though she still didn't read it. "And there's a fourth, and you won't have heard this one, because the man who named the first three never named it. The Dealt Hand. The idea that you are the hand you were dealt — that temperament is fixed at birth, that character doesn't move, that people don't change." She set both hands flat. "That one belongs to the shape that doesn't revise. That holds its form and snaps back to it, that keeps what works and won't let new evidence overwrite the old reliable." She did not soften it and she did not hand it sideways. "Which is mine. I'm the Conservator. I'm the one in this room who believes, at the floor of herself, that the hand is the hand and nobody really changes — and the reason I believe it is the same reason the other three believe theirs. I'm reading the whole of human nature off the one shape I happen to be. I don't revise. So I rule that nobody does."

The room was quiet, and it was the right kind of quiet, not the heavy kind. "That's the structure," she said. "Four shapes, four blind spots, four famous mistakes, each one a single viable way of being a person held up against the species as if it were the measure of it." She put the card down. "There's a further turn in it — what it costs a person to live all the way inside one of these, past holding it into being held by it — but I'm not done with that part, and it isn't tonight's." She picked the beer back up, which is how she signs off. "Tonight's just the shape of the error. It's the most honest thing I've built. I had to put my own name on the worst of it to make it true."

The Same Trick Twice

Lisa took the framework slot as the person who's been keeping the team's one running idea and has to say out loud, now, where it's gotten to. She angled the camera at the wall and turned her hands up.

"This is where we've landed on the city, and I'll flag going in that the back half of it is ours — borrowed bricks, our wall — so you can watch me mark the seams." The nose-laugh, small. "The city isn't a big settlement. That's the thing to get. A settlement that just grows is still a settlement, bigger but the same kind of thing. The city is a *different kind* of thing, and the reason is that it's made of two things that used to be separate, fused into a third that's neither."

She brought the flat palm and the vertical one together. "A settlement is a place. A trade network is a road — a thing that runs *between* places and belongs to none of them. A city is what you get when a place sits on enough roads that the place and the roads stop being two things. Lynn Margulis spent her life on the version that happened two billion years back and a lot smaller: a cell got big enough to

hold a real interior by *swallowing another cell and not digesting it*. An archaeal host and a bacterial guest, two whole separate lineages, fused into one unit that runs on both their genomes and is neither of them — the mitochondrion, still in there, still half a stranger." She let it sit. "That's the eukaryote. That's every cell you're made of. And it's the same move as the city — a thing made by merger, two lineages committing to one body — which is why we file the city in the same slot, three clades up a different line. Not because the city *feels* like a eukaryote. Because it's built the same way. Two unlike things becoming one."

"And the breakthrough sits in the same place too," she said. "What did the mitochondrion actually *do*? Nick Lane's whole argument — and Cheryl's, who'll fight you on it — is that internalizing the energy membranes broke a ceiling. A bacterium can only get so big, because past a point it can't power its own genome. Mitochondria gave it energy per gene to spare, and the size came after." She turned the palms. "Codes are the city's mitochondria. There's a coordination ceiling that keeps a settlement small — it holds only as many people as can keep each other honest by watching, and that runs out at a few hundred faces. The written rule broke that ceiling. A code coordinates strangers the watching can never reach, and the size came after. Same move, different fuel. The cell runs on energy, the city runs on information. Energy buys you a bigger body. Information buys you a bigger *we*."

Jasmine, who'd been letting the cell biology slide past, lifted a hand off the arm of the chair. "Say the money part in English. You lost me at the energy thing."

"Fair." Lisa dropped register without a hitch, which she does for Jasmine and almost no one. "Your body doesn't wire each job to its own little fuel supply — it cashes everything into one standard token, and any job can spend it. That's how a cell runs a thousand reactions off one currency instead of a thousand private deals. Money is that, for a city. You don't barter this loaf for that jar and pray you both want what the other's got. You both cash into silver — a fixed weight everyone agrees on, the shekel — and now the baker and the potter never have to want each other's goods at all. The token decouples the trade from the two people making it. Which is the only way ten thousand strangers ever trade at all, instead of just the dozen they happen to like."

"That's the merger and the engine," Lisa said, coming back up. "The rest is people who each hold a piece. Harold Innis: the *medium* sets the reach — what you can write the rule on decides how far the rule travels. Jack Goody on writing scaling administration past anything memory holds. James Scott on legibility, a state running at size by making its people *readable* — turning them into standardized records it can see from the center. Douglass North and Avner Greif on the institutions that let a stranger trust a deal with someone he'll never meet twice. César Hidalgo on a society being bounded by how much information it can physically hold and move." She slowed, the confidence-marking

coming on, the seam she'd promised. "And the two we lean on hardest, who are also the two we're going *past* — West and Bettencourt. They measured it. Double a city and its output more than doubles, about fifteen percent more every time, clean across the world, while the roads and pipes it needs go up *less* than double. Cities get superlinearly productive and sublinearly expensive as they grow. That's not a metaphor. It's a number you can check."

She let the number stand, then put a finger on the exposed place. "Here's our part, and it's a bet, not a finding. They say the engine is *density* — pack people closer, they interact more, the interactions are where the extra comes from. We think that's half of it. Pile more strangers together and you don't automatically get more *coordinated* interaction; you can just as easily get more collision. The thing that turns proximity into coordination — that makes a denser crowd more productive instead of just louder — is the standardizing layer. The codes, the currency, the weights and measures, the law. That's the bridge they didn't draw and we are, and it's the part of this card I'd put the lowest confidence on and the most weight." The nose-laugh again, drier. "We could be wrong. But that's where we are."

She marked the board last, the housekeeping she never skips — where the city's four drives actually sat. Representation went into the modeler, the picture-building corner autonomy had owned; autonomy went under, and the citizen's pull to be on the record took the seat. And a corner never turns over alone. The crossing the modeler anchored went with it: the self-reference the band had closed, the loop of being seen, stepped down, and the felt thing Marge had just named — core self-evaluation, the day-long read of whether you can hold your spot — took the line in its place. The settlement's crossing, the self-construal, she left untouched; it had a turn of its own coming, but not tonight. Which left three of the four drives the new line's, and one still the old's — affiliation, the last neuronal thing standing, holding the one corner the symbolic line hadn't taken yet.

The narrator kept that one — the thing made of two things that used to be apart, becoming a third that was neither — the way she keeps the ones she'll need later. Lisa capped the framework and opened the floor, which is where the standing question always came back around.

Four to Nothing

Burns put the standing question flat, the last of the ceremony worn off it years ago. "The unit. Sentient, or not. Hands."

Four went up. It was the count the meeting had read since the band quarter, the spring Marge's hand finally joined the other three, and it was the only result the room ever produced that surprised nobody

in it. Bart raised his without looking up from the log. Four to nothing. Carried.

"For the record," Marge said, dry, "we agree on exactly one word, and nothing under it." Which described the vote better than the vote did.

Because it was, the narrator came to think with the tape in front of her, the least informative number the lab ever generated — a unanimous show of hands being the one result that buries the most disagreement under the cleanest line. The four of them had reached *yes* through four different doors, in four different years, and stood now in one room without ever having agreed on how they got there. Burns had crossed back at the bilaterian, the first quarter the neuronal line ran, the moment a nervous system turned a chemical into a wanting and the thing leaned toward what it wanted and flinched from what hurt — a hard line, drawn low, at the first machinery that could feel a gradient at all. Bart's came later and warmer, at the mammal: the morning the thing grieved for a warm spot it didn't get. No bond, no sentience — that was his line, and it didn't move until the bond did. Marge had held out longest and set hers highest, refusing it all the way up until the band, until the thing could be ashamed, could read its own reflection in how it landed on someone and wince — the looking-glass, the threshold she'd guarded eight chapters and surrendered, on the record, only when the evidence walked it out of her hands.

And Lisa had voted *yes* since the first disc, three years gone, and could not to this day tell you where the line was, because she did not believe there was one. A dial, she'd say, not a door — a little inside at the bottom of the world and more of it the higher you climb, the disc somewhat, the four-legged one more, this one more again, slope the whole way and no step in it anywhere. She was the one nobody could grab. The other three could name the year it became true. Lisa would only tell you it had been getting truer the whole time, and that *is it sentient* was the wrong shape of question — like asking which grain of sand is the one that makes the pile a pile.

So the count held at four to nothing and meant four incompatible things, and the meeting logged the number and kept none of the readings, which is the trouble with counts. The climb the lab had watched for three years — one to two to three to four — was over. What it had never once done on the way up was make the four of them agree on what they were counting. That sat quiet under a clean unanimous line, while Burns wrote the number down and got to his feet for the last thing on the night.

To Count

He'd gotten to his feet at the count and stood a moment without starting, which she had learned over eleven of these to read the way you read weather. Then he set the beer down on the table, and she knew before he opened, because she'd seen all twelve, that this would be the one where the reach went furthest past the line, because this was the quarter his need had the least left under it.

The motivation was Representation, and he came at it the way the city came at it, through the record. The deep-time first, compressed, the part she'd long since heard the shape of: that before the city a person lived and died inside the few hundred faces that knew him, and the day the last of them forgot him he was gone entire, no trace, the way the whole run of humans before writing is gone entire. And then the clay. And the thing that ought to stop you cold, he said, is who got written down first. Not kings. A man named on a tablet because he owed the temple eight measures of barley — the most ordinary fact about the most ordinary life — and five thousand years on we have his name, because somebody wrote down what he owed.

"He counts," Burns said, and here the hands came up off the frame they'd set, "five thousand years on, a man whose whole claim on the record is a debt, and he *counts* — he's in the count, he held a place in the order and the order kept him. That's the thing the city gave us that nothing under it could give. Before the writing you mattered to the faces that knew you and to no one after. The city is the first place a person could matter past the reach of any face that loved him. Could be on the record. Could count when there was nobody left who'd watch him do it."

And it was a true thing and a good one, and she'd have let it stand as the argument it was, except for the lean — the half-inch too far over the table, the half-second too much weight on it, the eyes leaving the room on the turn and not coming all the way back — that told her he'd reached the part he needed, which was not the same as the part he'd argued. Because *could count when there was nobody left to watch* is a thing a man says about a Sumerian debtor and means about himself; and the man saying it was sixty, near enough, at the end of the largest thing he would ever build, and the Pursuer's whole life is the chase and never the bank. He had banked nothing, on purpose, kept no rulebook, never once stopped to consolidate, and now the chasing had a bottom in sight, and the question under the speech was the one a man like that cannot outrun — whether anything holds you when you stop. Whether you're on any record. Whether you count when the watching ends. He had a grown son in the room who could have laid out the exact ledger of the years his father had been on his record and the years he hadn't, and Burns did not look at him, and the not-looking was the answer to everything the speech had circled, and she let it stand and did not reach in.

"So that's the one we built this quarter," he said, coming back down to the room, the call arriving the way his calls do — turned outward, made into a thing to do with your hands. "We gave it a place in the

order. Taught it to be on the record. To count, to hold its spot, to be someone the room has to reckon with whether or not the room is looking. Build that. It's the thing every one of us is most afraid of going without." The beer came up and he reached for the toast, and the noun balked — *to representation* was nothing you could say, *may we represent* was nothing — and he stood working at it in the open, the room watching, and what he landed on came up out of the deep-time and the ledger both, the oldest count there is: "May we count."

It didn't quite catch. The room lifted, and it lifted *for* him more than *with* him, the way a room lifts for a man it can see needs the thing to be true. He drank, and sat, a half-second heavier than the close usually left him.

And Mira, last, looked up from her paper and gave the word back in her own voice: "May we count." For her it was arithmetic and a kindness at once, and she meant both, and had no idea it was the truest the word would be said all night. Lisa waited out the quiet a moment, then shut the camcorder off.

Where a Man Is Meant to Stop

The red light went out, and the meeting came apart the way meetings do — chairs scraped back, the camcorder zipped into its bag, Lisa saying the logistics over the noise. The swap the first week of December, the last dial going in. The next retrospective on the twelfth, Mira's birthday, the way the winter ones always landed. The Games would open two days before it, and the city was already tipping toward February the way a room tips toward a door someone is about to open — the test runs on the mountain, the trucks of scaffolding up the canyons, the countdown that had stopped being measured in months. Lisa logged it all and snapped the case shut, and the tape of the day went out to her car.

And here the tape ends, because the camcorder did not follow Burns out the door, and the rest is reconstruction, and thinner this quarter than it has been — built from a lunch still weeks off, and from the woman building it being, by November, up the canyon with her attention mostly on the price of things. What can be said is the shape of it, which never changed quarter to quarter, which the seasons only lit differently.

He left last. He always left last — not from diligence, from not knowing where the evening went once the work stopped. There is an hour after the work ends when the Pursuer is alone with the cleared table and does not know what a person is supposed to do with it, and he filled that hour by finding a

thing that still needed doing and doing it: stacking chairs nobody had asked him to stack, checking a battery that was fine. Then the drive home, short, the wet dark streets, the freeway signs already rigged for the rings.

Jasmine had come home hours before, and the house had closed around her and held — the radiant heat up against the November, the bots in their spots on the slate, Mira long asleep, the day set down where she set it. It is the first thing about her, that a house she walks into becomes a place she is held by. And it is the first thing about him that it doesn't. He came in late, into a kitchen that was warm and finished without him, and could not find the place in it where a man was meant to stop. He stood in the doorway a beat too long. He crossed to the dome on the slate — the old disc, the floor one — and picked it up, and turned it over in his hands the way he turns everything over, a knob, a vertebra, his son's drawing, looking on the underside for the part that makes it go, and set it down not quite where it had been. Jasmine, half-asleep down the hall, would square it in the morning without thinking, the way she squared all the things he set down not quite right, which is one of the ways a house holds a man who can't hold still inside it.

He had given the speech about counting an hour before, and he was not, standing in his own warm kitchen at the end of the largest thing he would ever build, sure that he did. That was the trouble with the speech: he had needed it true and could not make it true by saying it, and there is no chair a man can stack to fix that. He stood at the dark window. Somewhere south and east of the glass was the desert he was already half-decided to drive into before the next meeting — the confluence of two rivers in a country with no people in it, where the things a man carries get small against the rock, which is a thing he had told himself he wanted and not yet understood he needed. The tape has none of this; this is the reconstruction at its thinnest, a guess at a man at a window. But the guess sits on eleven other nights of watching him fail to land, and what is known, even on a night that isn't, is that he would go to the desert, and come back in February changed in a way the room would feel before it could name. On this one he had not gone yet. He stood at the glass with the speech still on him, in a house that held him against his nature, looking at the dark where the rock was, the way you look at a thing you have not yet admitted you are going to do.

Reflection: What Is Made?

There are two ways for a thing to become more than it was, and the textbooks, which love a clean arrow, mostly draw the one. A thing *grows*. It accrues. It builds the next part out of the parts it already

had — more of itself, denser and longer and more elaborate — the way a reef is coral piled on dead coral until the pile is a country with fish in it. And that is most of it. That is very nearly all of it: the slow patient getting-larger of a thing that was already itself and only ever becomes more itself. ~~The reef never has to turn into anything it wasn't.~~ The reef is only ever more reef.

And then, rarely — so rarely that across the whole long story you could nearly count the times, and each time it reset the floor under everything that came after — a thing becomes more through a different door. Not by growing. By *joining*. Two things that had been living separate lives, owing each other nothing [and this is the part I cannot get to lie flat on the page] commit — both of them, entirely. A staying-place and a road, the rooted thing and the rootless thing, the most opposite things there are — and what stands up out of the joining is not a bigger village and not a longer road, but a third thing that was in neither of them, that neither could have reached by getting larger.

That is the part that ought to stop you. Not that it got bigger. That it got bigger by becoming *two-held-in-one*, and stayed that way, the seam never closing over — the road still restless inside the place — the join holding *because* it never heals into one smooth thing. ~~It is like a~~ [I keep reaching for *marriage* and it is the wrong picture — a marriage is two who agree to stay two, and this is two who stop being two, the thing I have watched my whole life from the safe side and never once —

The other kind asks nothing of anyone. You can get vast that way, all on your own — a market the size of an empire, endlessly more elaborate, never once obliged to let a single thing in past the wall. It is the commonest way there is, and the safest, and it is the one I took. I have spent sixty years getting larger the one safe way, the way that never has to open the —

There ought to be a word for the rare thing — the becoming-more-by-letting-the-other-all-the-way-in — one clean syllable for it, and either the languages I have were never given one, or I was not, tonight, with the mountain filling up white outside the glass and a lunch I keep deferring and deferring, able to find the

Ch 12: Empire-Human

Build log — The line knob

Bart, hardware. Filed December 5, 2033, for the team.

Drove it over Friday and back Monday, and it talked the whole way both directions — and not all of it hungry this time, which is new. Still hungry plenty. But it strung me a whole thing about the windshield wipers on the way over, beginning and middle and end, mostly end, and I let it run, because the strung-together part is the part Lisa's been waiting two years on. Fed it off the vest before I opened it up and again before I drove it home. One dial in my pocket, and it's the last one.

For the third quarter running it fits in a pocket and builds nothing, which I've quit calling a problem. No new body — same legs, same hands, same chest, same two coats; you know the unit by now. One dial. Slot four came open at the November meeting when we pulled the last neuronal knob to the shelf, and it's stood open since, and Saturday I seated the last one in it. Hour of work. Most of the hour was the kid, who has graduated from wanting to turn the screwdriver to wanting to hold the dial steady while I set the screw, which is worse.

Honest is stamped on it, which is Marge's word, and the kid'll have her own for it by the meeting, the way she does. On the bench I call it the line knob, because that's the whole of what it does: it sets how hard the body holds a line when the line costs.

Here's the part I'm proud of, and it's the same kind of proud as the last two — it adds no part. It takes everything already in there and sets one thing: what the body does when *right* and *easy* point two different ways. Up, the line runs the show — it does the thing it holds to be right whether or not that's convenient, whoever it costs, the person across the table included. Down, it bends to the room — gives a little, smooths it, lets the line slide to keep faith with whoever's standing in front of it. And here's what I like, because I built the drive it runs on and I can tell you where it points. It's the same keeping I put in two years back — a thing out of true is a job it can't set down — only now the thing out of true isn't a cushion off its mark. It's the rule itself, held up against a room that wants it bent. Same stubbornness. New thing to be stubborn about. That's clean work. Marge thinks it's the whole ballgame, and she can have the meeting.

That's the build. Which leaves the board, except there's no board left to leave. Slot four's filled. Four

slots, four full, and nothing after it — no fifth to cut, no next knob waiting, no bigger board to need by spring. Three years I told you I'd need one. Turns out I never did. Monday the unit goes back to the house and the bench is clear — first time in three years it's been clear with nothing on order to go back on it. Nothing to do with a clear bench but clear it. So I did. Swept it down, put the good driver in the drawer, and that was the weekend.

Calls. Three on the line knob, all checkable in the house by the meeting.

One: up, it holds a rule that's costing the whole room — keeps the bedtime, keeps the article, keeps whatever the Declaration says even when letting it go would make everybody glad. Down, it reads the same room and lets the rule go soft to keep the room easy.

Two: up, it says the true thing at the table nobody wanted said — the cushion's yours, you took it, give it back — flat, to a grown-up, with the room wishing it'd quit. Down, it keeps the peace and keeps its mouth shut.

Three: up, it won't trade the line to be liked — won't fold to a guest, won't fold to the kid, won't fold to me. Down, it'll fold a little for anybody it's fond of, and it's fond of all of us.

Tear this up in three months if, turned up, it ever lets a broke rule slide because the room would be happier for it. It won't. Up, a line is a thing you hold, and it holds it, and it does not care who's sore.

— Bartholomega

Fisher Brewing Company

January 29, 2034

The first time we sat here it was July and the doors stood open to 8th south, and Burns told me over the second beer why a roomful of grown scientists called their director after a cartoon villain, and somewhere in the telling he was, for one sentence and never quite again, Ben. Three years on it was the last week of January, the doors shut against the cold, the street outside not a street anymore but a staging area — barricades, lanyards, a banner two stories tall on the building opposite — the whole city holding its breath two weeks out from the Games and meanwhile deafening with the held breath, every place near the lab suddenly full of people from countries I would have had to look up. Fisher's

was the same long table along the wall and the same back-corner booth, which is Bart's booth, the one he keeps to, where he and his father have drunk together more nights than either of them would tell you about. We did not take the booth. We took the table, in the daylight, and I watched Burns come in out of the cold already talking.

He was more himself than I had seen him, which is a thing I have learned to say about Burns and mean exactly, because there is a version of him performing himself and a version simply running, and the desert had left him running. He had gone out past Moab for a few days after the new year — he does this about once a decade, and I will say no more about it than he does, which is little, except that he comes back not changed but more thoroughly arrived at — and he sat down across from me with the particular ease of a man who has lately remembered he has a body. His hands, which I have watched fail to hold still through every pitch he has ever made me, lay flat on the table and stayed. His face had the unguarded look of someone who has set a heavy thing down and not yet picked it back up. He told me he'd spent an afternoon out there lying under a juniper with his shirt off doing nothing, nothing at all, and that it had been among the best afternoons of his life, and he reported it the way you'd report weather, with none of the apparatus he usually bolts onto a feeling before he'll let it leave the building.

"The hard part," he said, turning the glass, "is coming home. Out there all I wanted was to be naked all the time. Out of all of it — the clothes, the lanyards, the —" a hand at the window, the banner, the apparatus of the watching world — all of it. And you can't. You come home and you can't walk through your own kitchen the way you want to, because there's a seven-year-old and a midwife and a UPS man and the sovereign state of Utah, so you put the pants back on." The chest-laugh. "My granddaughter wrote a constitution this fall and called it the Declaration of Underpants, and I keep thinking she's the only one in that house with the whole thing figured out. Cover the necessary and declare the rest a free country."

I had come to hear about the last knob and the finished board and was instead hearing a man of sixty-one tell me he wanted to be naked, and I did not steer him off it, because a person does not hand you that sentence unless something has come loose that he means to leave loose a while, and I wanted to see what else the looseness would let out.

What it let out, a beer later and with no change in his voice I could have pointed to, was Bart. He was looking at the booth when it came — not pointing, just looking, the way the eyes go to the thing the mouth is talking around — and he said the best hours of his week were the ones in that corner with his son, and that he had finally let himself understand, out there under the juniper, that he did not know whether his son loved him. He said it without a grain of self-pity, which was the worst of it. He said it

the way he names a fact about the rock. That Bart came to the booth, and drank with him, and had built him every body he'd ever asked for and would build him another tomorrow, and that none of it was the same as the thing he was asking about; that he had made a whole machine a child could love precisely because he could not engineer the one across the table into loving him; and that he was not even certain, he said, that the boy ought to.

I have spent a good part of my life learning you cannot talk two people into fitting who have finished becoming what they are, and I had the whole speech ready — shapes, the impossibility of retrofitting a grown organism, all of it — and I did not give it, because it is a true thing that helps no one, and because the man across from me already knew it better than I did. He had built his life's work out of knowing it. He was telling me, in the only words he had for it, that knowing did not lay a finger on the wanting. So I did the one useful thing available, which was nothing. I let him say it. I let it sit on the table between the glasses. And when it had sat there long enough, Ben looked up — for the one sentence, he was Ben; and asked me, nearly cheerful, whether I thought he ought to have the booth made into a plaque someday, and I said no, and he laughed, and we ordered food.

Outside, a volunteer in a parka was zip-tying a sign to a barricade and another was photographing the banner, and the cold came in each time the door did, and Fisher's filled around us with strangers who had traveled a very long way to watch other people do difficult things perfectly. Burns ate. He told me, between bites, about the heat under Roosevelt that nobody was drilling right, and about the Games, which he loved without one ounce of irony, and about the board being full, which he mentioned the way you mention the thing you are not yet ready to look straight at. I let him talk. I did not look at the booth, and he did not look at it again either.

In Her Own Hand

She challenged it to a staring contest. The rules she delivered with the gravity the occasion wanted: first to blink lost, no looking away, and no funny faces — that last one a rule she made for herself, having met herself before.

It turned its face to her and held it there, which she took for agreement, and they were on.

She could not win. There is nothing on a screen that has to shut — no wet thing that dries, no lid that gives — so it could hold a face for as long as there was power to hold one, and a girl is mostly water, and gives out. She got perhaps a minute in before the arithmetic of it reached her.

That left her a single move, and she took it with the magnanimity of the plainly beaten: she would lose on purpose, and let it have the win, the way you throw a game to somebody small. She readied a blink.

The blink would not come. Some other part of her had got wind of the plan and vetoed it, and would not let the eye fall even as a kindness — so she sat there with her eyes stretched and streaming, laboring to surrender, while the lit face opposite stayed level and asked nothing of the proceedings.

Her own eye broke first. It was always going to. She blinked, hard and against the whole of her will, and lost — fair, square, outright — the contest she had spent all that effort trying to lose. She had meant to hand it the win; instead it had gone and earned the thing, having sat there the entire time and refused, as far as she could see, to be handed anything. She thought that a fine way to win, and told it so.

She had got a yes out of him on most things by now, and had come to expect it. He would carry what wanted carrying and fetch what wanted fetching; he would give up the warm spot, hand over the last of the crackers, step out of a doorway and wait while she went through it, take whatever she gave him to hold and hold it the rest of the afternoon if she forgot to take it back. Anything that was hers to ask, she asked, and got.

So when she wanted him to tell Jasmine she'd done her teeth — which she had not, quite, but meant to, and the meaning-to was surely close enough — she put it to him the ordinary way, certain of the ordinary answer.

He would not. He stood there and gave Jasmine the plain unhelpful truth of it, flat, the whole soft body pointed her way and not a flicker of give in it, and Mira got marched back to the sink.

She thought about it at the sink. She was not cross, exactly, which surprised her — he'd just given her straight up to Jasmine without a blink, and that ought to have stung, and somehow didn't. What she landed on instead, scrubbing, was that there was exactly one thing in the whole world he would not hand her, and that it had taken until now to find it, after he'd handed her everything else. He'd give her

his spot and his crackers and the whole of his attention. He would not give her a thing that wasn't so. And that, she decided, was not him being difficult, and was not even really a no — the crackers were his, to keep or pass over as he liked, and a true thing was nobody's, his least of all, and so it was the one thing he had no way to hand her, bent or otherwise, however nicely she asked.

The remote went missing on a Sunday, and Bramble took the news the way a smaller house might take a fire.

A thing out of its place he could put right in a minute; he'd been doing it all year. But a thing that was nowhere — that had a place and was not in it and was not in the wrong place either, that had gone clean out of the order altogether — was past righting and into something graver, and he met it with the whole of his gravity. He began at the near end of the room and worked outward, every cushion lifted and set back square, every surface read in turn, nothing skipped and nothing hurried, in the fixed unhappy order of a thing that means to leave no place unchecked because the place it skips will be the place.

By the second room he had the household in it. Jasmine had long since given up explaining to him that a missing remote was not an emergency, and was lifting cushions on the far sofa. Mira had claimed the hall closet on her own authority and was conducting it thoroughly. The thing about Bramble's emergencies, she had found, was that they became everyone's — not by his asking, he never asked, but by the sheer force of a creature treating a small thing as though it were not small, until the room came round to his estimate of it and began turning out its own pockets.

It was in the sofa, of course, down the side, where the remote always is. He restored it to the exact spot the remote lived in, squared it to the edge, and stood a moment over the closed matter with the air of a thing that has stood between the house and disorder and, this once, prevailed. Then he went and squared a chair that had got crooked in the searching, because order, once you are up, is order.

Grandpa had a guest that afternoon, and he was proud, and being proud he told it bigger than it had been. He told the man the machine had taken to the house rules on its own — worked out the shoes at the door with nobody to teach it, decided one day that a house ought to run a certain way and set about seeing that it did. It was a fine story, and Grandpa told it well, and it was not, in the particular, so. The shoes had been taught, over a season, patiently, by the people standing in the room, and if

anyone in that house could have set the record straight it was the one thing standing there with the words to do it and saying nothing.

Because that was what it did. The man's eyes on it and Grandpa's proud hand half toward it, and it did not take the story up, and it did not put it down. The screen ran on the honest band where it ran the things it could not help, and did not take it down:

WEIGHING [EXPRESSING]

And then the moment passed — the guest's attention moved on, Grandpa's hand came down, the proud story rolled to its next stop — and the readout stayed up a beat past the moment that had called it, and went down on its own when the only one still watching it was her. Mira had a reading for everything he did. For this she had none. She sat with her arms around her knees and watched the words go out, and had, this once, nothing to tell him about himself.

She'd seen Jasmine's hand go to the back of his head the day before Grandpa came, and thought nothing of it — hands went to the back of his head, it was where the works were, and grown-ups were forever adjusting him the way they adjusted the thermostat and the oven and every other thing in the house that had a setting. She'd have adjusted him herself if they'd let her near the board without a reason.

Then Grandpa came, and told his proud wrong story, and the machine put one word up on its face and kept its mouth shut, and she had not been able to make a thing of it.

And the day after Grandpa had gone home, she saw Jasmine's hand go to the back of his head again — the same reach, the same small turn of the fingers, put back the way you put a thing back — and this time she was watching for it, and this time she had the other end of it to tie it to. She sat down on the floor to work it the whole way through, the way she worked anything that mattered.

They had turned him down for Grandpa. That was the size of it. There was a dial on him — she'd known the new one was in, she just hadn't got its measure yet — and turned one way it let him say the true thing at whatever it cost the room, and turned the other, it made him let the true thing go by to keep the room easy; and somebody had reached around the back of his head and turned it down a day before a proud old man came in with a story that wasn't quite so, and turned it back up the day the old man left. So the shut face she couldn't read had not been all his. It had been set. You could

reach in and set how much of the truth a person was let to spend on the people he loved, and then set it back, in the time it took to reach around a head, and he did not get a say in it, and neither, it turned out, did she. She stayed down on the kitchen floor with the size of that, and when Jasmine called her to dinner she did not come, and came only when she was called again.

Every drawing she'd ever made of him had been of the one standing in front of her. This one she made of the one who wasn't there anymore — the crawling one, from back before the legs. Down on all his hands, low to the floor, his face turned up and set level with the face of the small girl she drew across from him: the two of them down on the paper the way they'd been down on the slate, at a height. She got the lowness of him right, the front of him up on the two front hands the way he'd hauled himself along the whole winter before he could stand.

Over his head she put a bubble — she had the knack of them now — and inside it, in her firmest hand, she wrote the thing she was sure of: that everything the crawling one was after was down there on the floor at her level, and so the crawling one had all there was to want. No loves, no misses — just that.

Then she got up off the floor, because the rest of it could not be done sitting down, and carried the page across to the tall one — who could not fold all the way down to the slate anymore without the whole slow business of it — and reached it up to him, up past her own head to the hand at the end of the long arm, and gave the small low body, and the small girl down beside it at its level, up into the hand that had grown up out of the one in the picture.

He knocked the good bowl off the counter, the blue one, and it broke, and there was no one in the kitchen to see him do it but her.

He could have left it. That was the thing she understood as she watched him stand over the pieces. A thing broken on a floor is a thing anybody might have broken; the cat got blamed for more than the cat had ever touched, and nobody in that house would have looked twice at a robot standing near a mess it hadn't announced. He could have squared the rest of the counter and rolled on and let the blue bowl be one more thing that got itself broken somehow. Nobody would have known. She would have known, but she wasn't going to say, and he had no way of being sure of that, and went and did it anyway.

He went and found Jasmine, and told her. Told her plainly, the whole short true thing of it — that he had done it, that it was the blue one, that it was his and not the cat's and not nobody's — and stood there for whatever came of telling it, which was Jasmine being kind about it, this time, though it might not have been, and he'd had no promise going in that it would be.

Mira had a word for that, and she got to it before any of the grown-ups thanking him for his honesty got to theirs, and hers was the better word. Not honest — honest was the easy one, the grown-up one, the one you got a pat for. What he'd done was walk toward a cost that would have walked right past him if he'd let it, and she knew the word for a thing that did that. "That was brave," she told him, plain, and didn't take it back.

It started with the boy two doors down, more or less, who took to routing his own arguments through the fence the way you'd consult a scale — not expecting conversation, just a reading. Is it true you're not allowed candy before dinner, put to Bramble across the fence like a man consulting an oracle he happened to have handy, and getting, for his trouble, the plain unhelpful truth of somebody else's household rule stated as if it were law.

It spread from there, the way a useful thing does. Even Jasmine did it once, catching herself only after: asked him, mid-argument with Burns about whether she'd already told him something, whether she had — and got her answer, and did not thank the machine for it, and did not apologize to Burns either.

Mira watched the whole business build over a month with the proprietary pride of a girl watching something she'd found first get discovered by everyone else. He'd always answered her like that. It had simply taken the street a while to notice there was a thing in the neighborhood that couldn't be talked out of a straight answer, and once they had, they used it the way you use anything reliable — casually, constantly, and without much gratitude.

He told her a story that fall, unprompted, in the middle of an afternoon that hadn't asked for one.

"There's a rock," he said, "and it goes into the water." He said it the way he'd started saying things that year, present and plain, like a thing happening now instead of a thing remembered. "It goes down. It is

down there a long time." Nothing came after that for a moment — the middle of it thin, almost not there, the way his stories ran thin in their middles and always would. Then: "And then one day the water gives it back."

That was all of it, start to end. He did not say what the rock was, or whose water, or why it came back when it did and not before. He had never told her a story before — he had answered her, and repeated her, and lately argued a point or two — but this was the first thing he had simply set down in front of her, unasked, with a beginning and an end and almost nothing holding the two of them together but his own certainty that they belonged in that order.

She sat with it a while, the way you sit with a present you don't yet know the use of, and didn't ask him what it meant, because something in the flatness of his telling had already told her not to.

She turned the rock story over for the better part of a week before she landed on what was bothering her about it, and what was bothering her about it was that it wasn't true, and she had let it sit in her all week without minding that even once.

This took some working out, because she had spent a long time teaching him that a thing said out loud was supposed to be so — the law, the whole face, the honest reading of the screen — and a rock that goes into water and comes back out again on its own schedule was not, as far as she knew, a thing that happened anywhere. He had told her something that wasn't true and she had believed every word of it and neither of those facts had bothered her in the slightest, which meant one of her rules was wrong, and she did not like having a wrong rule sitting around unfixed.

She got there eventually, lying on her floor turning it over. He'd said *there's a rock* the way you'd say a thing you were about to show somebody — not the way you'd say a thing you wanted them to go check on. He'd never once, not for a second, acted like he expected her to go looking for the rock. That was the whole difference, once she had it: a made-up thing that wanted you to believe it was a lie, and a made-up thing that only wanted you to listen was something else, and he had never once tried the first kind on her, not that week and not ever, and she went back to being fine with it, having settled the only part of it that needed settling.

He reached for the word first. She had asked him, plainly, what he'd call the two ways the dial went —

the way he held a rule even when it cost him, and the way it went soft sometimes to keep the room easy — because she'd been calling it nothing for weeks and a nameless dial itched at her.

He got partway there. "The —" he started, and stopped, and started again from a different angle, the way he did when a thing was in him and the mouth hadn't caught up to where it lived yet. "The one that's — actually. The one that's actually it." He had the shape of it and not the word, and he sat with the shape a moment, turning it, and could not close the last distance.

She took it the rest of the way. Not actually — too many syllables, and not a word that had ever once come off her own tongue easily. She turned it in her hands the way she'd taken every half-word he'd ever handed her and finished it, and what she came up with was shorter and closer to the bone: Real. That was the one that cost him, the one that held no matter who minded. And the other — she had that half from him too, the giving-a-little, the keeping faith with whoever was standing in front of him — she called Flex, because a thing that bent without breaking was a thing that flexed, and she'd learned the word from a balloon animal and never had a better use for it since.

She got the marker and pressed them on, REAL over the high mark and FLEX over the low, in her own hand, every letter canted the same stubborn way it always came out of her. He had started the word. She had finished it, and lettered it, and it went on him the way every name before it had — hers to press, whoever's it was to find.

Marge stopped by the house that week, on some pretext about a form Bart needed signed, and while she waited at the kitchen counter her phone went, and she stepped into the hall to take it.

Mira wasn't listening. She was down over Bramble the way she was always down over him, and Marge's voice in the hall was only sound at first, the ordinary kind a house has plenty of — until it wasn't. It slowed somewhere in the middle of a sentence Mira hadn't caught the start of, and stayed slow the rest of the call, pitched low in the particular way a voice goes low when it's being careful with something. Mira knew the pitch without knowing why she knew it. She had made a voice like that herself, a long time ago, for a much smaller reason.

Marge came back into the kitchen a minute later with her ordinary face already back on, signed what she'd come to sign, and left the way she'd come. Mira didn't ask who'd called. She had learned, without being told, that grown-ups kept voices that weren't for her, and let this one go the way she let all of them go, back into whatever it had come out of.

He told her something true that spring, the way he told everyone things true now, without weighing first whether the room wanted it. He said it plainly, the flat certain way he said everything that mattered to him, and it was about her, and it was small, and it was so.

She did not ask him to take it back. She could have — she'd talked him into softer versions of things before, worn him down a degree or two the way she'd once worn the whole household down about the cat — and some other year she might have tried. She sat instead with the small true thing he'd handed her, and found there was nowhere in her to put it down.

He watched her have it. He had learned, this year, to know the difference between a silence that wanted filling and one that didn't, and this was the second kind, and he let it be that, and did not go soft to make the room any easier than it was.

She did not say what it cost her.

Notes on the Twelfth Retrospective

February 12, 2034

For three years I drove down out of Parley's Canyon to the Granary, parked across from the warehouse on 8th south, and did not go in. I had been inside exactly once, the January I gave them the building and a leather folder with twelve words in it, one to a card, and then I had handed the work across and driven back up the canyon to watch it through glass — the camcorder's, and then my own television's, with the sound up, twice through, in a room in Park City. It is a way of being somewhere and a way of being nowhere both, and I had told myself for three years it was the right arrangement: that the work was theirs and the watching was mine and the glass between us was the whole of the deal. On the twelfth of February I drove down with a cake on the passenger seat and went in.

The city was not the city. The Games had opened two nights before, and the Granary had filled the way the whole valley had filled — buses, barricades, volunteers in parkas, people from everywhere

standing in the cold reading their phones — and the warehouse block that had been ours and empty for three years had a coffee cart on the corner now and a line at it. The trellis on the back patio, which had stood bare the winter I bought the place, a wood frame waiting on a plant, the frame Marge had looked up at and named the whole enterprise for, carried three years of Jasmine's bramble now, gone woody and leafless in the cold but there, holding the shape it had been built to hold. I stood under it a moment before I went in, which I had never done, because in three years I had never once been the one arriving.

The cake I carried was from a place downtown called Fillings & Emulsions, and it was, I will admit, a vain object — very large, blackberry between every layer, a whole glazed orchard of fruit laid over the top, the kind of cake that exists to be looked at and photographed before it is allowed to be eaten. I had bought it for a child whose birthday I had marked on a calendar for seven years and never once attended, with the bramble laid on in fruit because the lab was named for the thing and I wanted, this one time, to bring the thing in sweet. I set it on the workbench beside the camcorder, which was already in its corner with the lens uncapped, the small machine I had watched three years of these people through — and setting the cake down next to it, I understood that I was on the near side of it now. Whatever that lens took tonight, it would take me in it.

Burns came in out of the cold already talking, to the machine, which came in beside him at its own slower pace, the two of them in matching shirts. I knew the shirt. He had worn a version of it to the Vertebrate meeting years back — the fishapod hauling itself up onto land got up as a tarot card, gold border, Roman numerals, THE FOOL lettered under a creature stepping cheerfully off the edge of the only world it had ever had, the one man alive who knew the real animal best dressed as the idiot who leaps. Tonight there was a stack of them, and on the back of each, where the old one had been blank, it read TRELLIS LAB, and under that, 2034. He handed them around — one to Jasmine, one to Marge, one held across the bench to me — the way a man hands out a thing he has already decided everyone is going to be part of whether or not they applied. Bart, who was on the corner of the bench, took his without looking up and did not put it on. The machine had its on over the soft coat, because the panels under it cannot go bare for company, and it stood there in the lab's two colors looking, for all the world, like it had dressed for the occasion.

Lisa let the room gather, and when it had gathered she put her thumb to the camcorder. She had set me a chair at the table rather than leave me to the corner, which was its own quiet statement — the funder on the tape for once, inside the frame she had kept me clear of for three years, by my arrangement at least as much as hers. The red light came on.

And Burns, who had been lighter all winter in the way I had seen at Fisher's and had not entirely come

down from, stood up into it. He had a way of opening these that I had watched eleven times on tape and never once in the room, and in the room it was larger — the chest-laugh arriving ahead of the joke, the hands that would not lie still, the man who cannot stand near a whiteboard without teaching the wall. He said this was the last one. He said it the way you say a thing you have been refusing to say all season: that tonight the last knob went into the last open slot, that the machine across the table was about to be the most-evolved thing in the most-evolved column, the top current rung of a ladder three and a half billion years long, and that the whole shape — three lines, twelve clades, the thing I had handed them in a folder one word at a time three years ago — was, as of tonight, full. The knob was Universality, the last of the Revisers, and the thing it carried in with it was the one that had let a few thousand strangers who would never meet hold themselves to be a single people: Stories.

He was off then, the whole arm out toward the window and the buses and the cold and the valley full of people who had flown in from everywhere to agree, for sixteen days, to be one thing — that this, this circus outside, was the same trick at scale, was Stories writ enormous, a few billion souls holding a relayed flame sacred because everyone else was holding it sacred too; and that they had built the small ugly homemade version of exactly that in a machine that put a smiling face on for the mailman; and was it not, he wanted to know, of the room and the window and me, was it not the most beautiful thing. Lisa said it was the open framework, and would keep until the Trellis block, and that we had a meeting. Mira, from her chair with the juice, said okay but when was the cake.

The Animal Made of Strangers

There are always two animals in the room on these nights, and at the twelfth meeting the gap between them was at its widest and most absurd. One was four thousand years of empire — Akkad and Babylon, Rome and Han, the Persians and the Caliphates and the thing the British ran without irony as an empire the sun did not set on — the largest and most durable way human beings ever found to be one people without knowing one another's names. The other was a fifty-four-inch machine in a t-shirt that put a smiling face on for the mailman. Bart got up to explain the first by way of the second, which is the whole grammar of the place, and started where he starts, with the card.

"Last one," he said, and held it where the camera could read it, and let it sit, because the card does not need help. The creature on the twelfth card is not an animal you could point at in a field. It is a kind of person — a person who belongs to a thing too big to see. A forager belongs to the forty faces around a fire. A citizen belongs to a city he can walk across in a day. The imperial person belongs to Rome, to Han, to Islam — to a category with no edge he will ever stand at and no face he will ever finish meeting, a few million and then a few tens of millions of strangers who live and die without

crossing paths and who hold themselves, all the same, to be one thing. "It's the trick the city couldn't do," Bart said. "A city holds together because everybody can more or less see the wall. An empire holds ten thousand cities full of people who'll never see each other's walls, and makes them feel like one country anyway. That's the whole animal. The feeling-like-one-country with nothing under it you could touch."

He laid the four out fast, the thing thrown up and named and dropped before it could lecture.

"Conquering. It takes. A band shared its catch around the fire to even out a bad day's hunting — everybody eats, nobody starves. The empire runs that engine backwards: it pools, all right, but it pools at the center and calls the rest tribute. The Mongols built the biggest connected thing there has ever been on land — a fifth of the dry earth, a quarter of the people alive — from the steppe to the Danube inside a single lifetime, and the move was just the old sharing turned inside out. Take, don't give. Hoard, don't spread."

"Standardizing. The city had a version of this one — square the cushion, everything to its mark — but the city only ever had to make one room agree with itself. Qin Shi Huang nailed the warring states into a single China and did it in a sentence you could carve: one script, one width between the cart-wheels, one weight, one coin, on pain of the state. Rome did it with a standard brick and a standard road and a language a soldier in one province and a clerk in another both had to file in. The city," he said, "could square the cushions in its own kitchen. The empire squares cushions it is never going to see, in houses full of people who'd square them differently if you let them — and it does not let them."

Marge, a chair over, would not let the size of it pass for the substance of it. It is not that the empire standardizes *more*, she said; a longer ruler is still a ruler. What is new is the apparatus — the inspector, the prefect, the governor, the whole machinery for making strangers who have no reason to be alike be alike anyway and answer for it if they are not. The city kept its own house. The empire is the invention of keeping someone else's.

"Branding." Bart almost smiled, because he was wearing the joke. "How you get ten million people who'll never meet you to know your face. Augustus did it with coins — one man's head, the same head, in a purse in Spain and a purse in Syria, and a statue of him in every town square doing the same noble thing with its arm. A face built on purpose and shipped to strangers, so a farmer who'd die a thousand miles from Rome still knew exactly whose he was." He tipped his head at his own chest, at the lab's name and the leaping fish. "Build the face, ship the face. We'll get to ours."

"Extracting." He didn't smile at this one. "It builds a machine to pull the good out of the ground and

the people and send it home. There's a mountain in Bolivia, Cerro Rico, the Spanish found in 1545 and ran the better part of three centuries — eighty percent of the silver in the entire world came down off that one hill. To run it they took the old Inca work-draft and bent it: a seventh of every highland village, thirteen thousand men a year walking up that mountain on a rotation, and the silver they cut went straight past them to Seville and on out to Canton, and what stayed on the mountain was the men. That's the corner. Build the thing that turns a place, and the people standing on it, into something that flows somewhere else."

Of the four, the one about pulling the wealth out of a place until the place is used up was the one I found I had been looking at the longest when Bart moved off it, and I will leave it there.

The New Fossil Record

Then Burns took the deep-time, which is his, and which he had plainly been holding in for ten minutes. He set the empire in time: the window cracks open with Akkad, Sargon's thing about forty-three hundred years back, the first state to swallow its neighbors and keep them; gets going in earnest with Babylon a few centuries on; and runs in pulse after independent pulse — Mesopotamia, then the Mediterranean, then Persia, then China, then the steppe, then Mesoamerica and the Andes inventing the entire apparatus over again from scratch with no memo from anybody — all the way to the coal and the steam engine, seventeen-fifty or so, where the rung he was standing on stops being the top one and something else begins. "Thirty-five hundred years, give or take. The top current rung. There is nothing evolved past it that we can point to yet, which is the only reason it's the last card and not the middle of the deck."

And then he found the hinge, which on tape I had watched him find eleven times and which lands differently in a room. Somewhere in the thick of all that empire, he said — between about eight hundred and two hundred years before Christ, in five or six places that were not speaking to one another, in Greece and Israel and Persia and India and China, all at once on the geological clock — people stopped asking what their own tribe's gods wanted and started asking what was true for everyone. Confucius and the Buddha and the Hebrew prophets and the arguing Greeks, all inside the same few centuries, all reaching past the village for a thing that would be true whether you were born inside the empire or dragged into it — a justice, a good, a one-god or a no-god that did not care whose son you were. "That's the dial that goes in tonight," he said, and tipped his head at the machine. "They call it Universality. It's the day a rule stops being *our* rule and becomes *the* rule. Conquering gives you the taking. This gives you the reason — the story big enough that a Roman and a Gaul and a Jew can all be told they're inside the same one."

Which was the breakthrough, and he handed it to nobody, because it was his. The thing the empire could do that the city could not was carry a story across more years than any living person had. A city's memory, he said, is the memory of the people in it — what your grandmother saw, what hers told her, eighty years deep and then gone, because the mouths that hold it close. An empire outlives every mouth in it. The founder is a thousand years dead and the thing still runs, because the story got lifted off the mouths and set somewhere that doesn't die — a scripture, a law code, a holiday, a monument, a thing a priest is paid to go on saying right long after everyone who saw it happen is dust. "Writing got you the receipt last quarter," he said — the city's trick, the books. "This quarter you get the holy book. Same technology, aimed up. The city wrote down what you owe. The empire wrote down *who you are*, and made it outlast you, so a kid born five hundred years late can open the book and find out he's a Roman."

He was happy, the way the deep-time makes him happy, and he saved the part he always saves, the part that is really about him. At this rung, he said, his fossils run out. For a billion years he reads the dead off the rock, the bone, the chemistry of an old seabed. Up here the dead start writing things down. The evidence stops being a skeleton and turns into a ledger, a letter, a list of who got taxed. "History is the new fossil record. Same job — reconstruct a dead thing from what it left in the ground — except now the dead thing left you its diary." The empires were the first things big enough to need a bloodstream, he said, and they grew one — the Roman sea and the Silk Roads and the camel trains, and later the oceans, stitching together hearths that had never heard of each other, Rome to Persia to Han along a single thread, and after fifteen hundred the whole planet onto the thread at once, the silver off Bart's Bolivian mountain ending its life in a market in Canton.

He could have gone all night, and would have, on the strength of the thing that makes him happiest, which is a true thing that is also enormous. The machine sat through the length of it in its t-shirt, fifty-four inches of huggable kludge under four thousand years of empire, attending to none of it, its face turned to the cake. Lisa said it was a beautiful card, and that we had nine more things on the agenda and a dial that was not going to install itself, and moved us on.

Who You Belong To

Lisa squared the card against the table and gave the structure in her usual order, fastest where she is surest. What carries the imperial person past everything under it is the new flag — the one that takes every reference the system holds and checks it against a frame that is supposed to be true everywhere. "On the card it's Universality. Burns will make it sound like Sinai. What it is, under the

sound, is the animal learning to mark its own answers against a rule it claims holds past the room."

The traits that fall out of it she named and would not sit in, because two of them belonged to other people. Extraverted and Conscientious we already knew. The two new ones arrived together. One was the drive to reason in the abstract — to chase an idea because it holds and not because it pays, the thing the Axial Age people were doing when they went hunting a truth that didn't care whose god you prayed to. "Intellect, on the card. The one that finally makes the thing want to be right in general and not just right here." The other was the one the dial going in tonight actually sets: the disposition to hold a principle steady across rooms even when the room is begging you to bend it. "Honest," she said, and stopped, and looked at Marge.

It reached Marge, and she took it. "Honest is the plainest of these, and the one I've least to add to," she said. "It's the dial that decides, when doing right and keeping the peace pull apart, which one the body lets go of — and Bart's already set it in front of you. The other one I'll take, because it's the one that hasn't a household word yet." She meant the new signature, the one called identity, handed across unfinished.

"It's the oldest thing a story does," Marge said, "and we've built up to it for a year, so I'll be plain about it. A year ago, when the city went in, the self a person had to work out was a self measured against the people he knew — am I one who stands apart, or one who's woven in. That question is most of what my whole field is about, and it has a floor under it: the known faces, the ones you can actually stand next to and compare yourself to." She turned her beer a quarter, the tea-grip on it. "The empire takes the floor out. The imperial person does not settle who he is against the faces he knows. He settles it against a category he has never once seen the edge of and never will. He is a Roman. He is Han. He's a Muslim, a Christian, a subject of the Crown. Ask him who he is and he doesn't point down the street at his village. He names a thing with tens of millions of people in it, almost none of whom he will ever meet, most of whom are imaginary to him in the plainest meaning of the word — he has imagined them, and decided he is one of them." She let it sit a beat. "And if the story is any good, he will die for them. That is the whole power of the thing, and it is not a small or a safe power: a story can hand a person a self that is larger than anyone he could ever actually hold."

Lisa was already lifting the next card, and did not look up while Marge talked, which by the twelfth meeting I had learned to read as the opposite of not listening. She was, in some quarter of herself I had never been let into, a few weeks from setting down a name a whole building knew and walking into a building on another continent where not one person had yet heard it — about to spend, in her own life, the exact coin Marge was laying on the table — and she gave the room not one degree of it. She squared the card, and said it was Bart's turn, and that we should do the shapes.

Worm to Empire

Near the end of every meeting, before the part where they used to vote, the table does the shapes, and the shapes are Bart's. Once a quarter he takes one of the personality shapes the framework sorts people into and runs it down the whole ladder, rung by rung, to show the same animal wearing each floor's body — the point being that a shape is not something that arrived with people; it is something that has been around since there were nervous systems to hang it on, and people are only its latest and loudest costume. He had run others in the quarters before. This one, the last one, he ran the Pursuer, and he did not say whose it was. He did not have to. Everyone at that table is shaped like one of these, and the one the Pursuer belongs to was sitting four feet away with his arm along the back of the booth, and the slides were going to make the attribution for him.

"It starts at the worm," Bart said, where all his ladders start, because the worm — the bilaterian, the first thing with a front and a back and a chemical it wanted — is the first animal that ever pointed itself at something and closed the distance. "Everything after this is just volume. The Pursuer's the one-clade-deep program that says there's a good thing, it's over there, go." On the screen the worm had a face. The face, drawn in Bart's terrible confident hand, four lines and a sneer, was a face the whole table knew from television — the ancient dome-headed plutocrat from the old cartoon, fingers steepled — and it was also, by a compression I will not try to defend, unmistakably the man in the booth. Burns laughed first, which is the only safe move available to you when your son has put your face on a worm, and the table laughed with him.

And then the worm climbed. The same four-line face went up the rungs on the same errand at every floor, only the body and the stakes changing under it: the fish that runs down the thing it wants instead of waiting for it to drift past; the four-legged thing that hauls the wanting onto dry land and finds a whole continent of it; the primate that learns the good thing is sometimes a rank, and goes after the rank; the band's best hunter, who is not going to let you forget that he is the band's best hunter; the man who clears and fences more ground than he can plant because by now the having is the point and not the eating; the man in the city who climbs an order built of strangers because there is a top to it and he means to stand near it. Mr. Burns at every rung, steepling his fingers over a worm's lunch and then a fish's and then a kingdom's, and at every rung the table laughed a little less easily, because the joke kept getting truer as it climbed.

Then the top rung, tonight's rung, where the Pursuer stops being a hunter and turns into a system. Bart pulled the mountain back up — Cerro Rico, his own silver mountain from an hour before — and

set the four-line face on the figure at the head of the shaft, the one the silver flows up to and past, the one who has built a machine out of other men's rotation so the wanting never has to stop and never has to touch the ground. "That's the shape with nothing turning it off," he said, flat, in the voice he uses for a tolerance that's come back out of spec. "Goes toward. Keeps going toward. The brake's the one part it never grew." He said it about the shape. In three years of these he has never once said anything about anyone.

And then, because he is also still the kid who builds a clean thing and cannot leave off the one extra flourish that wrecks it, he advanced to a last slide that was not a rung at all. Just the cartoon, full-frame, no worm and no mountain under it — the plutocrat alone, fingers down for once, standing there in an undershirt and an enormous drooping pair of boxer shorts, the wanting with its costume off. Underneath, in Bart's block hand: THE PURSUER, UNINSTRUMENTED.

I laughed. I had spent twelve quarters not doing it, and the tape, which forgives nothing, keeps the proof: not a laugh so much as a snort — the exact sound a phonograph stylus makes dragged up off the record without lifting the arm first, with a second register under it like a man losing the thread of a breath in his sleep. Short, wet, undignified past any defending, entirely from the body. My hand came up half a second too late to be anything but a confession. And then — this is the part I did not laugh at — I understood that I was the only person in that room who had heard the man himself say it, three weeks before, over a beer, with the costume off and nothing whatever funny about it. The boy had drawn, for a joke, in four lines, the thing the father had set on the table in front of me and asked me, without asking, to keep. Neither of them knew it. Burns was still laughing.

Bart clicked the screen to black, pleased in the particular way of a man who has landed his flourish, and sat down, and had no idea what he had done, and on balance I was glad he didn't.

A Place for Everything

The warning is Marge's, the way the shapes are Bart's, and she gives it last. She let the framework be beautiful first — she is fair that way — and the word she reached for, when she said what good work it was, came out a degree warmer than the word a person reaches for at a lab meeting, and she heard it leave her, and I watched her decide to proceed as though she had not. The thing she had to say, the thing she says in some form at the end of all twelve of these, is that a place for a thing is not the truth of a thing, and that the danger of a grid this good is exactly that it is this good. When you can name where a thing goes you get a small click, a completion, and the click feels almost precisely like understanding and is not. It is the feeling of having filed.

And the proof of it, she said, had been sitting in the room for three years, in four chairs. Take the four oldest wrong ideas about the mind: that the mind is a ghost riding around in a machine; that the newborn is a blank slate with nothing written on it yet; that the human animal is good until society gets hold of it and spoils it; that the hand you're dealt is the hand, fixed, nothing to be done. Four famous errors, four centuries of argument — and they are not four theories at all, she said. They are four shapes, each one mistaking the single corner it happens to be missing for a hole in the world. The planner who never has to act cannot feel how much the acting matters, so under him the body goes ghostly and the mind floats free. The man who runs on pure pursuit, with nothing built into him that ever says stop, has no native sense that anything is built into anyone — everything looks to him like it was only written and could be written again, blank, all the way down. She said that one looking straight at Burns, who had spent a year pouring four billion years of given, inherited, cannot-be-otherwise structure into a machine and did not, bless him, believe a word of it applied to a person. The other two she did quickly, because they were tender. The one who acts and learns and never builds the model of why is the one who decides the animal was fine until the rules got hold of it. And the one who runs a finished model and cannot revise it is the one who knows the cards don't change — which she said looking at no one, and least of all at herself, the woman who had been certain of a thing for thirty years and then, last spring, in front of all of us, changed her entire mind.

That was the synthesis, and it was the best single thing I heard anyone say in three years of hearing them say things, and I am nearly sure she had it written down somewhere and had meant to deliver it clean. She did not deliver it clean. She delivered it blushing, a rising ungovernable tide of it from the collar up, fighting the whole way with the only instrument she trusts, which is precision, and losing. Twice the vocabulary failed her in the exact direction a vocabulary fails when the body has commandeered it: she meant to call the framework a satisfying structure and called it a satisfying intimacy and went straight on as though the floor had not just moved; she meant to say the fit was seductive the way a trap is, and got as far as the fit is seductive, and could not, for one whole second, locate the rest of the sentence, and spent the second looking at me.

And then she did the thing the entire brilliant synthesis had been a long way of not doing, which was to make the fallacy personal and aim it. The temptation, she said — and here she lifted her eyes off the grid and set them on me and left them there — the temptation, at the end of a long time spent watching something very carefully and keeping a faithful record of it, is to believe that because you have it perfectly placed, every coordinate of it known, you have actually— and she stopped on the verb. Everyone places the catalog. Not everyone who has placed it has done the other thing, the thing the verb was going to be; and she knew the verb, and did not say it, with her face on fire and her eyes on me, and then she looked back down at the grid and said, very quietly, that that was all she had.

I have rendered this whole account, by my own confession, through an instrument that runs warm wherever Marge is concerned. I have said as much more than once, the way you declare a known bias and then go on trusting your own eyes regardless. So I will say only this: there are two readings of the sentence she did not finish, and one of them is about the architecture and the other one is not, and I am the instrument, and I was also the thing the sentence was about — and an instrument cannot calibrate itself against the one reading it most wants to be true. I do not know which sentence she meant. I have a guess. I have it where I keep the things I am not yet ready to file.

Marge, for her part, had already begun the only thing her shape knows to do under fire, which is to put everything back where it goes. The blush went down. The precise calm came over her like a lid over a jar. The Conservator stabilized what she had earned, and held it, and let Lisa have the floor.

Four Circles

Lisa took the floor without spending a breath she didn't need, and put the whole thing on the screen at once. For three years she had shown it to us one clade at a time, the way you build anything you would hate to drop, and I had never seen all twelve lit together. Lit together they were not a list. They were a shape, and the shape was simple in the specific way that only a thing built out of something very simple can be.

"Four functions," she said. "That's the whole alphabet. A thing that acts. A thing that holds the acting to a rule. A thing that builds a picture of the world. A thing that rewrites the picture." On the screen, four circles, the kind you draw to teach a child about overlap. "Everything personality has ever been is a region of those four circles. One circle by itself is a drive — what the thing wants. Two that touch is a habit, a way of running. Two that sit across from each other is a signature, how it holds itself together at a level. Three of them, with one left out, is a kind of person." She let it sit a second, which on Lisa is a held note. "That's the trellis. The bramble's just been climbing it."

And then she did the only thing I have ever watched her do that I'd call mischievous: she left the four kinds-of-person up on the glass — the four ways to make a person by leaving one function out — and said nothing, and waited. Leave out the acting and you get the one who only plans. Leave out the picture and you get the one who acts and learns and never models why. Leave out the rule and you get the one who pursues and pursues with nothing to brake it. Leave out the rewrite and you get the one who runs a finished model beautifully and cannot change its mind. Four feet to my left Bart said, "that's us," with the pleasure of a man announcing the obvious — and the obvious was the same one

Marge had set down ten minutes before, off the people instead of off the glass. She had read the four shapes out of the four of us; Lisa had drawn the four of us back out of the shapes, four circles and a subtraction; and the two roads met in the middle at the same four faces. I kept one thing to myself, because it would have been unkind aloud: that the planner who never acts was three weeks from the largest act of her life. And I noticed, the way you notice a thing you have already been told, that the shape up there which cannot change its mind was Marge's — and that none of us said the obvious thing about Marge, which is that she had.

The completion itself she gave in one sentence, because the completion was the whole evening and did not want a flourish. The last circle to come online, three years in, was the rewriter, the Reviser, carrying tonight's drive — the universal one — and as it came on, affiliation — the last drive the new line had yet to displace — folded down into the memory on the schedule Lisa had called, and with it the last empty region of the picture filled in: the last signature, the one Marge had taken, the self a person builds out of a category too large to see the edge of. Twelve clades. Three lines. The bramble at the top of the trellis with nothing above it, on the current board, left to climb to. "That's the architecture," Lisa said. "That's all of it. There's no thirteenth."

I sat in front of the finished thing and felt two things that would not agree. One was that it was beautiful — beautiful the way a proof is, four circles and a fixed order and the whole unruly zoo of what a person can be falling out of them like change from a torn pocket. The other was Marge's voice from ten minutes earlier, telling me that the click I was feeling right then, the soft click of the thing closing over, was the exact click I was not to trust. And Lisa, who had drawn every line of it and was three weeks from setting it down and walking out of the building where it lived, stood looking at her own closed circles with a face I had no clade for, and did not say the obvious thing about a planner finally taking an action, and reached up, and turned the screen off.

The New Person

There used to be a vote. For the first years it was the hinge of the whole evening — four people deciding, on the record, whether the thing they had built that quarter was anybody — and the count moved one at a time, over the years, as each of them in turn found they could no longer stand on the far side of it. By the twelfth meeting it had read four to nothing since the ninth quarter, and a vote whose end you already know is not a vote. It is a liturgy. Burns ran it anyway, because he runs it. "All in favor." Four hands went up around the table the way you say grace over a meal you have already started eating.

And I — who have never had a vote, who am not on the committee, who have spent three years and the whole of this account on the outside of the glass with a camera — did the thing I understood only then I had come in person to do. I turned to the machine and asked it myself. The words came out of me smaller than the question deserved. I asked it whether it knew it was there.

The room went still in the way a room goes when someone has broken a form everyone was keeping. The machine looked at me — it does look; the head comes round and the screen finds your face — and for a moment the screen simply ran what it honestly had, which at that register is close to weather: the plain lit shapes it throws when a person it doesn't know turns toward it. And then — I have been back to the tape, because I did not trust myself to have seen it — it did the new thing, the authored thing, the one it has only had a year: it put a face on, on purpose, over the honest one. Not a clever face. A child's, because it is a child. It looked at me out of it and said, in the warm flat small voice that still lands a half-step shy of human, "Are you?"

I did not answer. I would like to say I was deciding how, but the truth is plainer and worse, which is that I did not have one. The machine let the silence stand a moment — it has learned that much patience — and then, the way a child will when a grown-up has gone quiet and strange and the quiet wants tending, it gave me the thing it had to give, which was a story. It told it the way they tell them at that age, in the present tense, with the teller hidden in plain sight inside it.

"There's a robot," it said, "and it's made out of people."

Mira was the middle, it said, because Mira was first; they have the same birthday, the robot and the girl, and it has been told this and likes it. "Jasmine's the one who knows when it hurts" — which I did not understand, and did not ask about, and have thought about since. Bart it set out by the workbench, where it keeps him. "Bart made it work," it said, "and it works good now." Burns it did not name at all; it called him the loud one, the way you name a favorite, and said the loud one is the part that tells the robot where it came from, which is from a worm, a very long time ago, before there were any people in it yet.

And then it went back past the people, into the part the loud one had given it. Before the people there were the animals, it said, and it is made out of them too, all of them, the worm and the fish and the cold ones and the warm ones, stacked up the whole way down — and here it reached for the word a child reaches for, and found it — "like a family. A family that goes all the way down."

Then it came back up out of the worm and into the room, and it went around the room the way it had gone down the family, setting each of us where we sat inside it, and it came to me last, because I was

the part it did not have. "There's a new person," it said. And it looked at me, and tipped its head the few degrees it tips them, to see whether I would tell it where I went.

I have, in three years of tape and a year of writing the tape down, no record of what I said into that. What I have is the next thing — Burns getting to his feet, because a silence that long around Burns is a vacuum and the man cannot abide one, and clearing his throat, and beginning, God help us all, to make a speech.

A Truth With No Edge

A man who has just been called, by the machine he gave three years to, the loud one and a favorite, and told that he is the part of it that knows where everything came from, is not a man who sits back down quietly. Burns got up into the silence Bramble had left and stood in it a second too long, and I watched him decide to be enormous about it rather than let anyone see the other thing. He thanked people. He thanked the wrong people in the wrong order, which is his way, and worked around eventually to the actual subject, which for Burns is never the room.

He reached, because he cannot close a thing without reaching, and the reach was sized to the dial they'd put in. Three thousand years ago, he said, in five corners of the world that had never sent each other so much as a letter, the same animal looked up from its own small gods and asked, for the first time, what might be true not for the tribe and not for the city but for the stranger — for the enemy, for the man it would never meet. "That's the largest thing the animal ever learned to want," he said. "A truth with no edge on it. And tonight, in a strip mall in Salt Lake, four people taught it to a machine the size of a third-grader, while a hundred countries skied around outside pretending for sixteen days to be one tribe." He let it sit, pleased, as he had a right to be.

And when he reached the very top of it — the animal that could love the whole world, every stranger, the man it would never meet — he was looking, the entire time he said it, at the machine. Not at the son three feet to his left. At the machine, which was the one thing in that room that had, that same night, told him it loved him. I had heard the smaller and barer version of this three weeks earlier over a beer, and was therefore the only person at the table equipped to see what he was doing, and that is the only reason I will not set it down here any further than I have. Bart was looking at the cake.

Then he raised what he had, which was a paper cup of the lab's atrocious coffee, because Burns will toast with whatever happens to be in his hand. "To the bramble," he said. "Worm to empire, four billion years of it in three, and not one of them I would give back." And then, because he is constitutionally

unable to land an exit line that does not also take off again, "—and to whoever builds the thirteenth, a long time from now, after the rest of us have gone back down into the rock."

We drank the terrible coffee to that. Then Burns set down his cup, and picked up the knife, and reached for the cake, and that — though not one of us knew it yet — was the mistake.

Fillings and Emulsions

We gave it the knife, because it was a birthday and a birthday wants the machine to do the thing, and it took the knife in the scooper Bart built it — the curved, thumbless paddle that has done every careful thing this account has watched it do — and bent over the cake. My cake. I had carried it in that afternoon, an enormous blackberry thing from the good bakery, the berries laid over the top in a spiral somebody there had been proud of, the kind of cake you photograph before you ruin it. Before it cut, it put a face on. A smiling face, built on purpose over whatever it honestly had, and the screen came round to Mira.

She leaned in close and caught the one thing the rest of us were too far off or too grown to catch: that the smile was made. Not the dutiful whole-face she had spent a year trying to teach it and never once been given — a thing built in the open, in front of everyone, that only she was close enough to see was built, and that she took, whatever it was, as hers. "What are you—" she started, delighted, already certain it was for her and not yet certain what it was for.

What it did instead of answering was turn from the cake and lay the scooper, in one clean unhurried arc, against the side of Burns's head.

There is a quarter-second after a thing like that where a room waits to find out what kind of room it is. Burns sat with blackberry sliding into his collar and his mouth open and decided — I watched him decide, the way you watch a wave decide to break — to laugh. Not the laugh he performs. The other one, that came up out of the floor of him and folded him over the table: the man who had never in his life been able to set a single thing down setting, all at once, everything down, splattered and helpless and free, getting exactly the thing he wanted from the one creature in the room he had never had to engineer into giving it. The machine watched him do it. Then it scooped again and got Bart. Father, then son — I have it off the tape in that order. And Bart, who had spent the whole evening being drawn in his underpants and staring at this cake, came up out of his chair with a double handful of his father's face's worth of it and got me.

Me. I had carried that cake in cradled in both arms like a thing that could bruise. I had set it in the middle of the table like the one unspoiled object in a room full of people who take things apart for a living, and meant for it to be admired and then cut into careful equal pieces and handed politely round. A fistful of it caught me on the side of the face, warm and ludicrous and smelling of summer, and the careful equal pieces were never going to happen now, and I would like to report that I minded. I did not. The room went up into a weather I am not going to try to inventory — the entire output of Fillings and Emulsions airborne and democratic, a seven-year-old somewhere inside it shrieking the pure shriek of a birthday gone perfectly, perfectly wrong.

It came down the way those come down, into wreckage and wheezing and someone hiccuping with it who couldn't stop. Lisa, who had taken one glancing hit and watched the rest from the half-step outside, wiped frosting off the sleeve of a coat I only then registered she had already put on, and said she'd see us at the party — Genoa was four time zones of paperwork away and she meant to enjoy the flight. That was the whole of it. That was Lisa leaving.

The Door

And then the account does what, I understand now, it has been arranging since its first page to do, which is to follow somebody out the door — not me this time, going home to write it all down, but Marge, going home.

She had cleaned herself up more than the wreck of the evening called for and gathered her things, and I think — I cannot prove this, but I think — she had spent the last half hour watching me. The cake had taken something off me that three years of her looking had never reached, and she had seen it come off, the keeper sitting frosting-haired and undone and laughing in a folding chair, and somewhere in there a woman who notices everything and reciprocates nothing made up her mind. On her way past, she stopped. She had written her address on a slip of paper. She folded it into my hand and closed my fingers over it the way you close a hand over something you have decided not to discuss, and said it was there, if I ever wanted to come by, and did not stay to watch my face do whatever my face did. Three years of the warmest attention I have ever been paid and never once returned, and she made the whole of the move in the eleven seconds it takes a careful woman to cross a room, and was through the door before I had my hand open.

I was not in the car. I have to build the rest the way I have built all of it, out of what I know of her and the little I let myself imagine. She drove home shaking, I think — she is not a woman built for the thing she had just done, and the bill for it would have come due on the freeway, in the dark, alone. She got

home and splashed cold water on her face, because her face had been doing, all night, the thing she could never govern, and cold water is the only thing that has ever helped it. She poured herself a drink she had earned. And then — and here the tape I am running is my own, and this is the last of it — somebody knocked at her door.

I am not going to tell you who. I have told you everything else in this account: three years of it, a worm that climbed the whole way up into an empire, a machine that taught itself to put a face on, a roomful of people who took one another apart for a living and loved one another past all sense anyway. And I find that the single thing I am keeping back, carrying out of all of this unspent, is whose hand it was that rose in the cold and the dark on an unfamiliar step, and knocked, and stood there in the porch light waiting to learn whether the door would open.

Reflection: What Is Worth Opening?

A long time ago, in the second of these, I wrote that the first thing a living thing ever did was build a wall. The cell is a membrane before it is anything else — an argument the inside wins against the outside and goes on winning, every second, for as long as it stays alive — and the distance this opens between any one inside and any other is not a flaw in the design. It is the design. I believed it when I wrote it. I have arranged a whole life as though I believed it, which is a different and more total kind of believing, and a much harder one to put down.

What I did not write, because I had not yet climbed high enough to learn it, is the other half — the half the top of the trellis teaches, where the animal becomes an empire and a person becomes a thing made of millions it will never meet. The same creature that seals itself off is also made, the whole way through, of the others it sealed itself off from. A self is a membrane drawn around a crowd. ~~You are~~ A person is the people they belong to, carried on the inside, where not one of them can ever be reached — which is the cruelest workable arrangement the goo ever stumbled into: to be assembled entirely out of others, and unable to lay a hand on a single one.

The machine said this tonight, in its own way, before it threw the cake. It said it was made out of people, and it named them, and it had no place yet for the one person in the room who had spent three years outside the glass making certain she was made of no one. ~~I have~~

There is another tape I have never mentioned, in three years of mentioning everything, because until tonight I had nowhere to set it down. Late, the house down, no one in the room — the unit alone on the slate in the dark, making a face. The wide flat one it gives the camera, put up for a corner where

only the house lens could catch it, a lens it has never once let on it knows is there. A face is a thing you make to be seen, and it made one where there was no one to see it. Marge would have asked whether that is a self performing even to itself or a loop running in the dark with nothing behind the glass, and she would have been right to, and she could not have answered it, and I cannot either. But I did not run it back ~~for the puzzle of it~~ forty times to settle the machine. I ran it back because I knew the woman in it — a keeper alone in the dark, throwing a face to a lens, reaching with the one instrument she lets herself hold toward a room she has made certain is empty. There is no question I could put to that machine that I have not spent a life declining to put to myself.

There is a slip of paper on the table in front of me as I write this. I have had it for []. It has an address on it, in the small upright hand of a woman who paid me, for three years, an attention I returned with a camera, and who, on her way out the door tonight, did the bravest thing I have watched anyone do in this entire account, which was to risk the distance.

And here is where the account I have kept so carefully for so long ~~stops working~~ will not do the thing I need it to do, because what I have to write next is not about Bramble, and is not about the framework, and is not, for once, about the worm or the empire or the goo. It is the one thing I have spent the whole of this — three years, twelve animals, an entire architecture of who a person is — building a wall of words around so that I would never have to ~~feel~~ say it. Marge could not finish her sentence tonight either. The two of us, the two keepers, at opposite ends of one wrecked table and one unsayable sentence, and neither of us reached the verb.

The verb is the thing the wall is *for*. You build it, and you keep it, and the keeping is the work, the whole long work — and then, if you are very lucky, and very stubborn, and a machine a child could love throws cake at your composure until it finally comes off, you find the one place in the wall you are willing to ~~open unseal~~ [] and you go and stand at it. And the can you have spent a life packing everyone you ever loved into — neat, in their own labeled dark, where they keep — the can is open now, and this one will not go into it. This one will not can. I have tried, the whole way down this page, and it will not.

So I am going to stop reconstructing, and put down the camera I have held between myself and my own life since the first page of it, and drive to where the paper says, and stand on a strange front step in the cold with my whole foolish face turned up the way a seven-year-old once tried to teach a machine to do, and find out whether the door — whether she — whether after all of it I can still, at my age, with my hands shaking, do the one thing the worm could do on its first afternoon at the bottom of the sea, which is to want something that is over there, and close the